深度学习平台与应用

第十四讲：视觉-语言模型

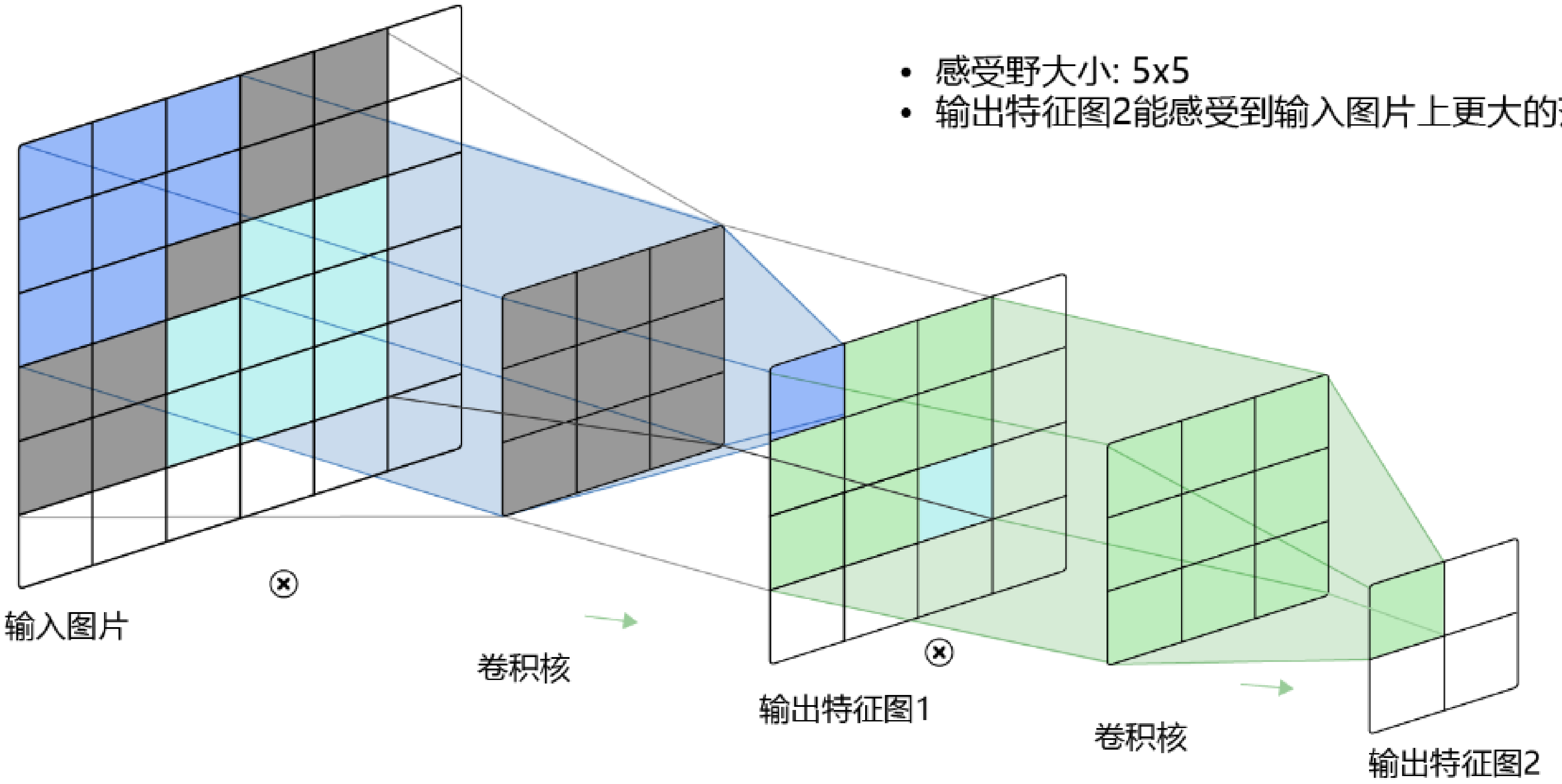
大 纲

- **视觉（-语言）基础模型预训练**
- **通用视觉（-语言）模型**

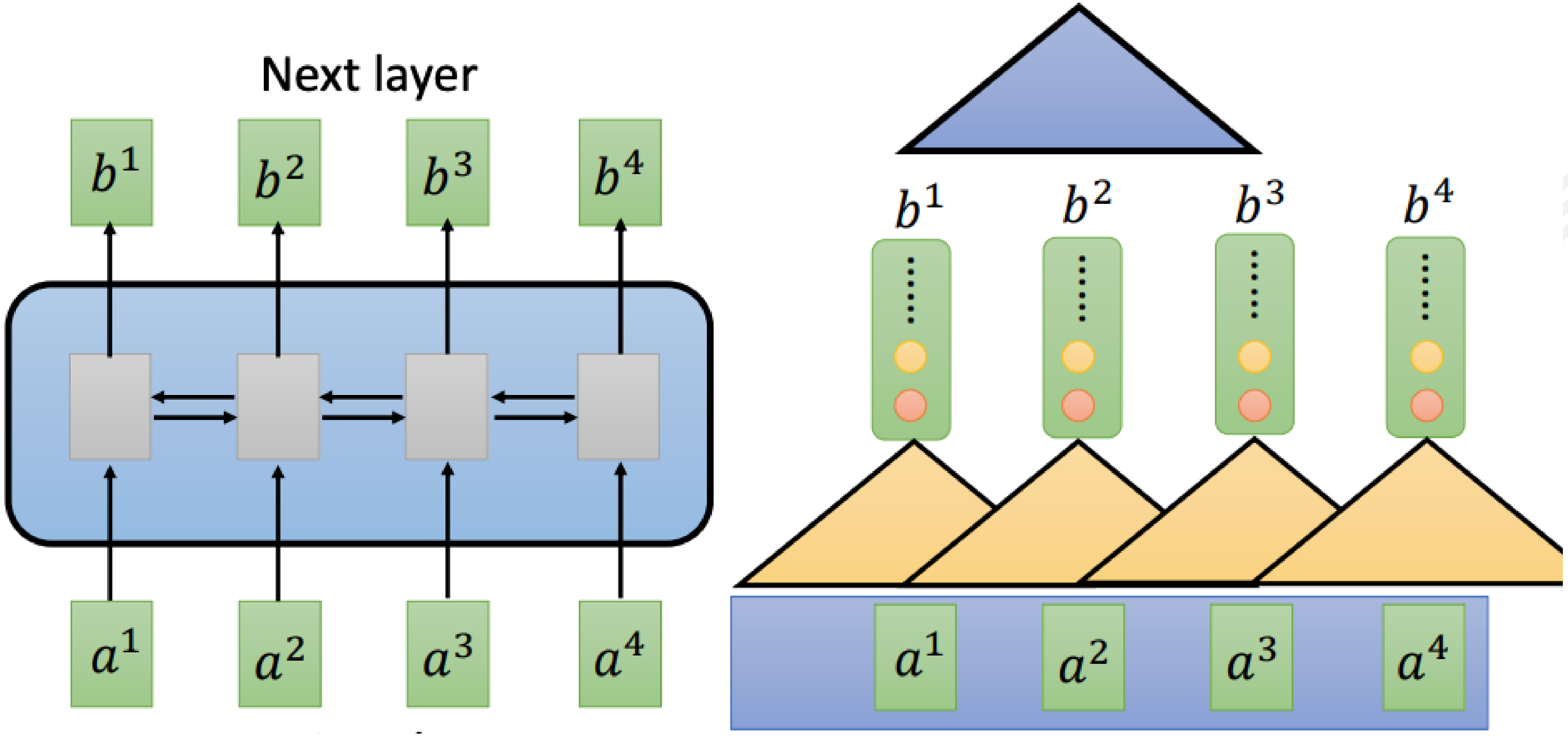
卷积神经网络

- 感受野大小: 5x5
- 输出特征图2能感受到输入图片上更大的范围

两层卷积

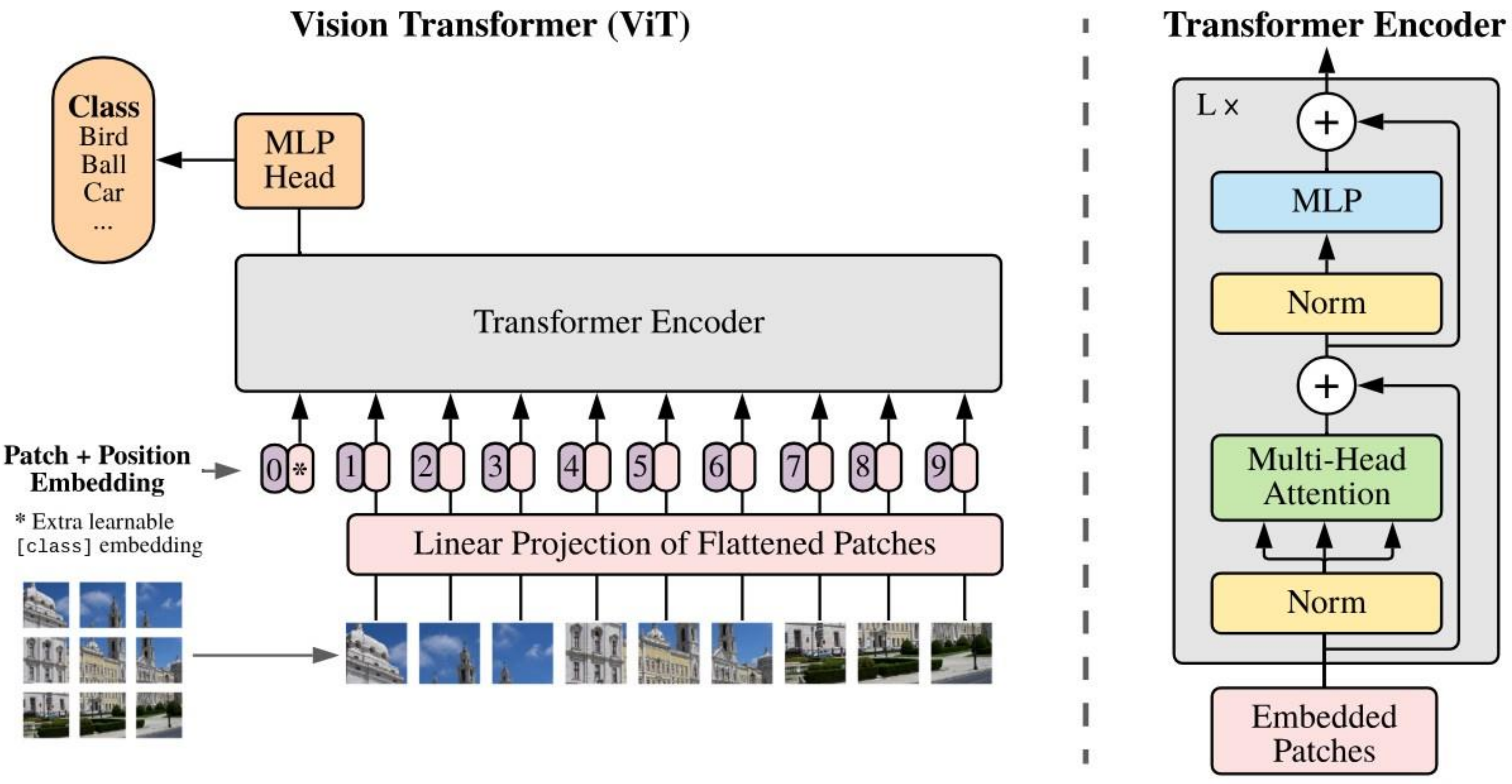


“深度”学习



RNN 可以看到更多内容，但是难以并行化，并存在遗忘问题

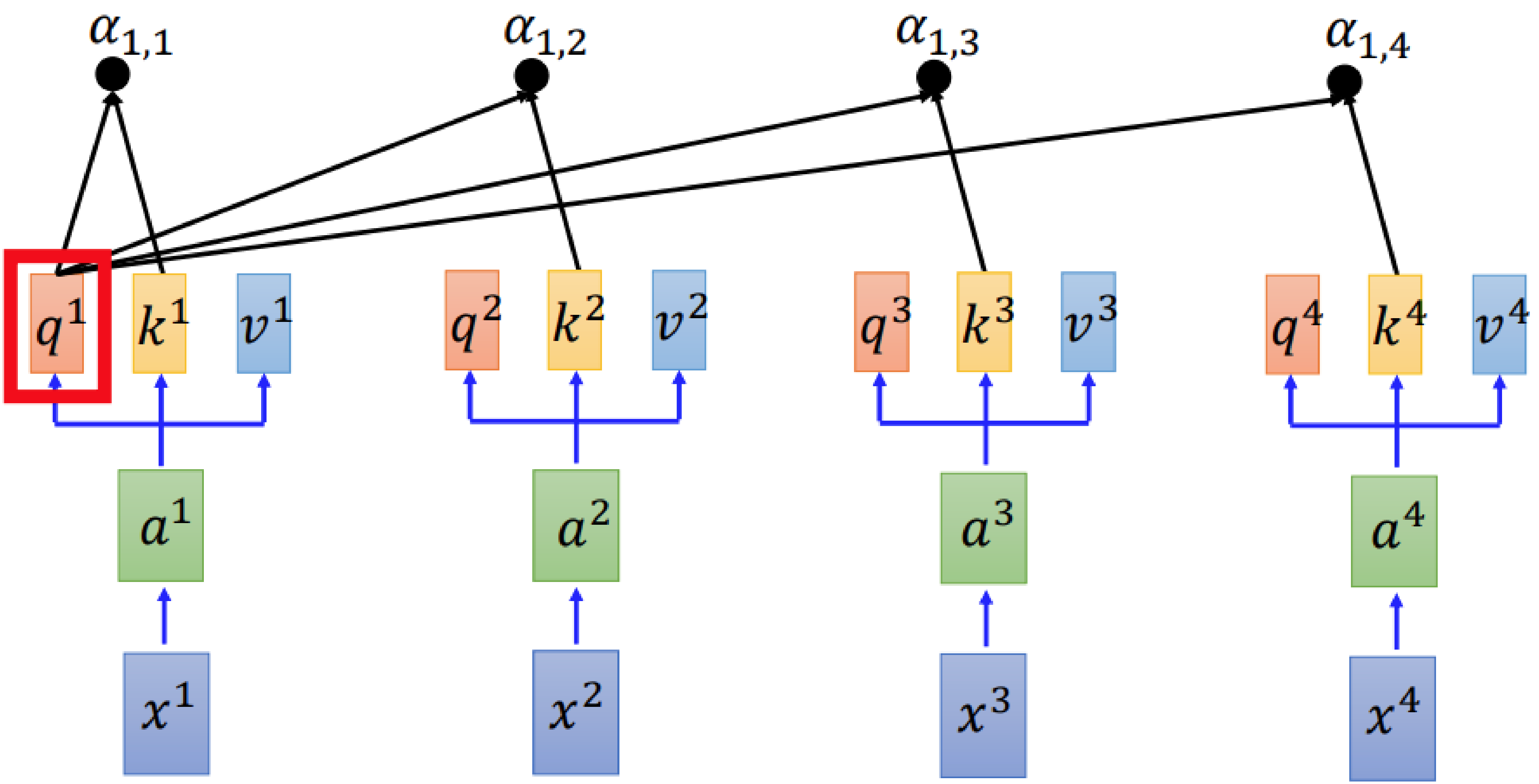
Vision Transformer



Vision Transformer

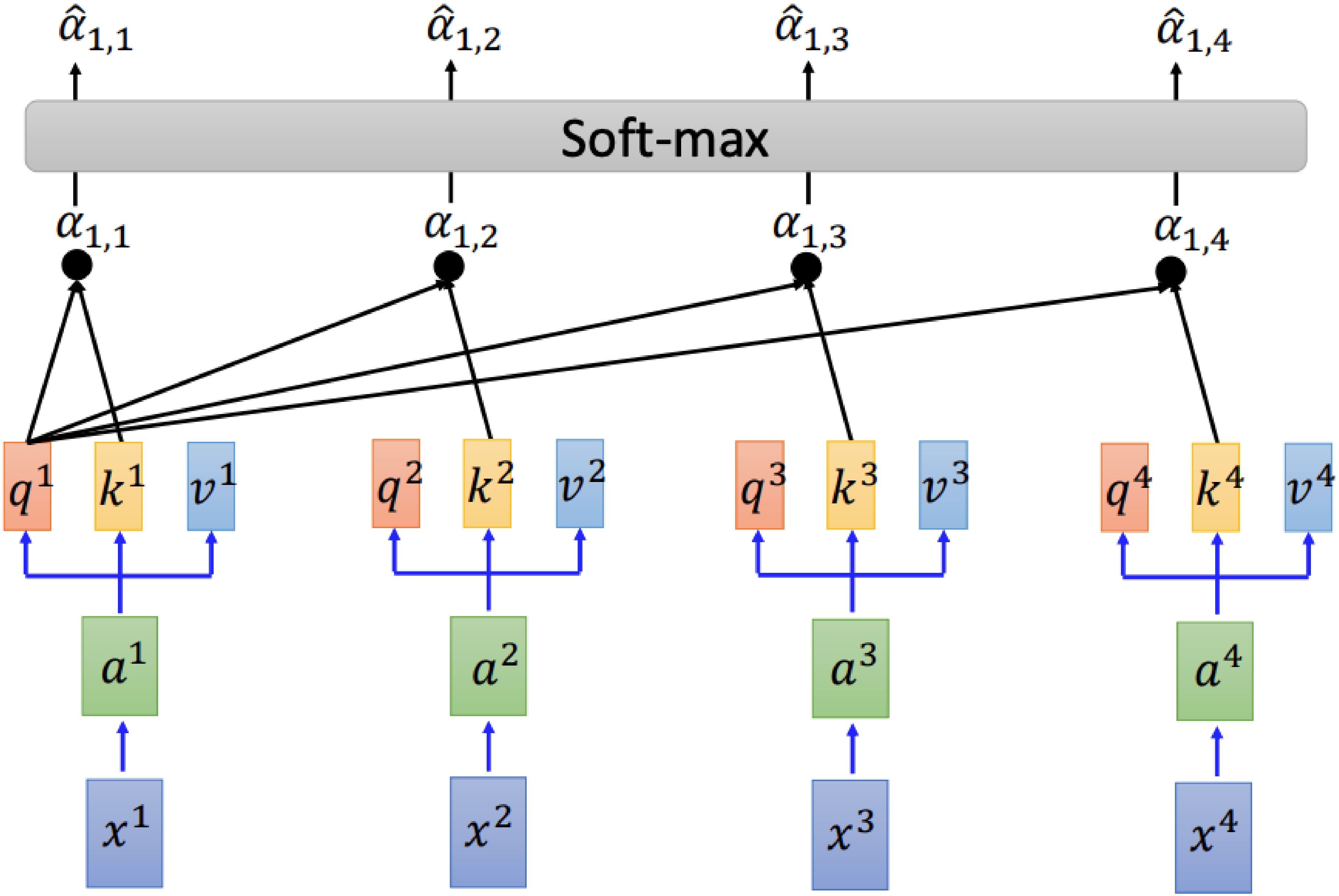
Self-Attention

Scaled Dot-Product Attention: $\alpha_{1,i} = \underbrace{q^1 \cdot k^i}_{\text{dot product}} / \sqrt{d}$



Self-attention

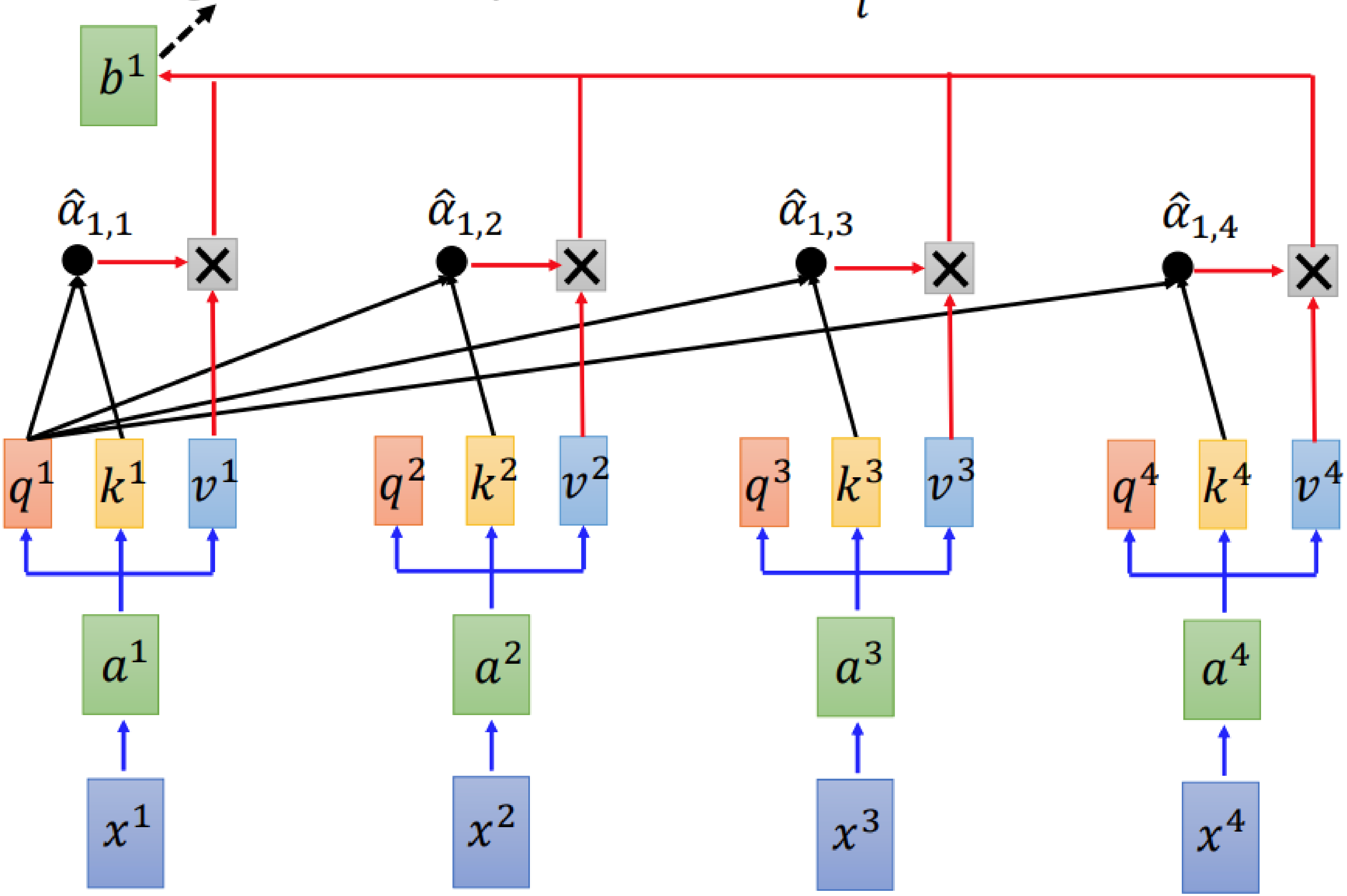
$$\hat{\alpha}_{1,i} = exp(\alpha_{1,i})/\sum_j exp(\alpha_{1,j})$$

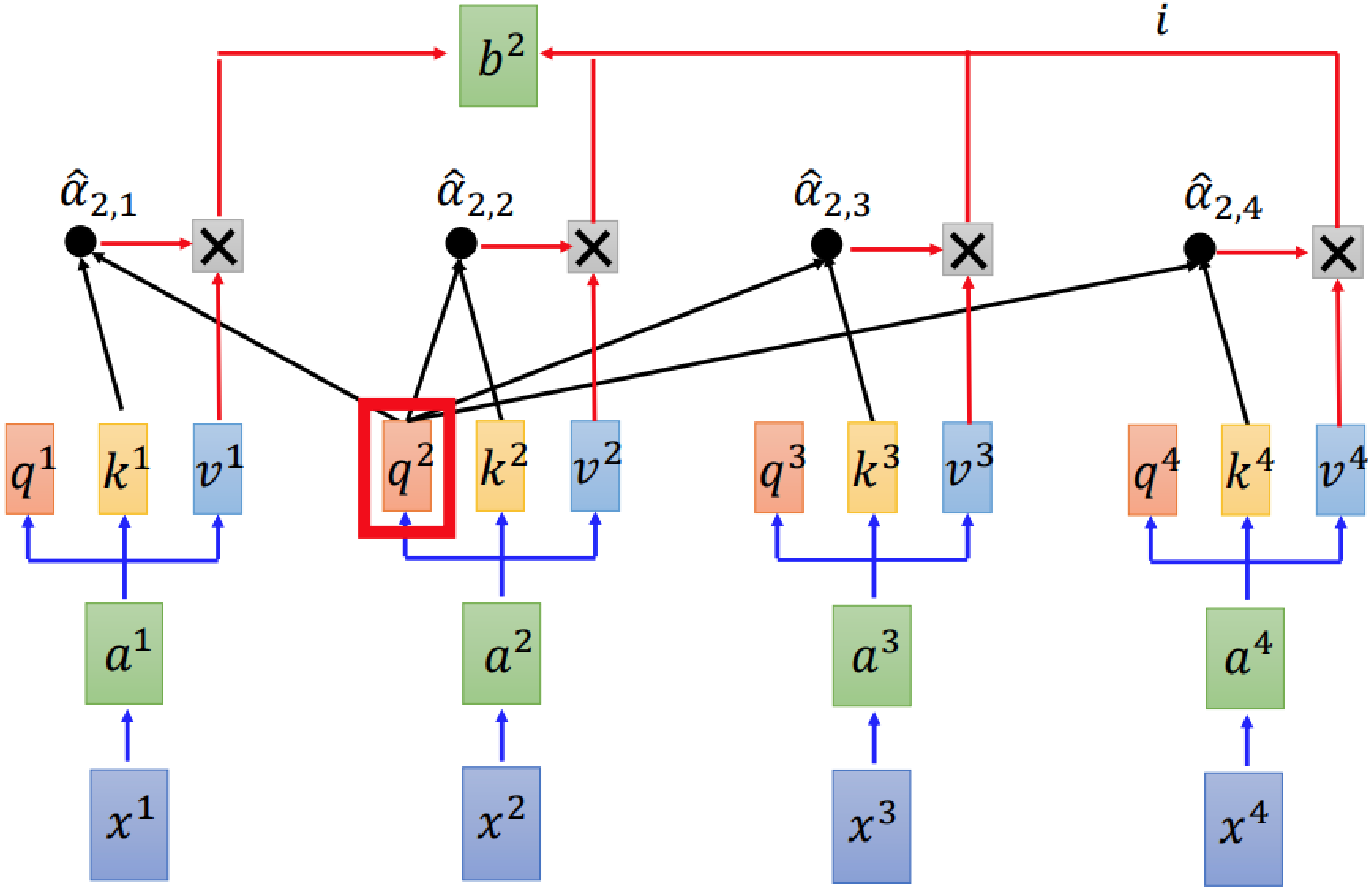


Self-attention

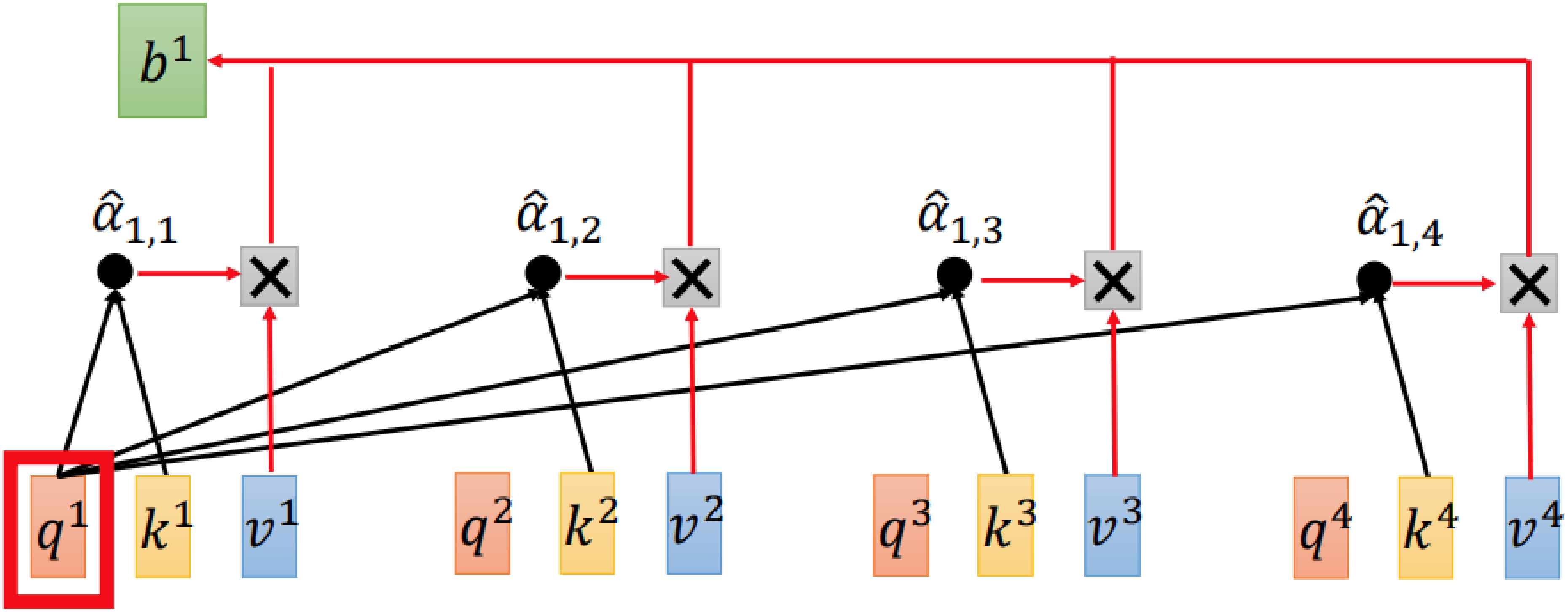
Considering the whole sequence

$$b^1 = \sum_i \hat{\alpha}_{1,i} v^i$$




b^2
i
$\hat{\alpha}_{2,1}$
$\hat{\alpha}_{2,2}$
$\hat{\alpha}_{2,3}$
$\hat{\alpha}_{2,4}$
q^1 k^1 v^1
q^2 k^2 v^2
q^3 k^3 v^3
q^4 k^4 v^4
a^1 a^2 a^3 a^4
x^1 x^2 x^3 x^4

Self-attention



$\alpha_{1,1} = k^1 \; q^1$ $\quad \alpha_{1,2} = k^2 \; q^1$

$\alpha_{1,3} = k^3 \; q^1$ $\quad \alpha_{1,4} = k^4 \; q^1$

(ignore $\sqrt{d}$ for simplicity)

$$\begin{bmatrix} \alpha_{1,1} \\ \alpha_{1,2} \\ \alpha_{1,3} \\ \alpha_{1,4} \end{bmatrix} = \begin{bmatrix} k^1 \\ k^2 \\ k^3 \\ k^4 \end{bmatrix} q^1$$

Self-attention

$$b^2 = \sum_i \hat{\alpha}_{2,i} v^i$$

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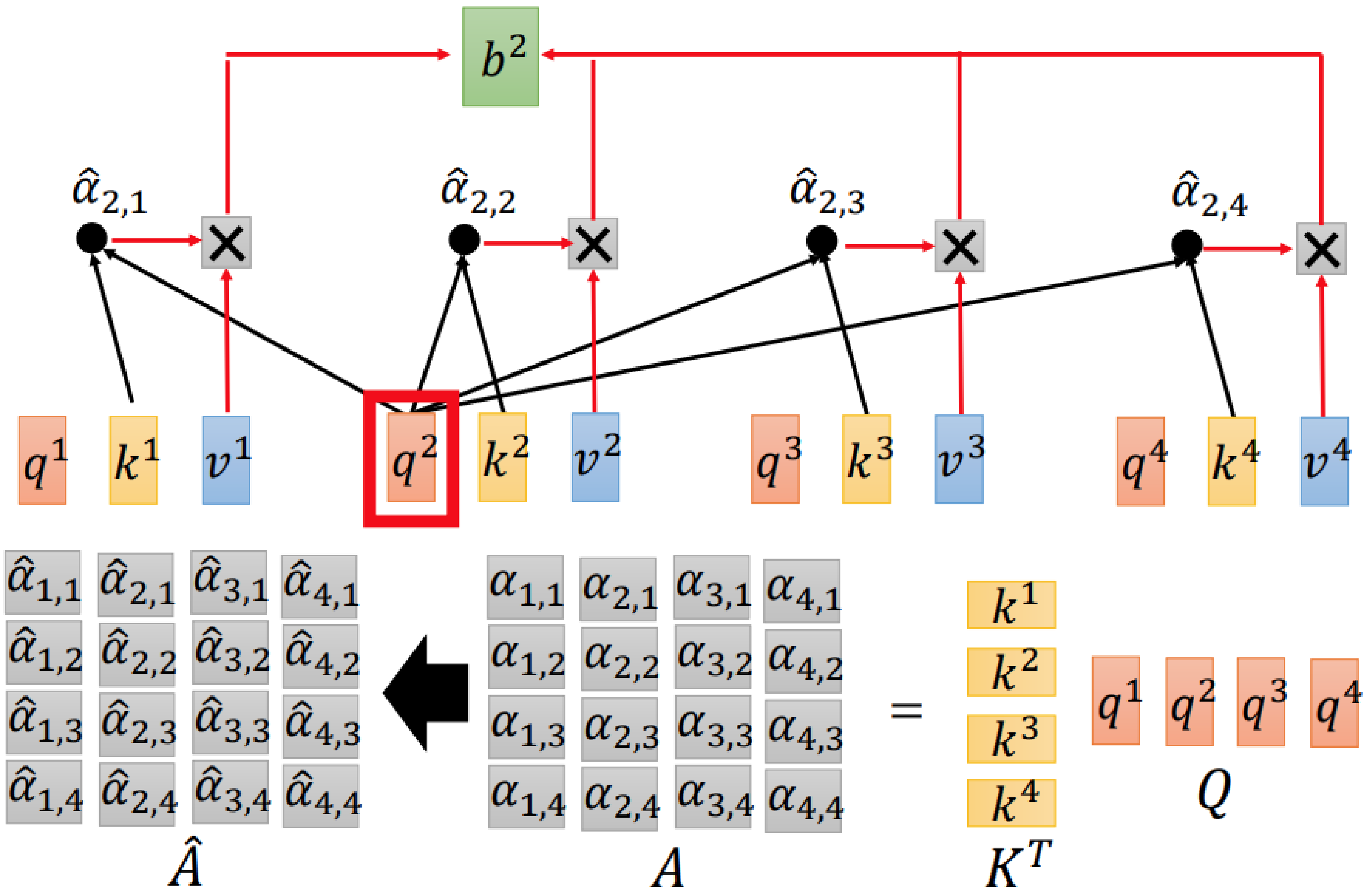
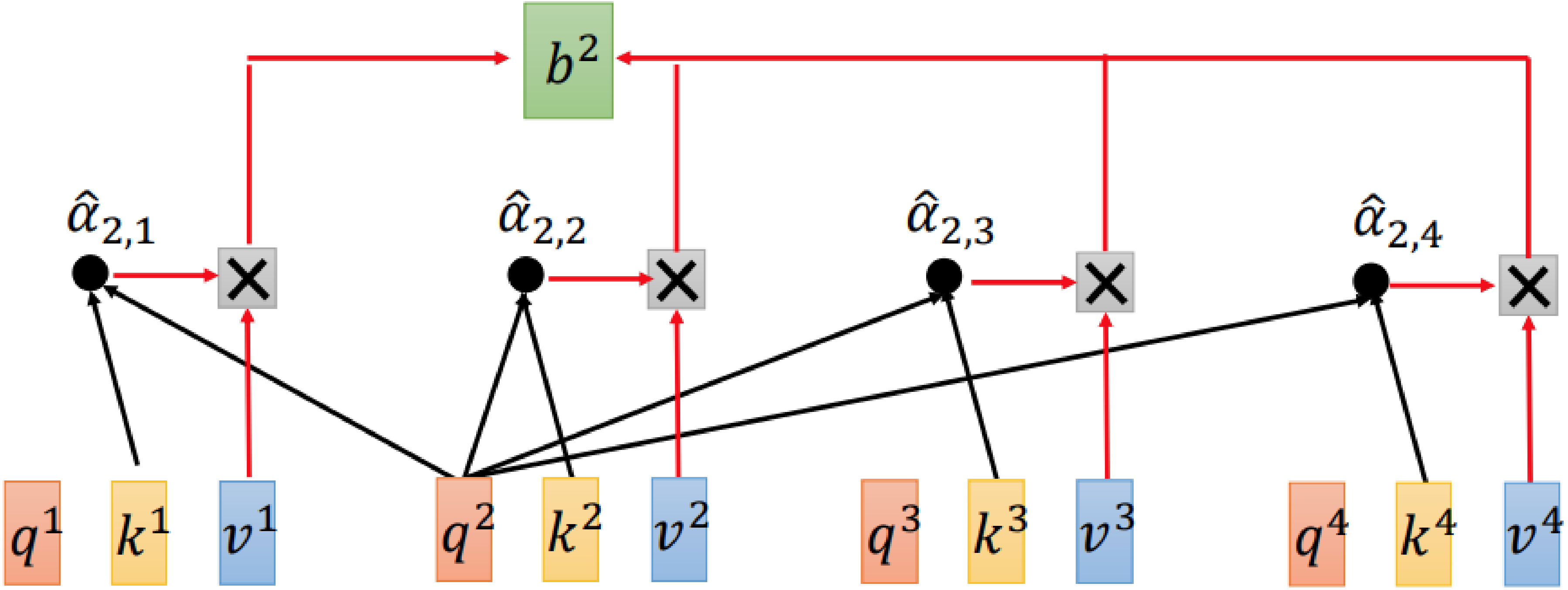


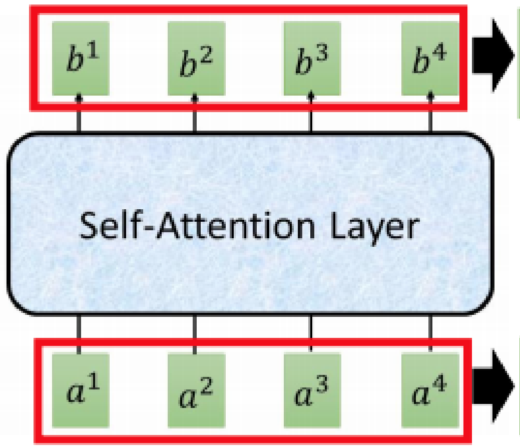
Self-attention

$$b^2 = \sum_i \hat{\alpha}_{2,i} v^i$$



$$\underbrace{\begin{bmatrix} b^1 & b^2 & b^3 & b^4 \end{bmatrix}}_{O} = \underbrace{\begin{bmatrix} v^1 & v^2 & v^3 & v^4 \end{bmatrix}}_{V} \underbrace{\begin{bmatrix} \hat{\alpha}_{1,1} & \hat{\alpha}_{2,1} & \hat{\alpha}_{3,1} & \hat{\alpha}_{4,1} \\ \hat{\alpha}_{1,2} & \hat{\alpha}_{2,2} & \hat{\alpha}_{3,2} & \hat{\alpha}_{4,2} \\ \hat{\alpha}_{1,3} & \hat{\alpha}_{2,3} & \hat{\alpha}_{3,3} & \hat{\alpha}_{4,3} \\ \hat{\alpha}_{1,4} & \hat{\alpha}_{2,4} & \hat{\alpha}_{3,4} & \hat{\alpha}_{4,4} \end{bmatrix}}_{\hat{A}}$$

Self-attention



$$b^1\ b^2\ b^3\ b^4 \Rightarrow O \qquad a^1\ a^2\ a^3\ a^4 \Rightarrow I$$

$$Q = W^q I$$

$$K = W^k I$$

$$V = W^v I$$

$$\hat{A} \Leftarrow A = K^T Q$$

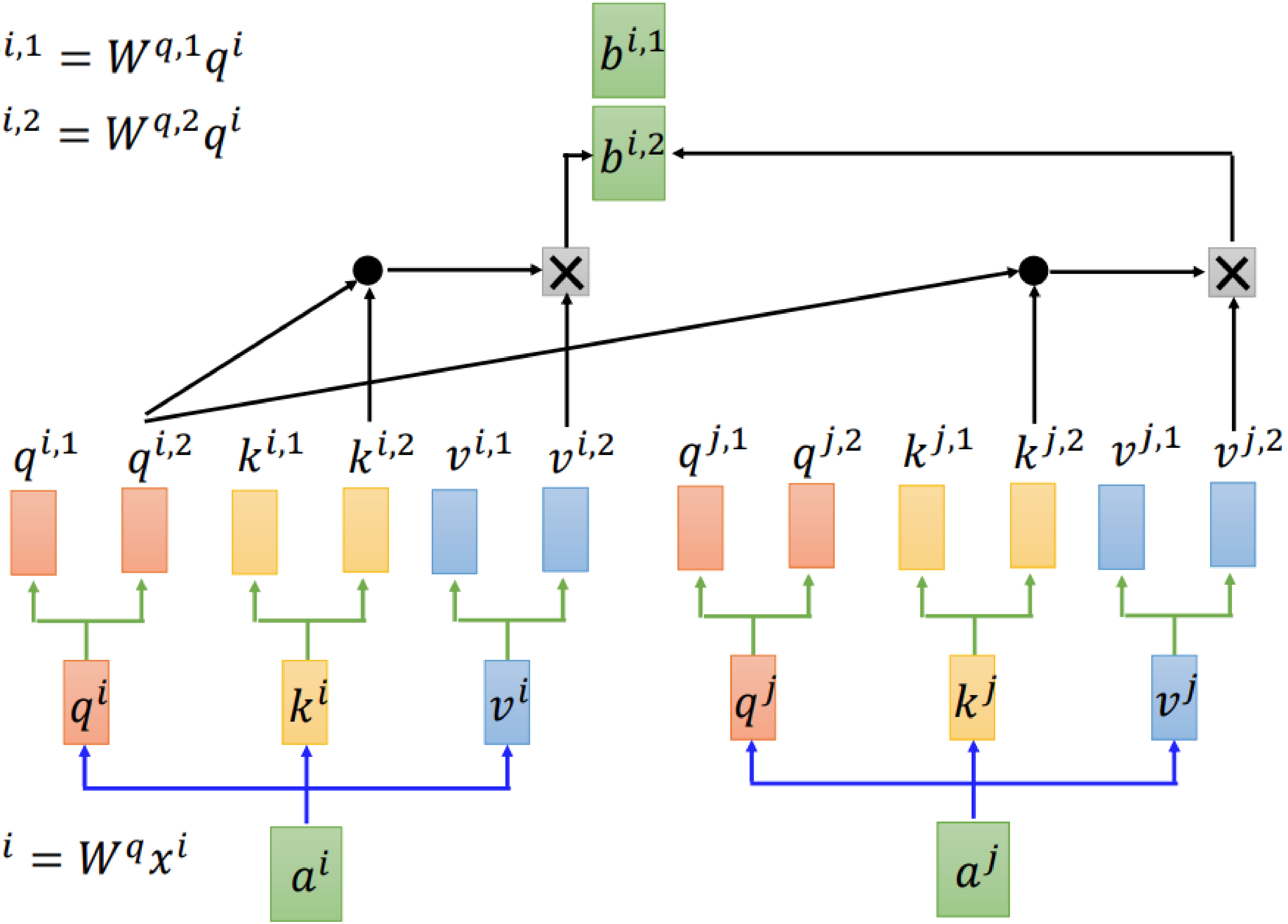
$$O = V \hat{A}$$

矩阵乘法，可以用 GPU 并行计算和加速

Multi-head Self-attention (2 heads as example)

$$q^{i,1} = W^{q,1} q^i$$

$$q^{i,2} = W^{q,2} q^i$$



$$q^i = W^q x^i$$

模型预训练方式

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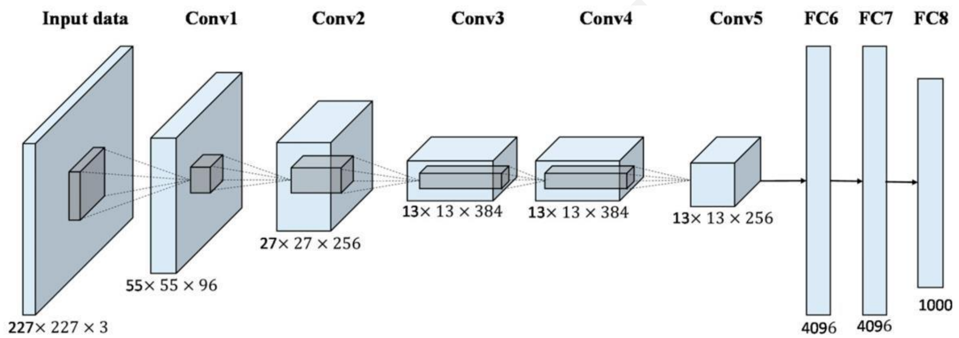
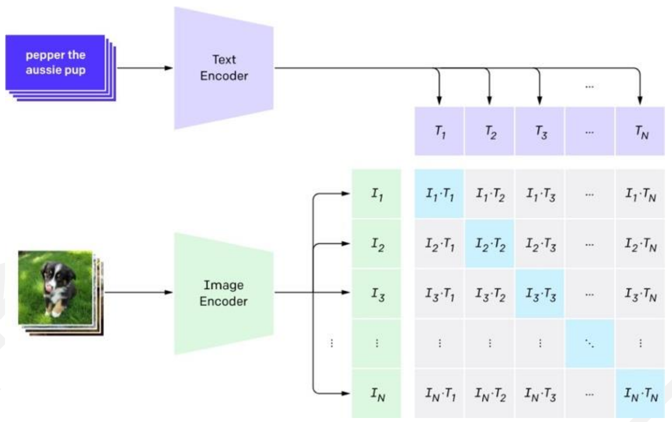
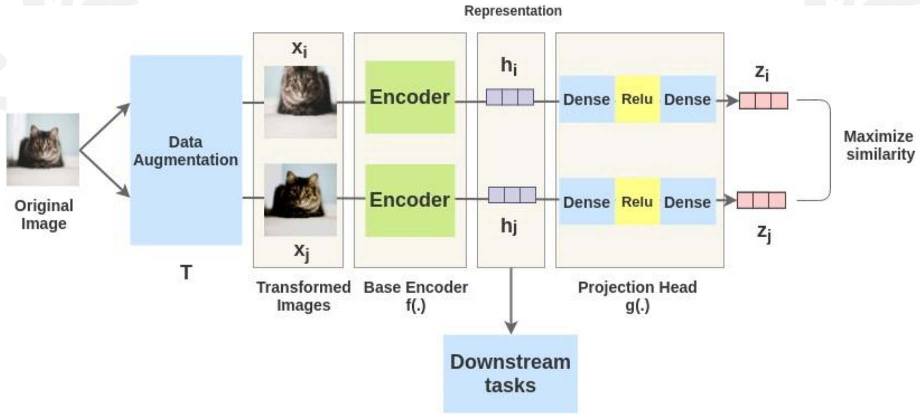
全监督学习



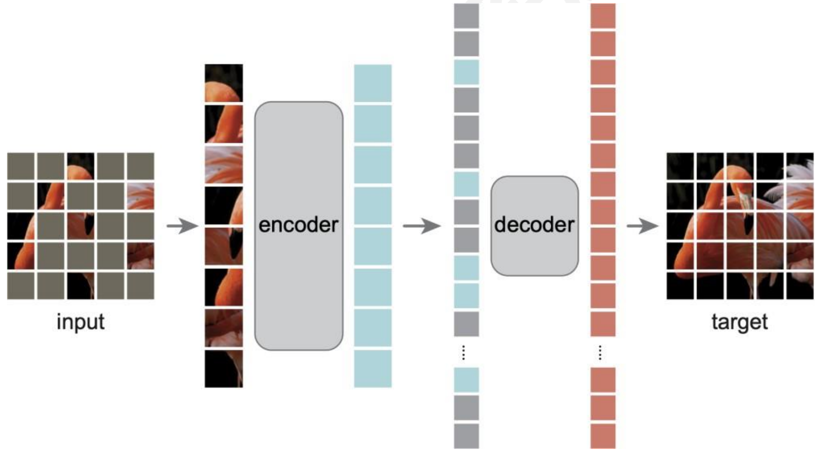
语言-图像对比学习（CLIP）



图像对比学习



掩蔽自编码器



全监督学习

- 将图像映射到与视觉概念相关的**离散标签**
- 人工标注很昂贵，标签也可能有限

MNIST

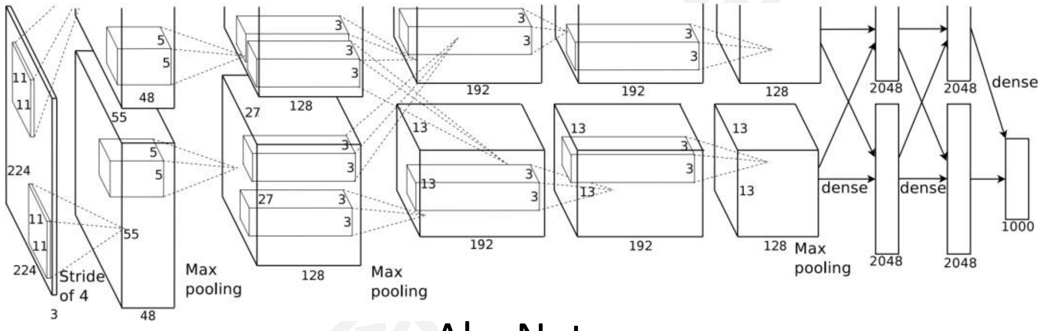
CIFAR-10

ImageNet

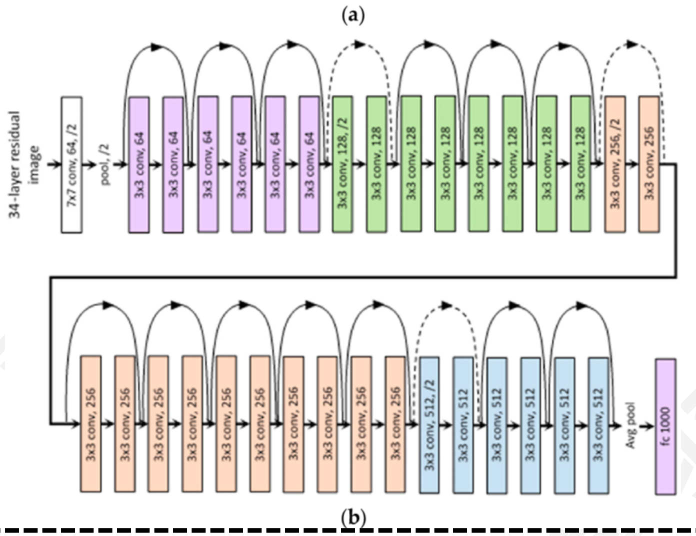
1 Scaling vision transformers, CVPR 2022
2 Revisiting weakly supervised pre-training of visual perception models, CVPR 2022

全监督学习

■ 各种强大的视觉模型

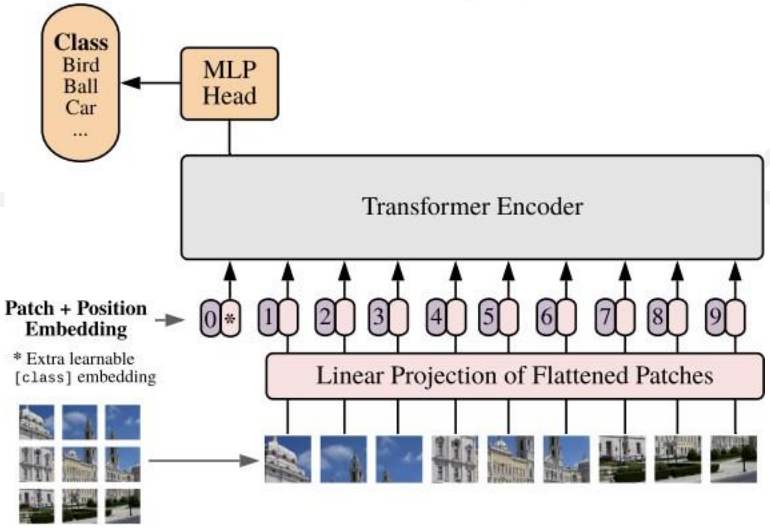


AlexNet

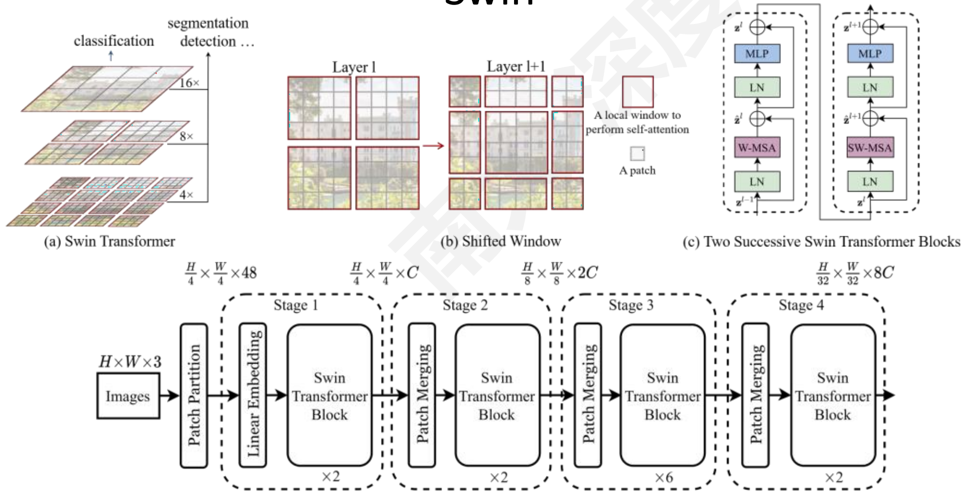


ResNet

Vision Transformer (ViT)



Swin

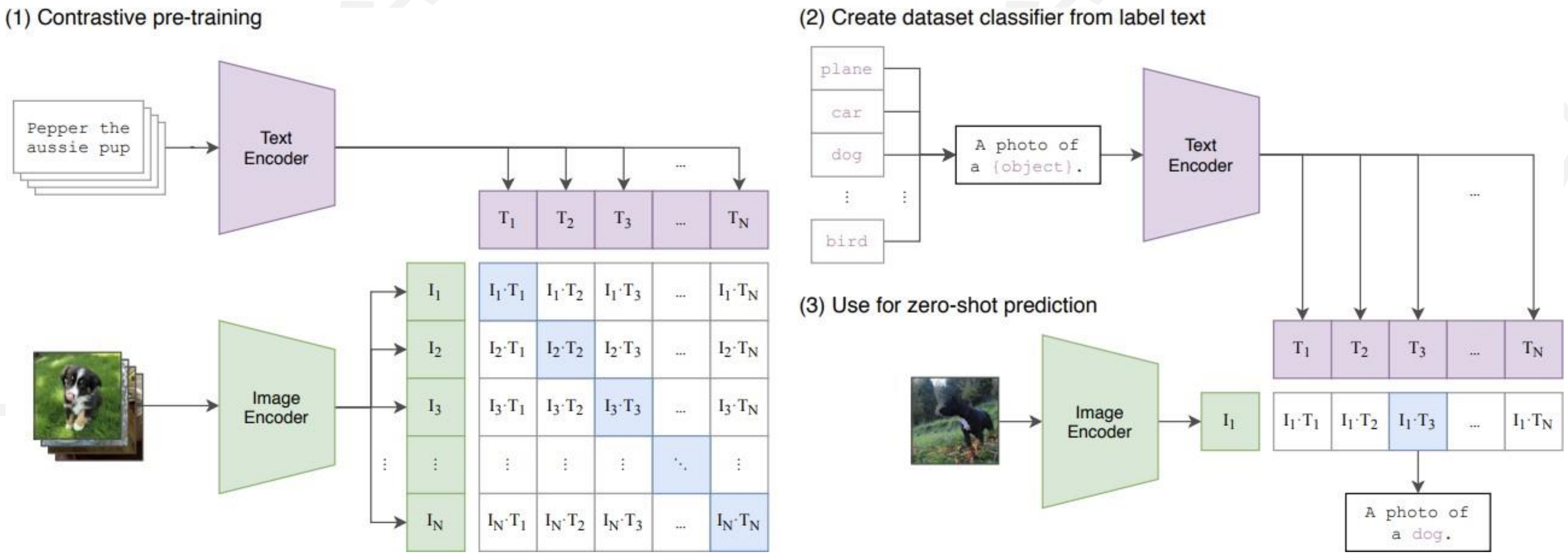


(a) Swin Transformer (b) Shifted Window (c) Two Successive Swin Transformer Blocks

(d) Architecture

语言-图像对比学习

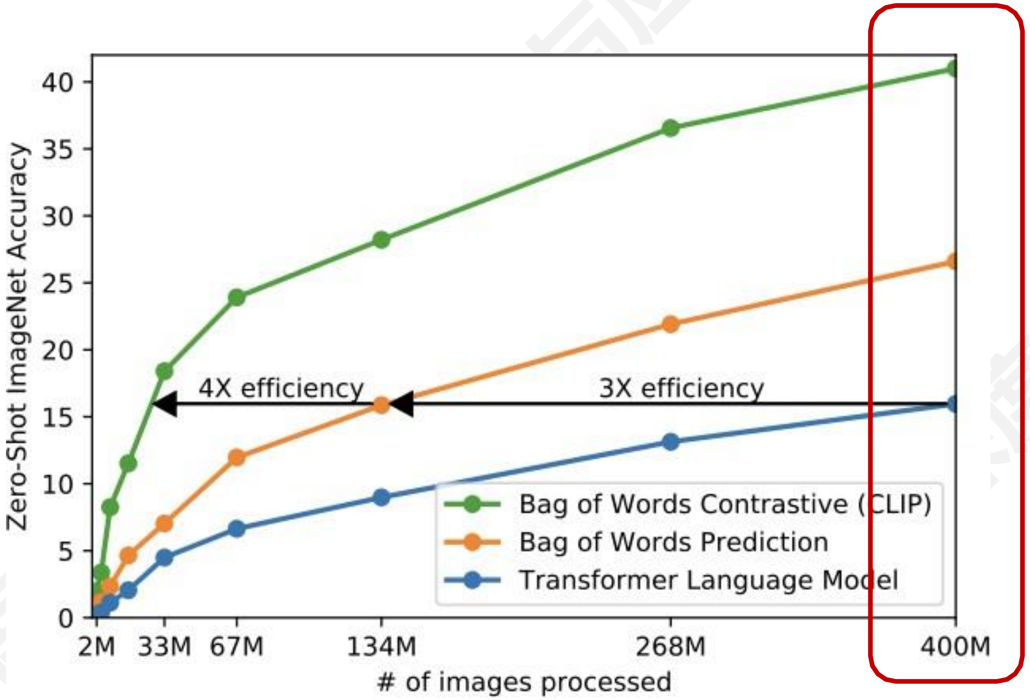
- 从海量的图像文本对中学习
- 简单的对比学习训练方式，但是**数据海量**



1 Learning transferable visual models from natural language supervision, ICML 2021
2 Scaling up visual and vision-language representation learning with noisy text supervision, ICML 2021

语言-图像对比学习

■ 数据量非常重要（CLIP[1] and ALIGN[2]）



Language is a stronger form of supervision than classical closed-set labels. Language provides rich information for supervision. Therefore, *scaling*, which can involve increasing capacity (model scaling) and increasing information (data scaling), is essential for attaining good results in language-supervised training.

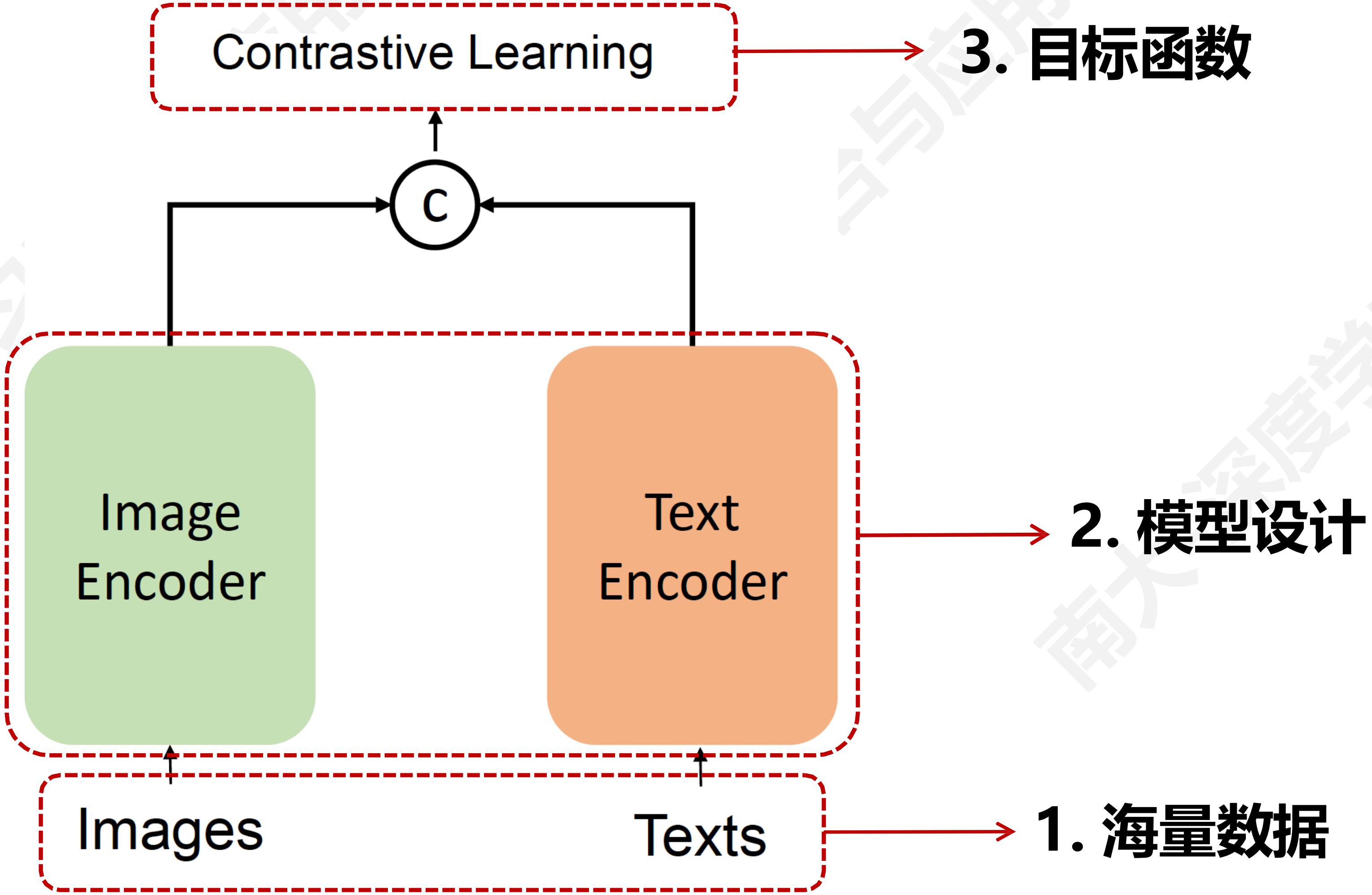
CLIP [52] is an outstanding example of "*simple algorithms that scale well*". The simple design of CLIP allows it to be relatively easily executed at substantially larger scales and achieve big leaps compared to preceding methods. Our method largely maintains the simplicity of CLIP

Quote from the FLIP paper

1 Learning transferable visual models from natural language supervision, ICML 2021
2 Scaling up visual and vision-language representation learning with noisy text supervision, ICML 2021

语言-图像对比学习

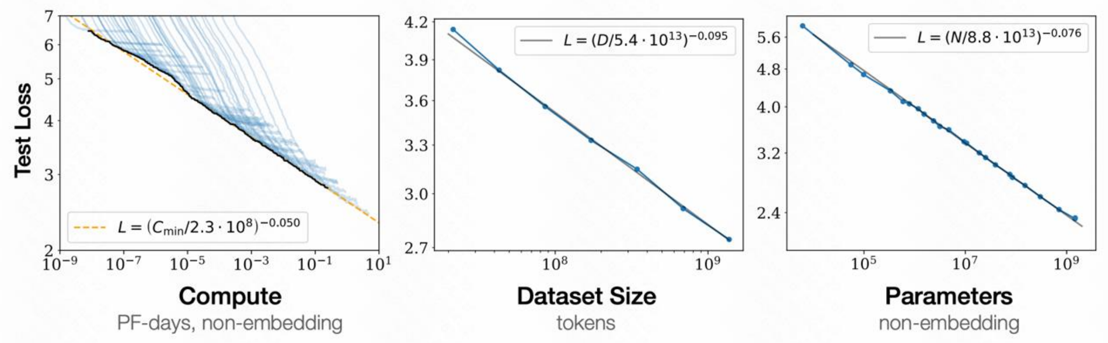
CLIP 的关键设计



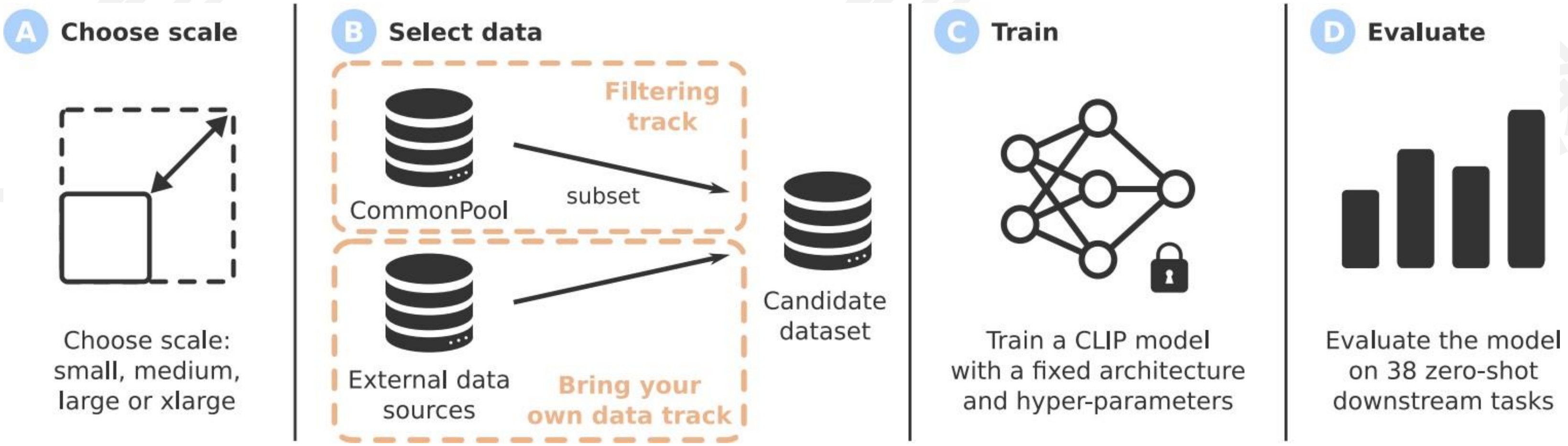
海量数据

■ **可复现的 CLIP 尺度定律（scaling laws）**

	Data	Arch.	ImageNet	VTAB+	COCO
CLIP [55]	WIT-400M	L/14	75.5	55.8	61.1
Ours	LAION-2B	L/14	75.2	54.6	71.1
Ours	LAION-2B	H/14	78.0	56.4	73.4



■ **DataComp：选择合适的数据集**



1 Reproducible scaling laws for contrastive language-image learning, CVPR 2023
2 Datacomp: In search of the next generation of multimodal datasets, 2023

模型设计：图像端

- **FLIP**: 通过掩蔽自编码器（MAE）扩展 CLIP 的视觉编码器训练
  - 训练：依旧使用 CLIP Loss，不使用 MIM Loss
  - 方法：随机掩蔽图像区域，只编码可见的图像区域用于训练，从而提升训练效率

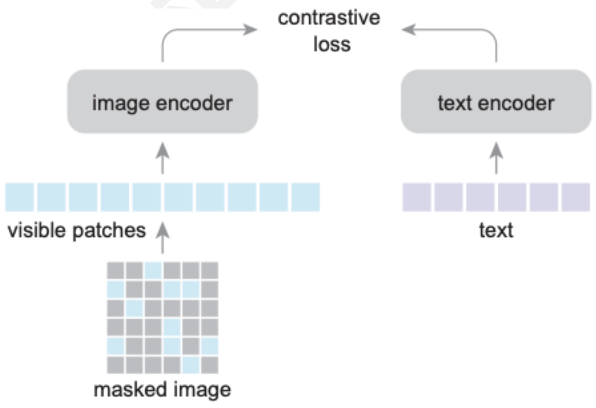
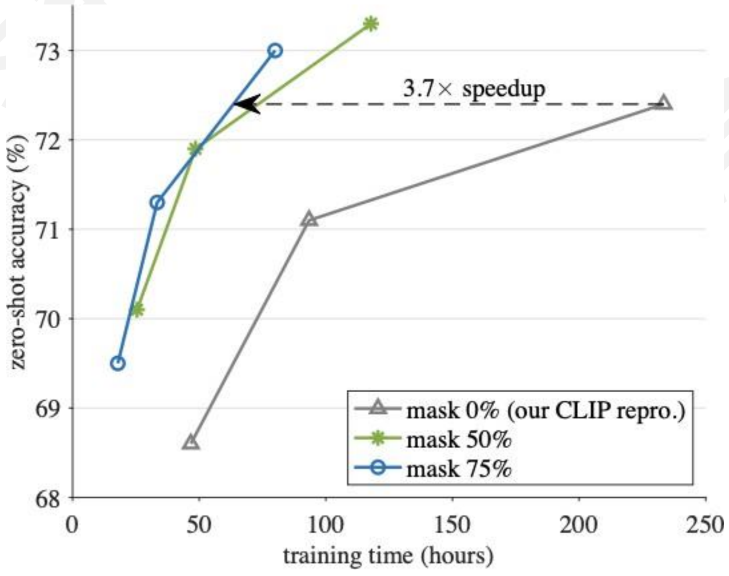


Figure 2. **Our FLIP architecture**. Following CLIP [52], we perform contrastive learning on pairs of image and text samples. We randomly mask out image patches with a high masking ratio and encode only the visible patches. We do not perform reconstruction



[1] Scaling language-image pre-training via masking, CVPR 2023

模型设计：语言端

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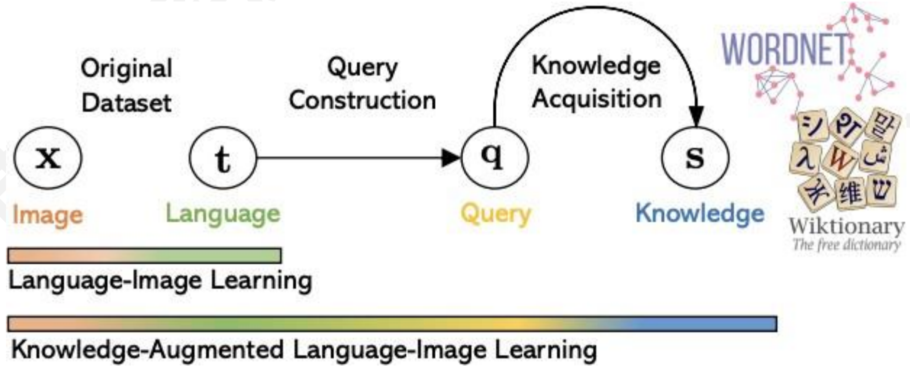
■ K-Lite: 使用外部知识（字典定义）对语言端进行增强

Figure 1: Motivating examples: knowledge explains the content of the rare dish concepts.

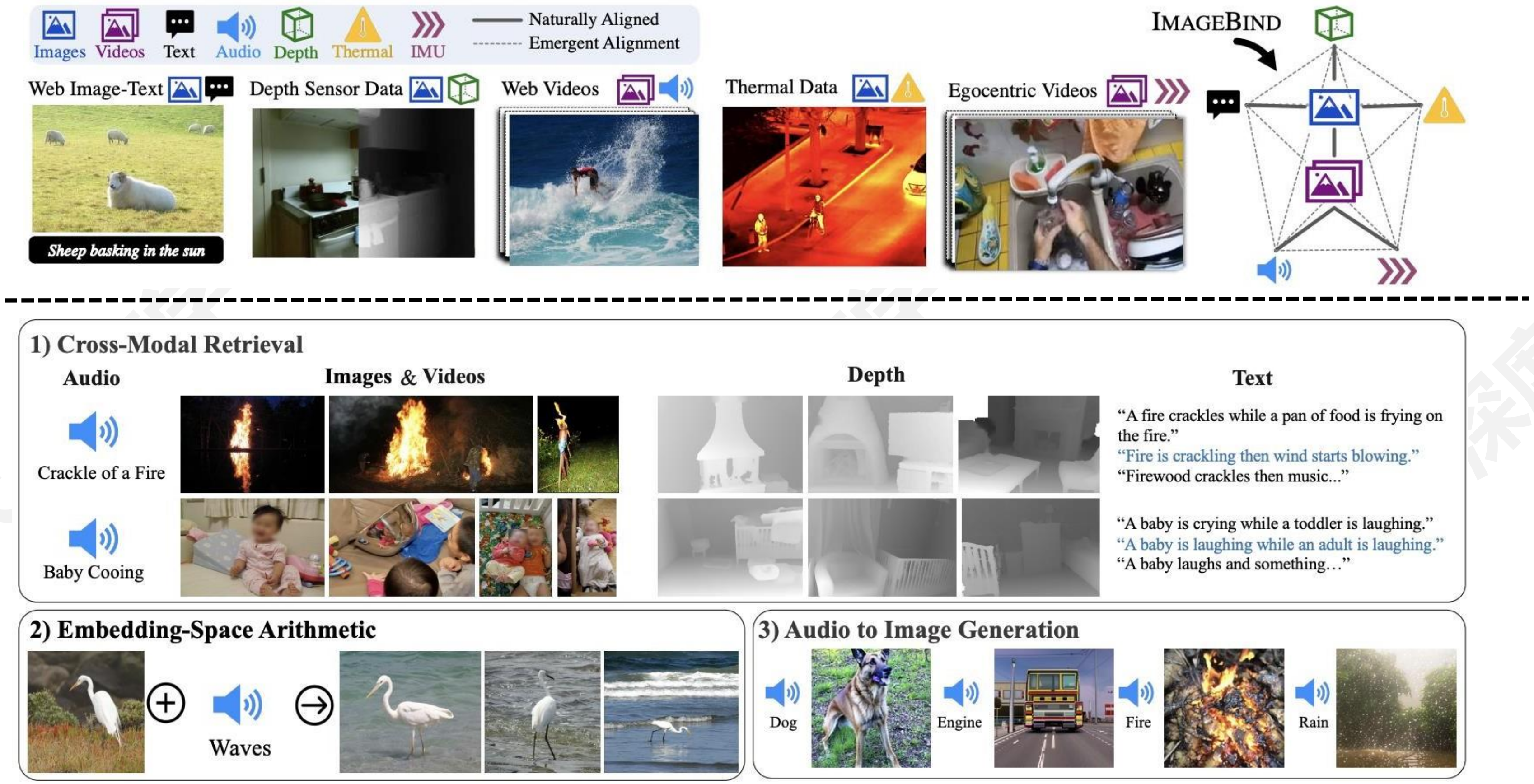


Dataset	Training Data # Samples	Method	ImageNet-1K Zero-shot	ICinW (20 datasets) Zero-shot	ICinW (20 datasets) Linear Probing	ICinW (20 datasets) Fine-tuning
ImageNet-21K	13M (full)	UniCL	28.16	27.15	53.07 ± 4.15	55.96 ± 2.50
	13M (full)	K-LITE	**30.23**	**33.44**	**53.92 ± 1.05**	**57.81 ± 1.48**
YFCC-14M + ImageNet-21K	14M (half)	UniCL	34.43	34.30	53.50 ± 2.22	56.45 ± 2.48
	14M (half)	K-LITE	36.67	36.50	49.48 ± 2.23	55.88 ± 1.64
	14M (half)	K-LITE$^\diamond$	42.36	36.50	54.28 ± 3.66	52.11 ± 4.90
	27M (full)	UniCL	43.06	35.99	55.96 ± 3.38	58.25 ± 2.98
	27M (full)	K-LITE	**45.67**	**38.89**	**57.06 ± 1.48**	58.24 ± 2.36
GCC-15M + ImageNet-21K	15M (half)	UniCL	41.64	36.31	53.86 ± 2.73	59.04 ± 3.13
	15M (half)	K-LITE	44.26	39.53	55.91 ± 2.53	58.20 ± 3.39
	15M (half)	K-LITE$^\diamond$	47.30	40.32	57.38 ± 2.70	60.72 ± 2.29
	28M (full)	UniCL	46.83	38.90	57.92 ± 3.31	60.99 ± 2.74
	28M (full)	K-LITE	**48.76**	**41.34**	**58.56 ± 3.12**	**63.39 ± 1.74**

[1] K-lite: Learning transferable visual models with external knowledge, NeurIPS 2022

模型设计：更多模态

■ ImageBind: 使用一个特征空间连接所有模态



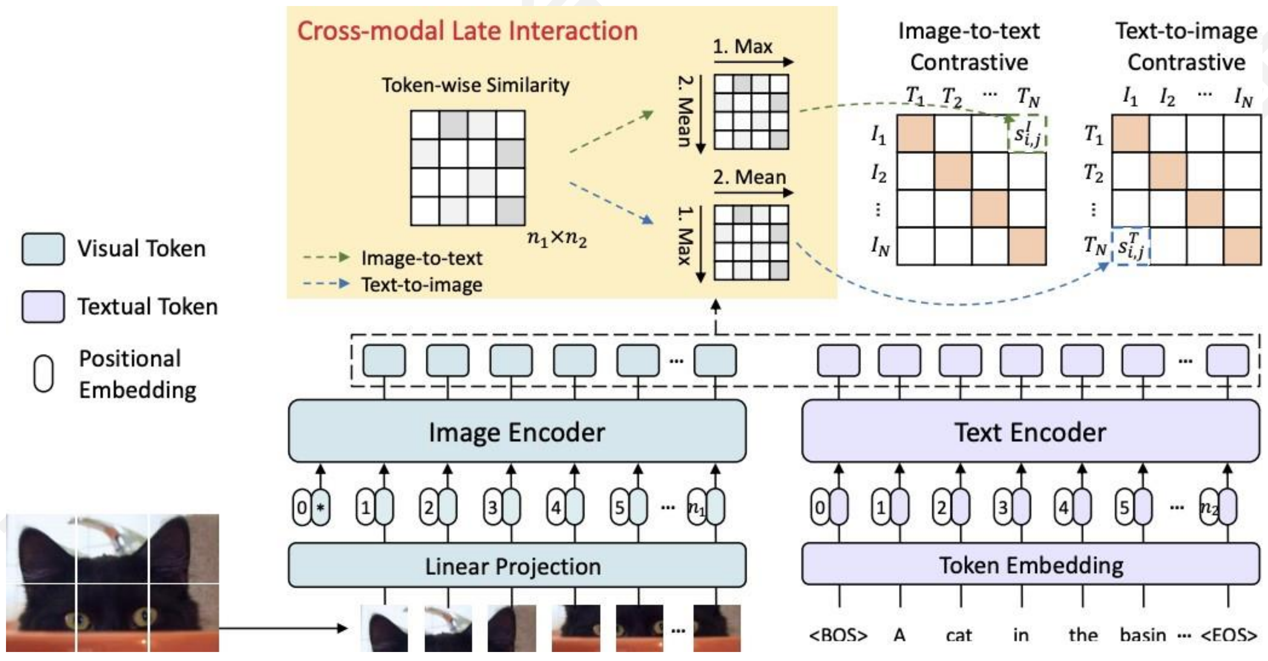
- 使用预训练的CLIP并保持冻结
- 训练其他模态的编码器来对齐CLIP特征空间

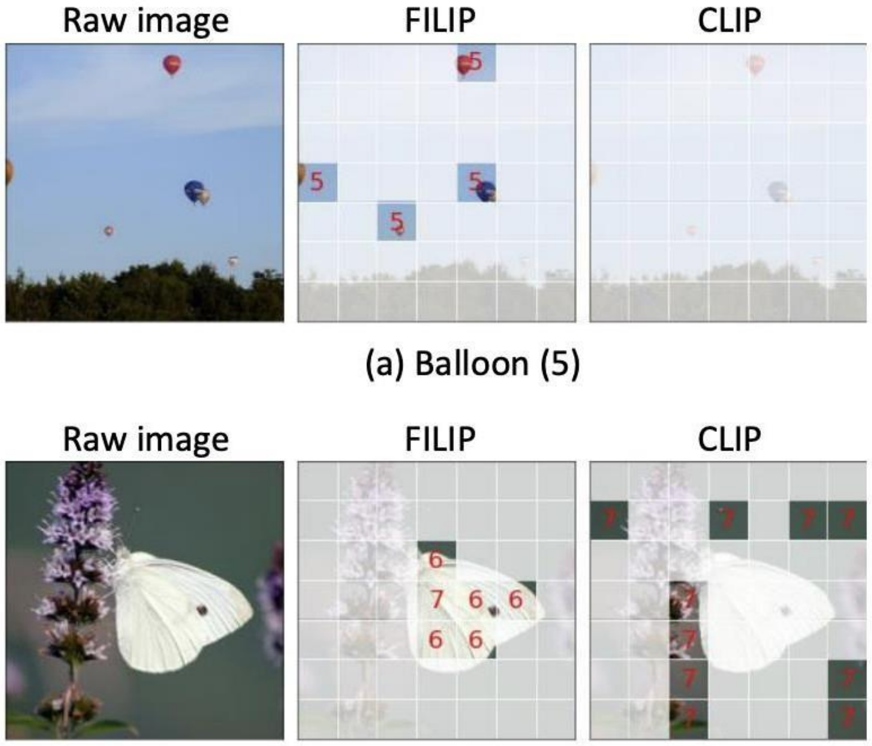
[1] IMAGEBIND: One Embedding Space To Bind Them All, 2023

目标函数：细粒度的监督

FILIP: 细粒度的监督

- 学习文本与图像块之间的对齐关系





(a) Balloon (5)

(c) Small white butterfly (5, 6, 7)

[1] FILIP: Fine-grained Interactive Language-Image Pre-Training, ICLR 2022

目标函数：加入生成模型监督

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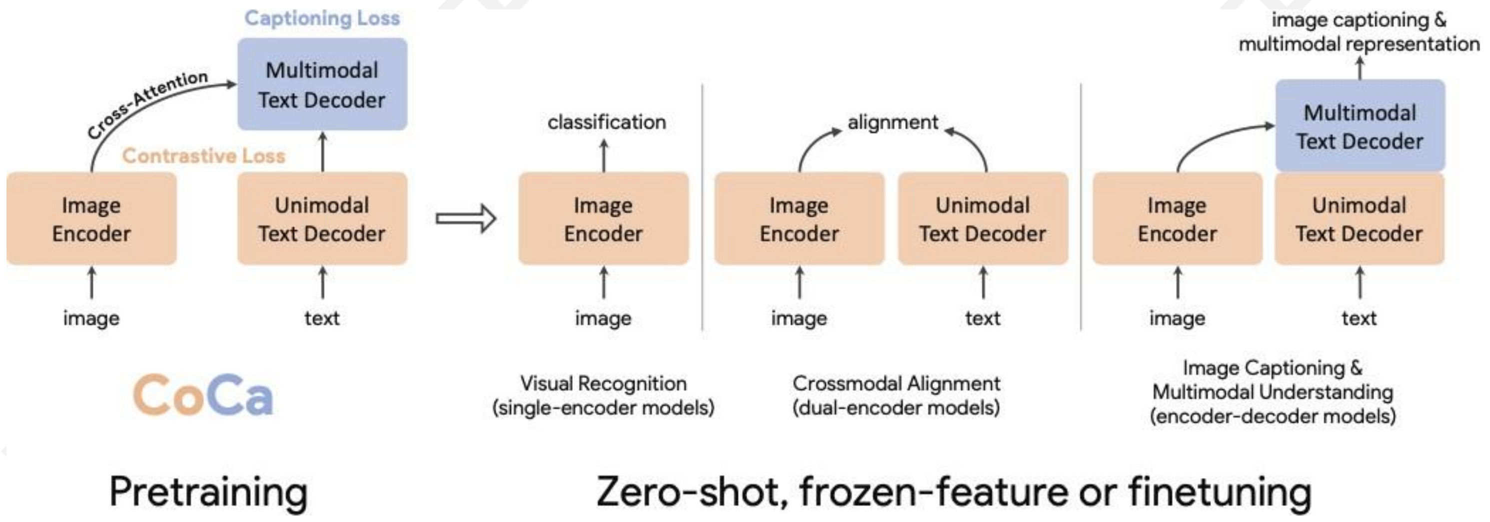
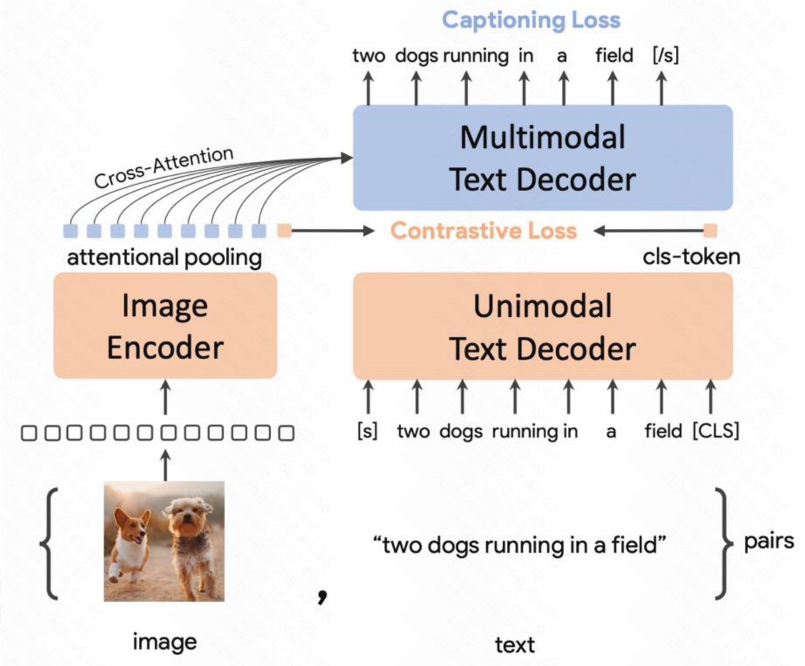
- **CoCa: 加入生成模型分支的监督损失（Captioning Loss)**
  - **提高性能，并可以执行文本生成任务**





[1] Coca: Contrastive captioners are image-text foundation models, 2022

将 CLIP 与其他学习方法结合

Contrastive Language-Image Pre-training

+ Supervised Learning = ?

+ Image-Only (Non-) Contrastive Learning = ?

+ Masked Image Modeling = ?

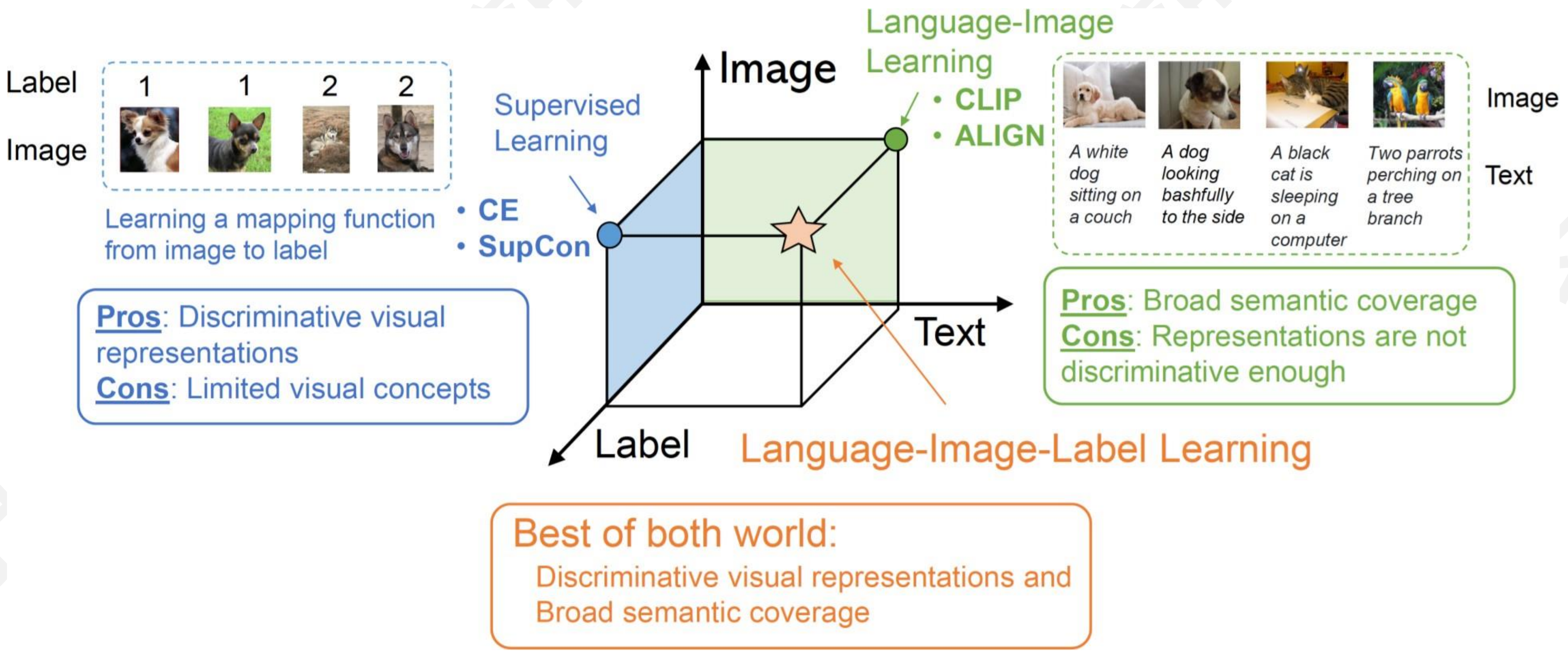
Image-Only (Non-) Contrastive Learning + Masked Image Modeling = ?

将 CLIP 与其他学习方法结合

Contrastive Language-Image Pre-training	+	Supervised Learning	=	?
	+	Image-Only (Non-) Contrastive Learning	=	?
	+	Masked Image Modeling	=	?
Image-Only (Non-) Contrastive Learning	+	Masked Image Modeling	=	?

图像标签+文本标签监督

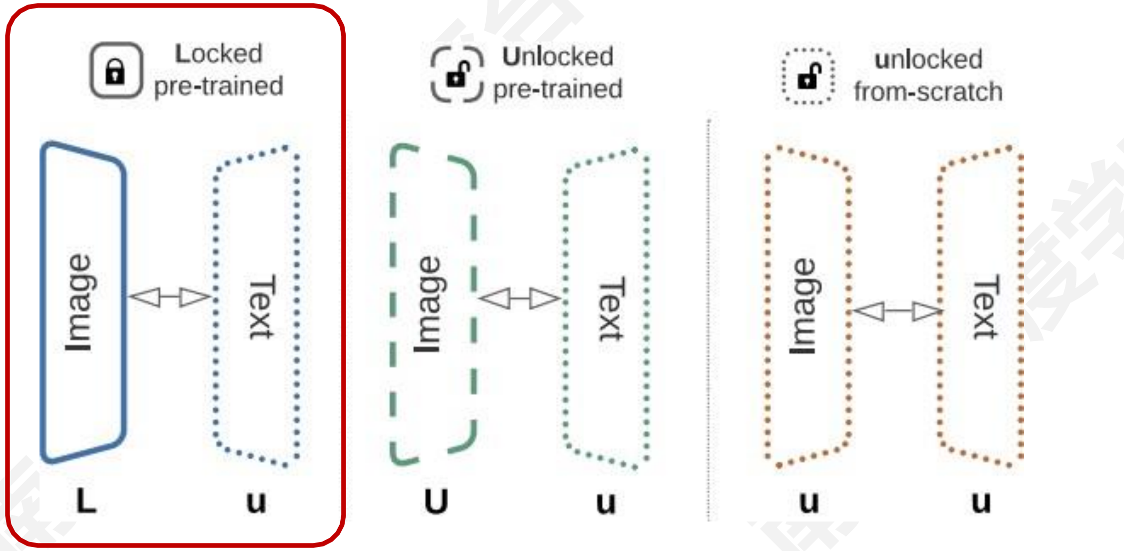
■ UniCL: 图像标签-文本标签混合监督

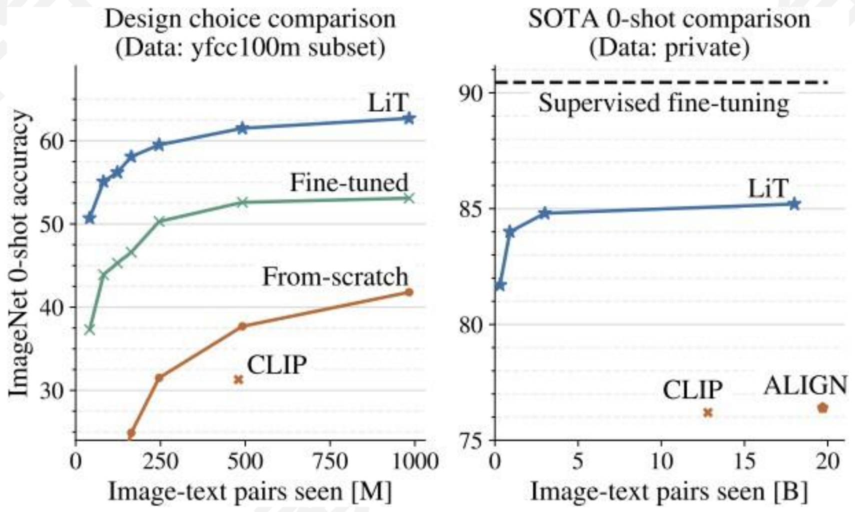


1 Unified contrastive learning in image-text-label space, CVPR 2022
2 Florence: A new foundation model for computer vision, 2021

图像标签+文本标签监督

- **LiT: 用冻结的预训练好的 image encoder 训练 text encoder**





[1] Lit: Zero-shot transfer with locked-image text tuning, CVPR 2022

图像标签+文本标签监督

■ MOFI: 用带噪声的 entity-annotated 数据训练图像特征

Model	GPR1200 mAP@1k (%)	ImageNet-ZS Acc@1 (%)
CLIP-L/14$_{\text{OpenAI}}$ [37]	72.19	75.27
MOFI-L/14	86.15	77.17
MOFI-H/14	86.66	78.46

(a) Comparison of MOFI and CLIP on GPR1200 image retrieval and ImageNet zero-shot classification tasks.

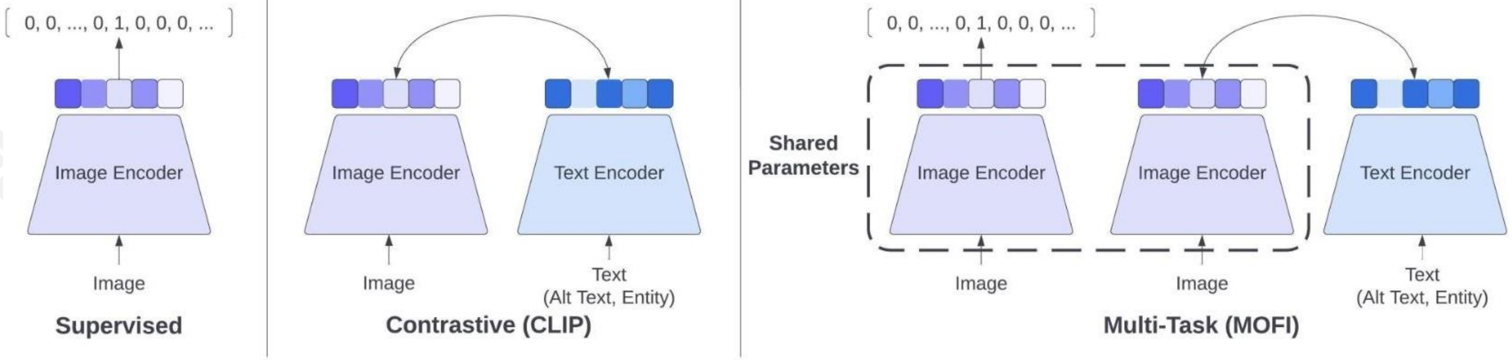
Dataset	# Images	# Classes
ImageNet-1K [41]	1.2M	1K
ImageNet-21K [40]	14M	21K
JFT-300M [48]	300M	18K
JFT-3B [62]	3B	30K
IG-3.6B [46]	3.6B	27K
I2E (Ours)	1.1B	2M

(b) I2E dataset and existing large-scale image classification datasets.

Table 1. MOFI is trained on the newly constructed Image-to-Entities (I2E) dataset, which has 66x more classes than the previous datasets. MOFI achieves significantly better performance on the image retrieval tasks when compared with CLIP.

Why?

Figure 1. **Examples of the I2E dataset.** Each caption is formatted as Entity_id (Entity_name).[4]



[1] MOFI: Learning Image Representations from Noisy Entity Annotated Images, 2023

将 CLIP 与其他学习方法结合

Contrastive Language-Image Pre-training

+ Supervised Learning = ?

+ Image-Only (Non-) Contrastive Learning = ?

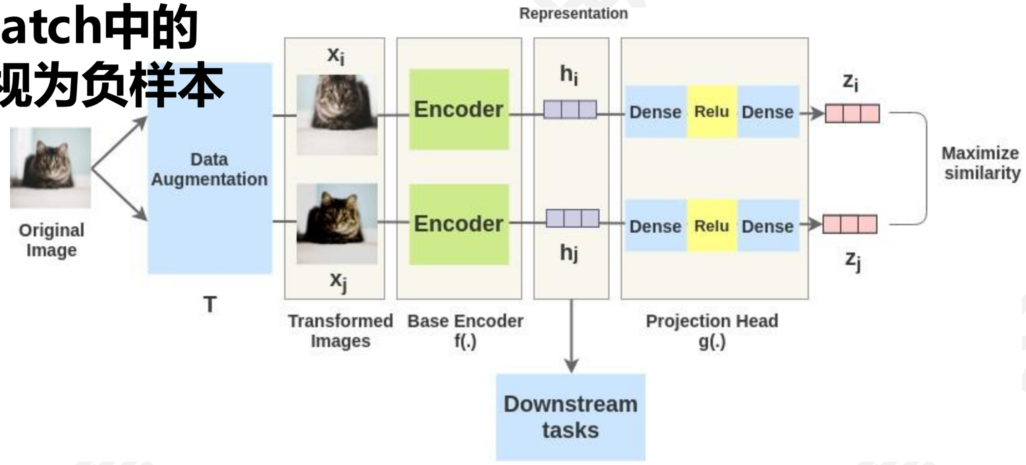
+ Masked Image Modeling = ?

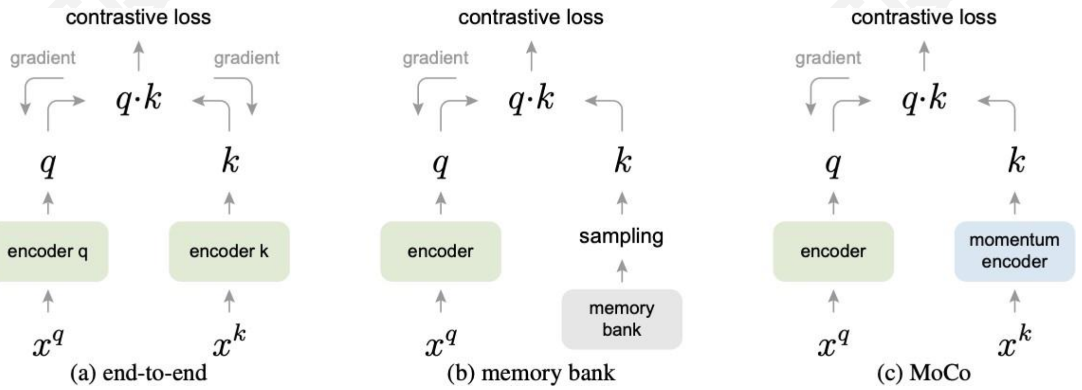
Image-Only (Non-) Contrastive Learning + Masked Image Modeling = ?

图像对比学习

■ SimCLR/MoCo: 图像对比学习

将mini-batch中的其他样本视为负样本





Algorithm 1 Pseudocode of MoCo in a PyTorch-like style.

```
# f_q, f_k: encoder networks for query and key
# queue: dictionary as a queue of K keys (CxK)
# m: momentum
# t: temperature

f_k.params = f_q.params # initialize
for x in loader: # load a minibatch x with N samples
   x_q = aug(x) # a randomly augmented version
   x_k = aug(x) # another randomly augmented version

   q = f_q.forward(x_q) # queries: NxC
   k = f_k.forward(x_k) # keys: NxC
   k = k.detach() # no gradient to keys

   # positive logits: Nx1
   l_pos = bmm(q.view(N,1,C), k.view(N,C,1))

   # negative logits: NxK
   l_neg = mm(q.view(N,C), queue.view(C,K))

   # logits: Nx(1+K)
   logits = cat([l_pos, l_neg], dim=1)

   # contrastive loss, Eqn.(1)
   labels = zeros(N) # positives are the 0-th
   loss = CrossEntropyLoss(logits/t, labels)

   # SGD update: query network
   loss.backward()
   update(f_q.params)

   # momentum update: key network
   f_k.params = m*f_k.params+(1-m)*f_q.params

   # update dictionary
   enqueue(queue, k) # enqueue the current minibatch
   dequeue(queue) # dequeue the earliest minibatch
```

bmm: batch matrix multiplication; mm: matrix multiplication; cat: concatenation.

1 A Simple Framework for Contrastive Learning of Visual Representations, ICML 2020
2 Momentum Contrast for Unsupervised Visual Representation Learning, CVPR 2020

图像对比学习

降低对负样本的依赖

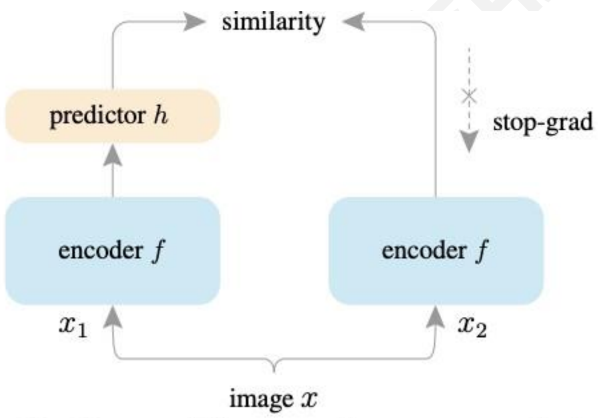


Figure 1. **SimSiam architecture**. Two augmented views of one image are processed by the same encoder network f (a backbone plus a projection MLP). Then a prediction MLP h is applied on one side, and a stop-gradient operation is applied on the other side. The model maximizes the similarity between both sides. It uses neither negative pairs nor a momentum encoder.

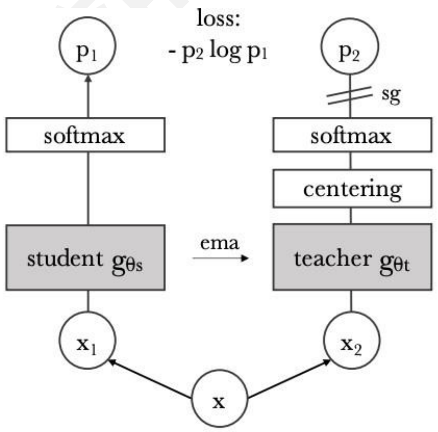


Figure 2: **Self-distillation with no labels.** We illustrate DINO in the case of one single pair of views (x_1, x_2) for simplicity. The model passes two different random transformations of an input image to the student and teacher networks. Both networks have the same architecture but different parameters. The output of the teacher network is centered with a mean computed over the batch. Each networks outputs a K dimensional feature that is normalized with a temperature softmax over the feature dimension. Their similarity is then measured with a cross-entropy loss. We apply a stop-gradient (sg) operator on the teacher to propagate gradients only through the student. The teacher parameters are updated with an exponential moving average (ema) of the student parameters.

1 Bootstrap your own latent-a new approach to self-supervised learning, NeurIPS 2020
2 Exploring simple siamese representation learning, CVPR 2021
3 Variance-invariance-covariance regularization for self-supervised learning, ICLR 2022
4 Barlow twins: Self-supervised learning via ´ redundancy reduction, ICML 2021
5 Unsupervised learning of visual features by contrasting cluster assignments, NeurIPS 2020
6 Emerging properties in self-supervised vision transformers, ICCV 2021

将 CLIP 与其他学习方法结合

Contrastive Language-Image Pre-training

+ Supervised Learning = ?

+ Image-Only (Non-) Contrastive Learning = ?

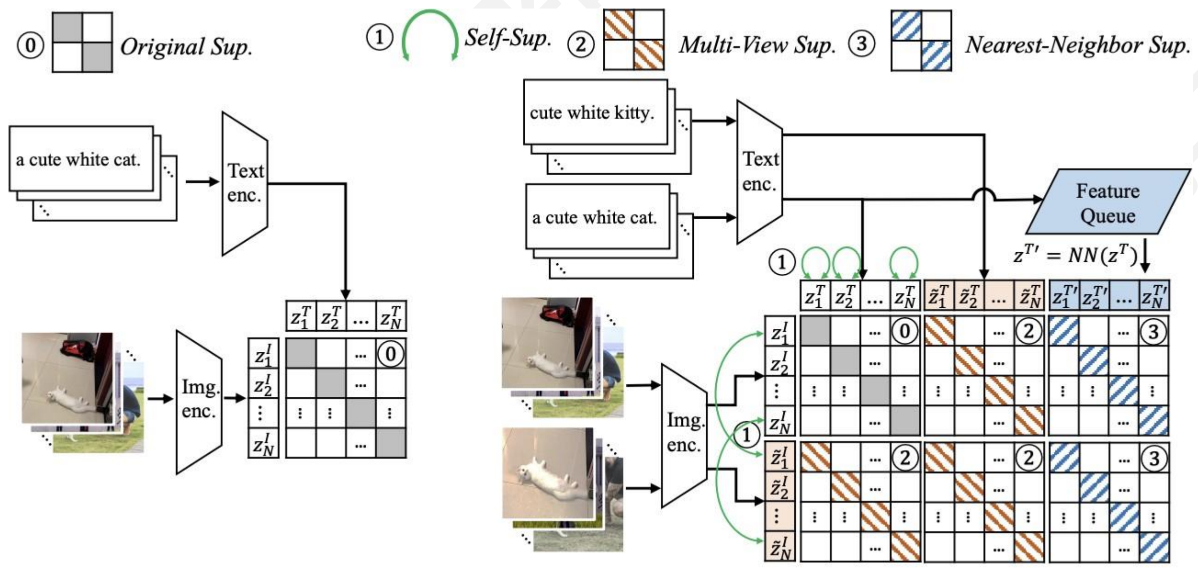
+ Masked Image Modeling = ?

Image-Only (Non-) Contrastive Learning + Masked Image Modeling = ?

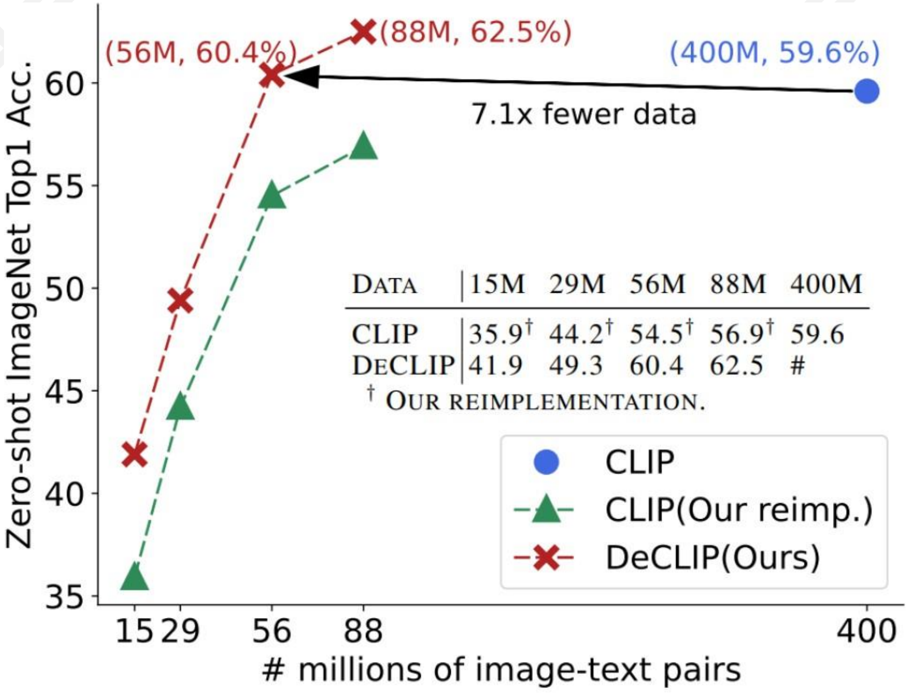
将 CLIP 与图像对比学习结合

- **DeCLIP: 使用多种监督**
  - **Image, Text, Image-Text**

Figure 4: (a) CLIP and ALIGN jointly train an image encoder and a text encoder to predict the correct pairings of a batch of (image, text) training examples. (b) Our DeCLIP overview. ① means Self-Supervision(SS). For image SS, we maximize the similarity between two augmented views of the same instance. For text SS, we leverage Masked Language Modeling(MLM) within a text sentence. ② represents cross-modal Multi-View Supervision(MVS). We first have two augmented views of both image and text, then contrast the 2 × 2 image-text pairs. ③ indicates Nearest-Neighbor Supervision(NNS). We sample text NN in the embedding space to serve as additional supervision. The combination of the three supervision leads to efficient multi-modal learning.



(a) CLIP & ALIGN (b) DeCLIP



Combining vision-language and self-supervised learning improves data efficiency significantly

[1] Supervision exists everywhere: A data efficient contrastive language-image pre-training paradigm, ICLR 2022

将 CLIP 与图像对比学习结合

■ SLIP: 图像自监督+语言-图像监督

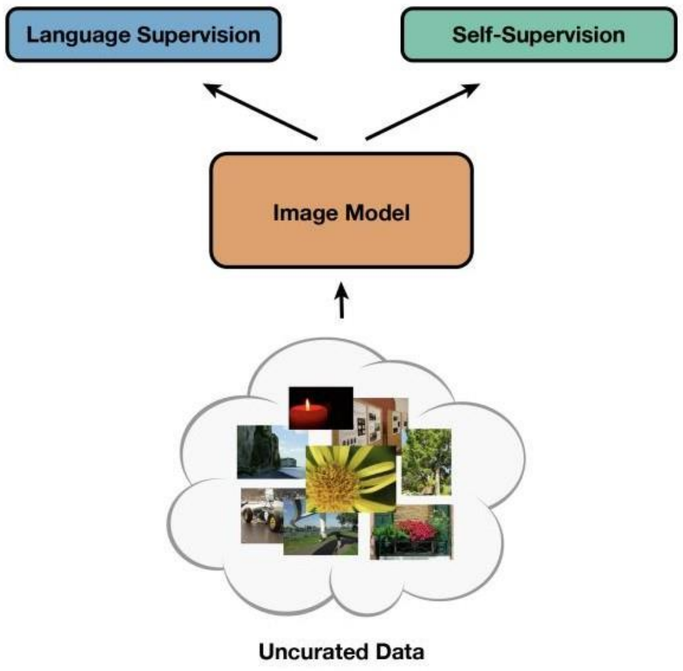


Figure 2. Illustration of SLIP, our multi-task framework. An image model has access to and can be trained with both language supervision from captions and self-supervision on images.

Figure 1. **SLIP pre-training on YFCC15M.** Combining image-only self-superivsion and image-text supervision simultaneously improves zero-shot transfer and linear classification on ImageNet.

[1] Slip: Self-supervision meets language-image pretraining, ECCV 2022

将 CLIP 与图像对比学习结合

xCLIP: 去除训练对负样本的依赖

借鉴图像对比学习中的方法

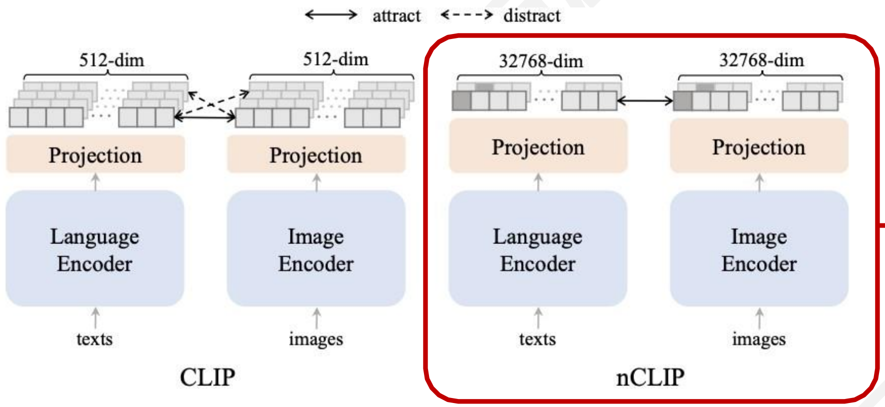


Figure 1. **Architecture comparison between CLIP and nCLIP.**

$$\text{Let } \boldsymbol{p} = \operatorname{softmax}(\boldsymbol{g}) \quad \boldsymbol{q} = \operatorname{softmax}(\boldsymbol{h})$$

$$\mathcal{L}_{\mathrm{CE}} = -\boldsymbol{p}^{\mathrm{T}} \log(\boldsymbol{q}).$$

$$\mathcal{L}_{\mathrm{EH}} = -\boldsymbol{p}^{\mathrm{T}} \log(\boldsymbol{p}), \; -\mathcal{L}_{\mathrm{HE}} = \overline{\boldsymbol{p}}^{\mathrm{T}} \log(\overline{\boldsymbol{p}}),$$

$$\mathcal{L}_{\mathrm{nCLIP}} = \mathcal{L}_{\mathrm{CE}} + \lambda_1 \cdot \mathcal{L}_{\mathrm{EH}} - \lambda_2 \cdot \mathcal{L}_{\mathrm{HE}},$$

Model	Food101	CIFAR10	CIFAR100	Birdsnap	SUN397	Cars	Aircraft	VOC2007	DTD	Pets	Caltech101	Flowers	MNIST	FER2013	STL10	EuroSAT	RESISC45	GTSRB	KITTI	Country211	PCAM	UCF101	Kinetics700	CLEVR	HatefulMemes	SST	**ImageNet**	**Average**
CLIP	61.2	79.7	50.6	23.7	56.5	15.9	5.8	46.4	27.6	54.7	71.3	48.9	10.5	37.3	91.3	24.5	39.7	13.1	31.6	9.1	50.0	45.0	32.4	12.8	**53.0**	49.1	45.7	40.3
nCLIP	28.4	79.5	49.1	11.3	57.0	5.9	4.5	51.4	22.9	14.6	65.0	23.1	9.9	13.5	**94.8**	15.1	21.2	2.7	35.4	5.8	**51.2**	42.0	28.4	12.4	52.7	50.0	37.0	32.7
xCLIP	**65.8**	**83.4**	**54.5**	**25.1**	**59.9**	**18.0**	**5.8**	**52.2**	**33.2**	**57.1**	**73.9**	**50.0**	**12.3**	**39.0**	92.8	**40.0**	**43.6**	**16.3**	**39.8**	**9.3**	51.1	**49.8**	**35.4**	**18.4**	52.5	**50.2**	**48.8**	**43.6**

Table 1. **Zero-shot classification.** We report on a variety of classification benchmarks with ViT-B/16 pre-trained on IT35M. Detailed protocols for each dataset strictly follow CLIP [78]. xCLIP achieves a consistent performance gain compared to CLIP in a wide range of classification datasets. Best results of each column are **bolded**.

需要将 nCLIP 与 CLIP 结合使用

[1] Non-Contrastive Learning Meets Language-Image Pre-Training, CVPR 2023

将 CLIP 与其他学习方法结合

Contrastive Language-Image Pre-training

+ Supervised Learning = ? ✓

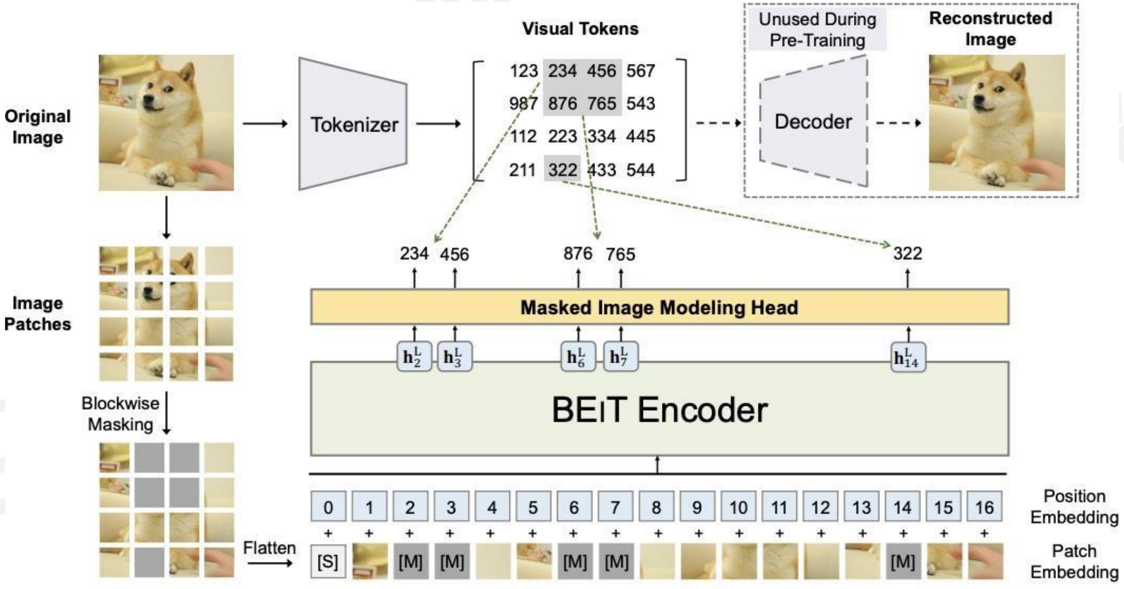
+ Image-Only (Non-) Contrastive Learning = ? ✓

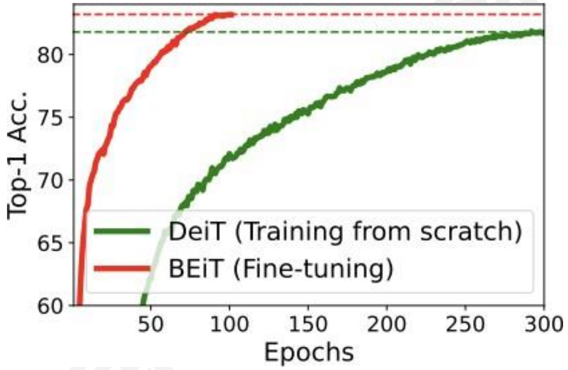
+ Masked Image Modeling = ?

Image-Only (Non-) Contrastive Learning + Masked Image Modeling = ?

掩蔽自编码器

■ **BEiT: 使用离散的 visual tokens 用于掩码图像块恢复**

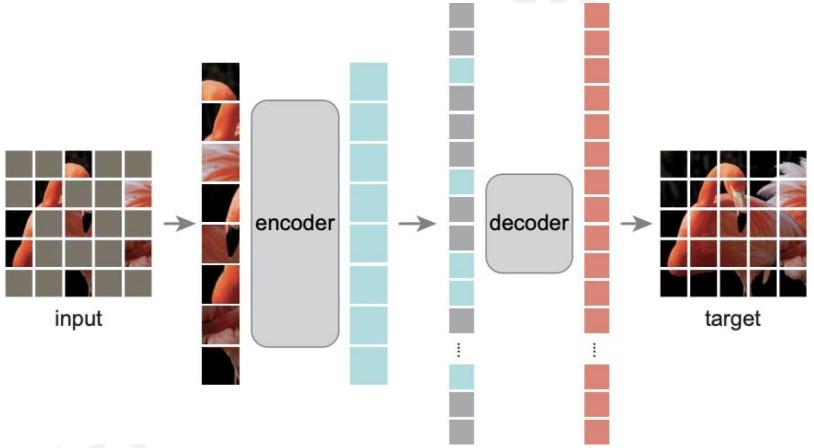




1 BEiT: BERT Pre-Training of Image Transformers, ICLR 2022
2 iBOT: Image BERT Pre-Training with Online Tokenizer, ICLR 2022

掩蔽自编码器

- **MAE:** 使用原始图像块作为掩码图像块恢复的目标
- **MaskFeat:** 使用图像特征作为掩码图像块恢复目标



method	pre-train data	AP^{box} ViT-B	AP^{box} ViT-L	AP^{mask} ViT-B	AP^{mask} ViT-L
supervised	IN1K w/ labels	47.9	49.3	42.9	43.9
MoCo v3	IN1K	47.9	49.3	42.7	44.0
BEiT	IN1K+DALLE	49.8	**53.3**	44.4	47.1
MAE	IN1K	**50.3**	**53.3**	**44.9**	**47.2**

Table 4. **COCO object detection and segmentation** using a ViT

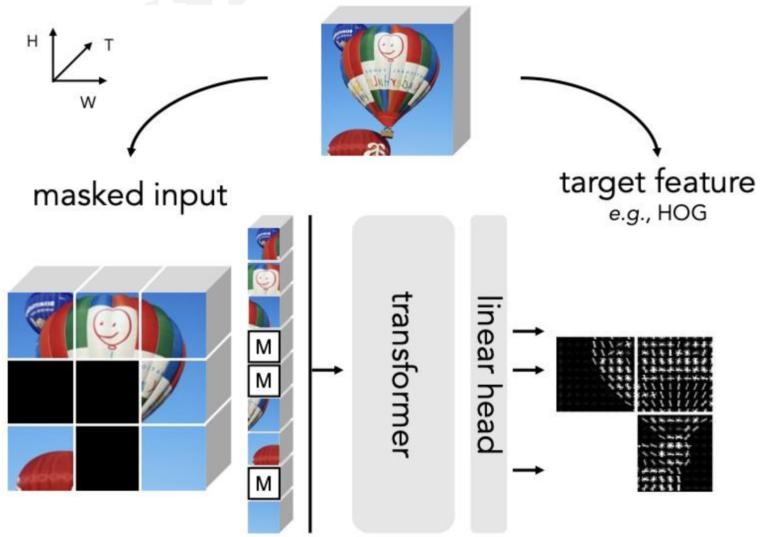


Figure 2. **MaskFeat pre-training.** We randomly replace the input space-time cubes of a video with a [MASK] token and directly regress features (e.g. HOG) of the masked regions. After pre-training, the Transformer is fine-tuned on end tasks.

1 Masked Autoencoders Are Scalable Vision Learners, CVPR 2022
2 Masked feature prediction for self-supervised visual pre-training., CVPR 2022

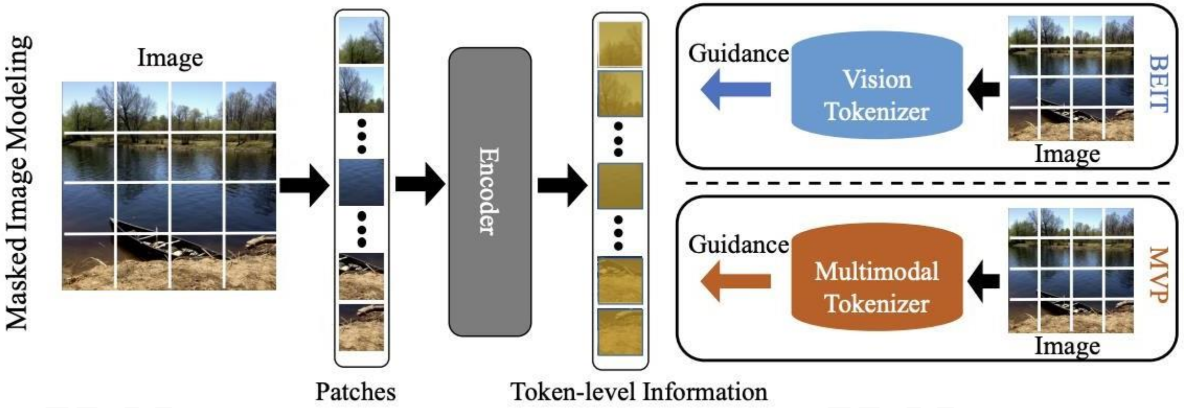
将 CLIP 与其他学习方法结合

Contrastive Language-Image Pre-training
- + Supervised Learning = ?
- + Image-Only (Non-) Contrastive Learning = ?
- + Masked Image Modeling = ?

Image-Only (Non-) Contrastive Learning + Masked Image Modeling = ?

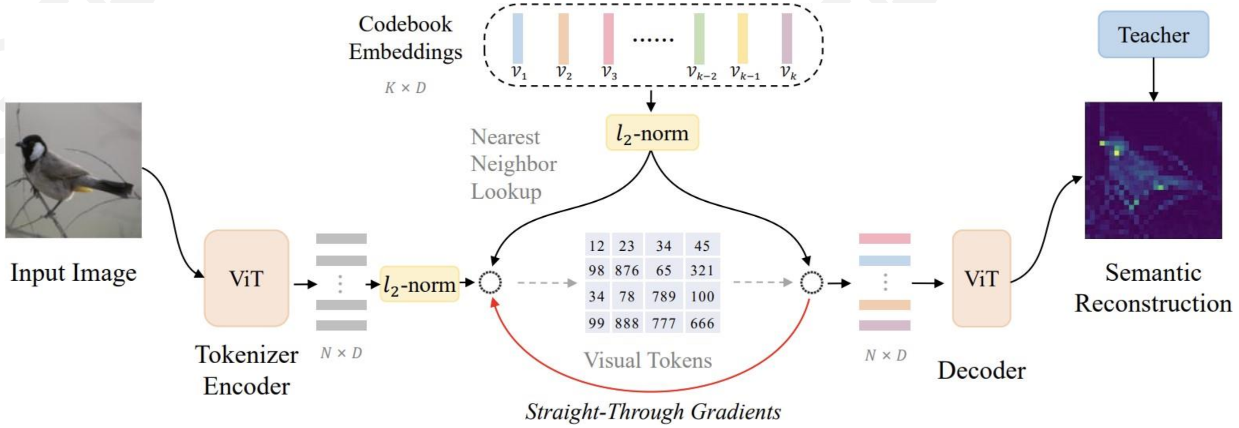
将 CLIP 与掩蔽自编码器结合

MVP/BEiTv2: CLIP 的图像特征（Semantics）利于 MIM 训练



- **MIM: Masked Image Modeling**

Approach 1 (MVP): regress CLIP features

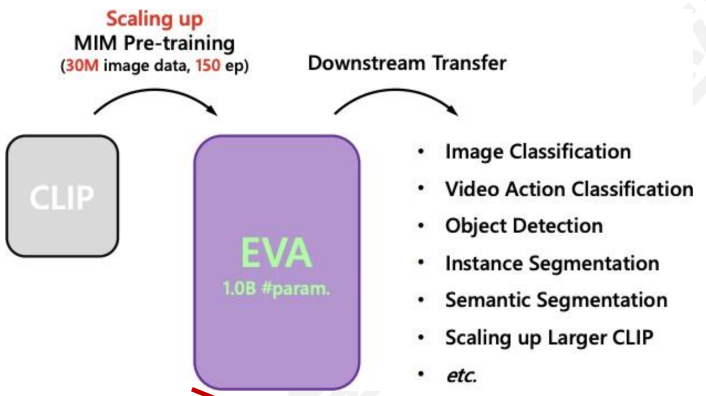


Approach 2 (BEiT v2): compress the information inside CLIP features into the visual tokens (Teacher Model), then perform regular BEiT training

1 MVP: Multimodality-guided Visual Pre-training, ECCV 2022
2 BEiT v2: Masked Image Modeling with Vector-Quantized Visual Tokenizers, 2022

将 CLIP 与掩蔽自编码器结合

■ EVA: CLIP 的图像特征（Semantics）利于 MIM 训练



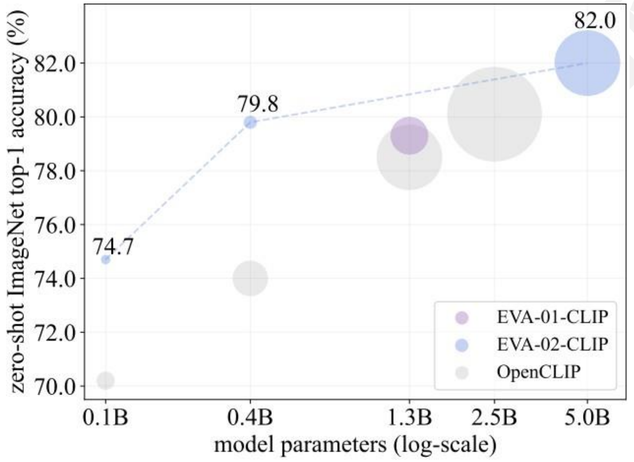


Figure 1: **Summary of CLIP models' ImageNet-1K zero-shot classification performance.** The diameter of each circle corresponds to forward GFLOPs x the number of training samples.

只训练 MIM Model

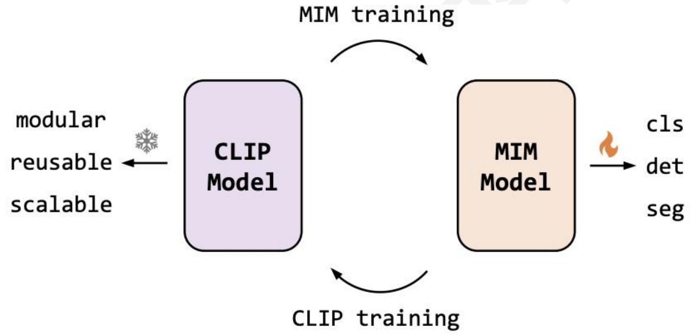


Figure 3: **Alternate learning of MIM and CLIP representations.** Starting with a off-the-shelf CLIP(*e.g.*, OpenAI CLIP [95]), alternate training of the pure MIM visual representations as well as vision-language CLIP representations can improve both MIM and CLIP performances in a bootstrapped manner. The MIM representations can be used to fine-tune various downstream tasks while the (frozen) CLIP representations enable modular, reusable and scalable next-gen model design.

1 EVA: Exploring the Limits of Masked Visual Representation Learning at Scale, CVPR 2023
2 EVA-CLIP: Improved Training Techniques for CLIP at Scale, 2023
3 EVA-02: A Visual Representation for Neon Genesis, 2023.

将 CLIP 与掩蔽自编码器结合

- **BEiT-3: 使用一个 transformer 将多种模态融合**
  - **图像、文本、图像-文本对**

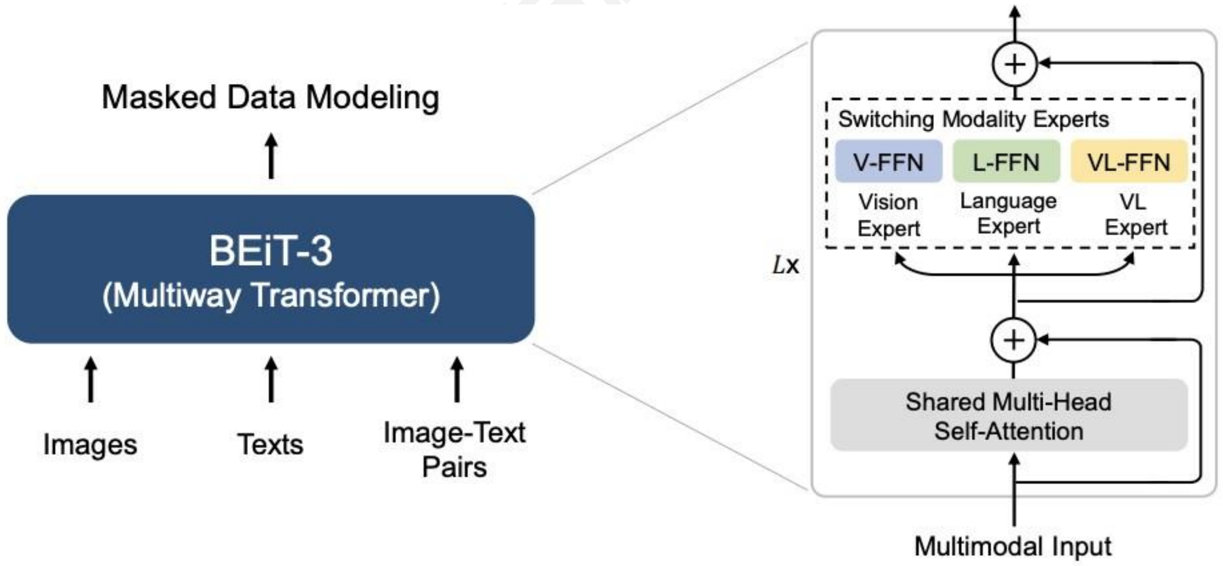
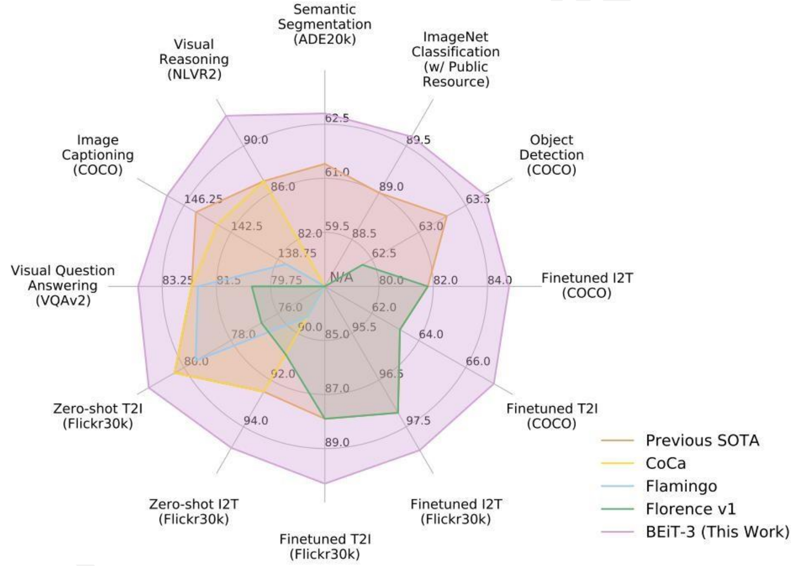


Figure 2: Overview of BEiT-3 pretraining. We perform masked data modeling on monomodal (i.e., images, and texts) and multimodal (i.e., image-text pairs) data with a shared Multiway Transformer as the backbone network.



[1] Image as a Foreign Language: BEiT Pretraining for All Vision and Vision-Language Tasks

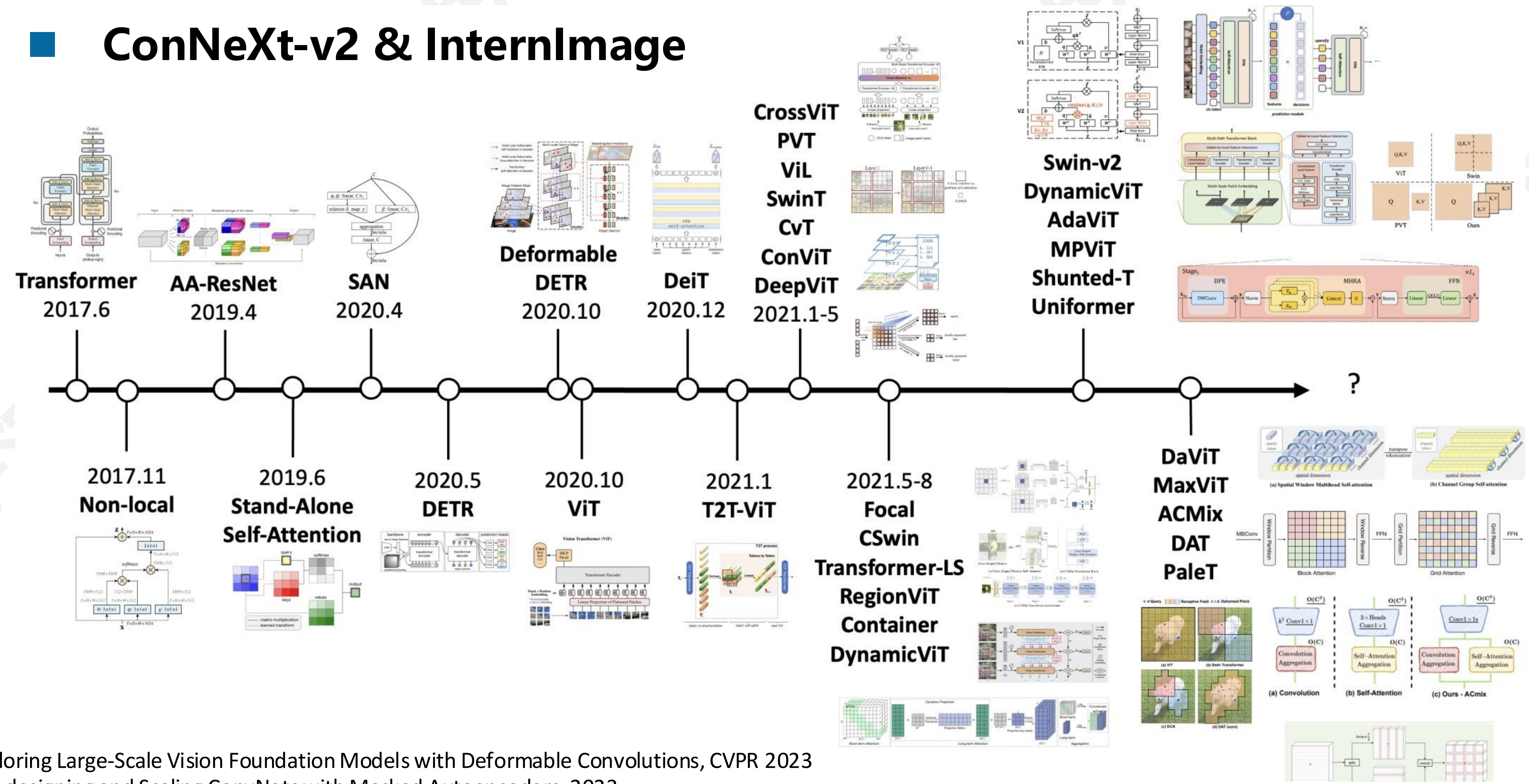
将 CLIP 与其他学习方法结合

Contrastive Language-Image Pre-training
- \+ Supervised Learning = ?
- \+ Image-Only (Non-) Contrastive Learning = ?
- \+ Masked Image Modeling = ?

Image-Only (Non-) Contrastive Learning + Masked Image Modeling = ?

ViT 以外的骨干网络

- **除了 ViT，其他模型也能进行参数缩放，构建基础大模型**
  - **ConNeXt-v2 & InternImage**



1 InternImage: Exploring Large-Scale Vision Foundation Models with Deformable Convolutions, CVPR 2023
2 ConvNeXt V2: Co-designing and Scaling ConvNets with Masked Autoencoders, 2023

大　　　　纲

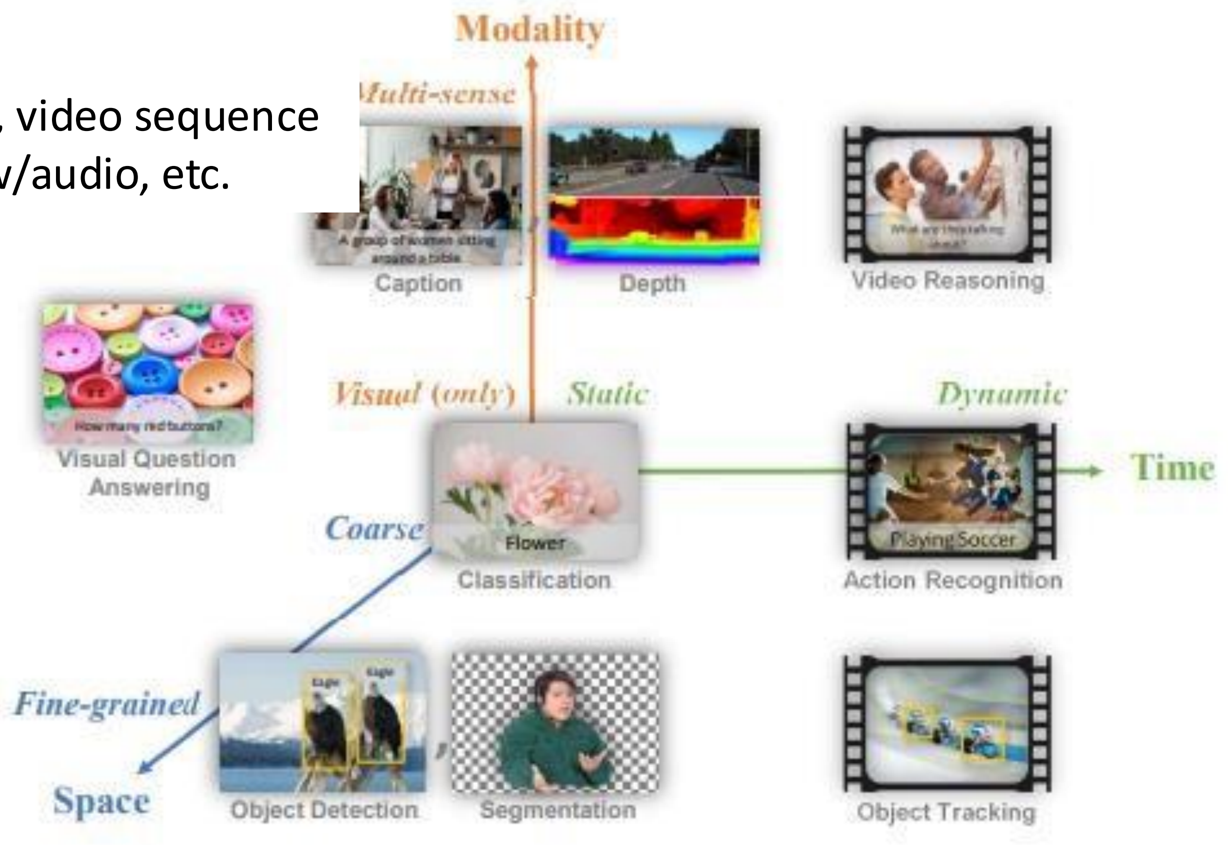
- 视觉（-语言）基础模型预训练
- 通用视觉（-语言）模型

视觉中的挑战：建模

a) 不同类型的输入:

Temporality: static image, video sequence

Multi-modality: w/text, w/audio, etc.

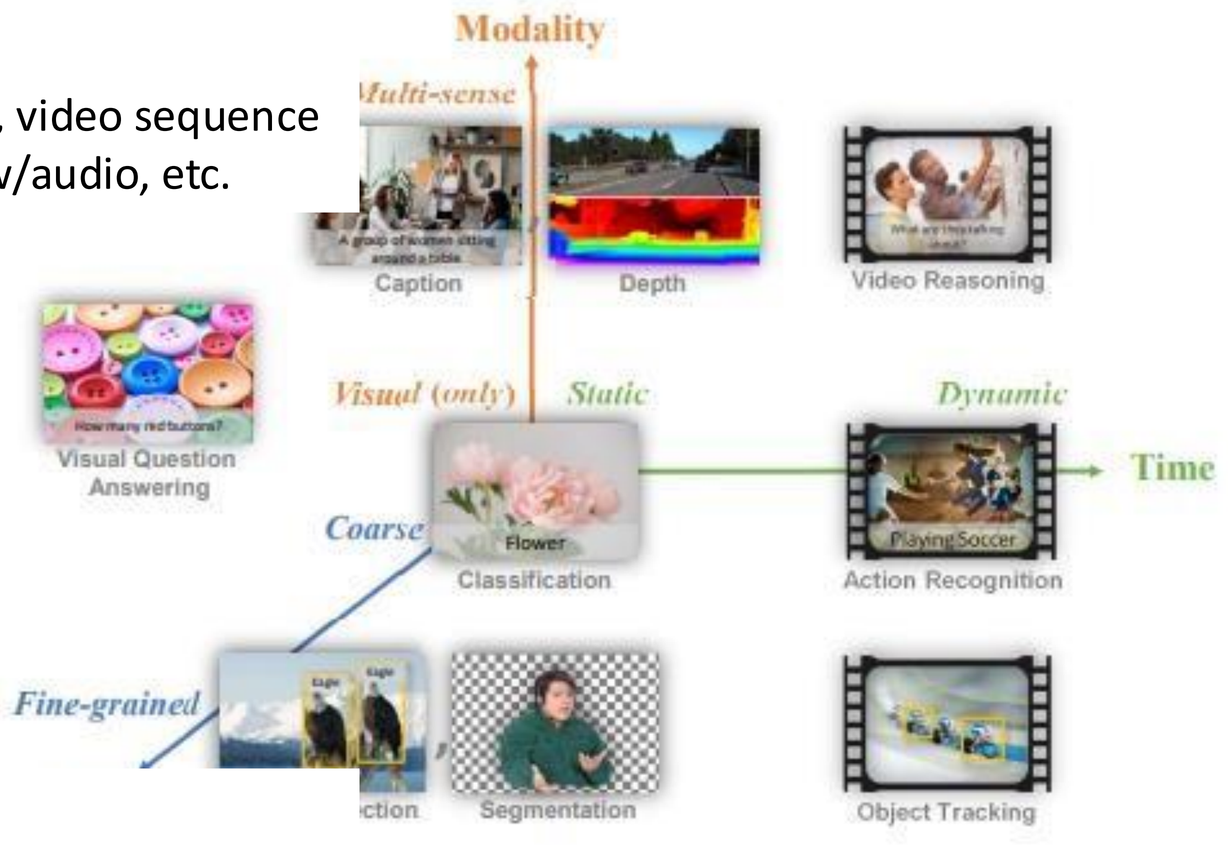


视觉中的挑战：建模

南京大学 NANJING UNIVERSITY

a) 不同类型的输入:

Temporality: static image, video sequence
Multi-modality: w/text, w/audio, etc.



b) 不同类型的任务粒度:

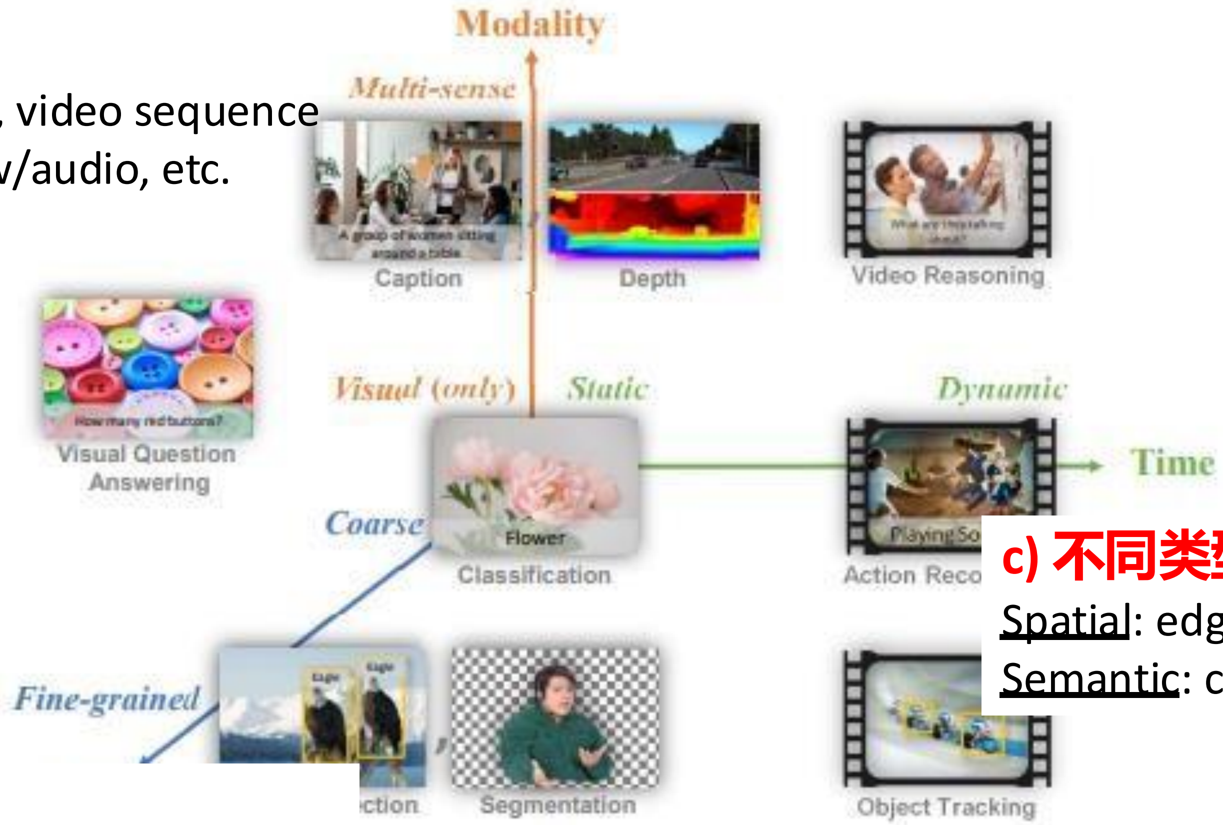
Image-level: classification, captioning, etc.
Region-level: object detection, grounding, etc.
Pixel-level: segmentation, depth, SR, etc.

视觉中的挑战：建模

a) 不同类型的输入:

Temporality: static image, video sequence

Multi-modality: w/text, w/audio, etc.



c) 不同类型的输出:

Spatial: edges, boxes, masks, etc.

Semantic: class labels, descriptions, etc.

b) 不同类型的任务粒度:

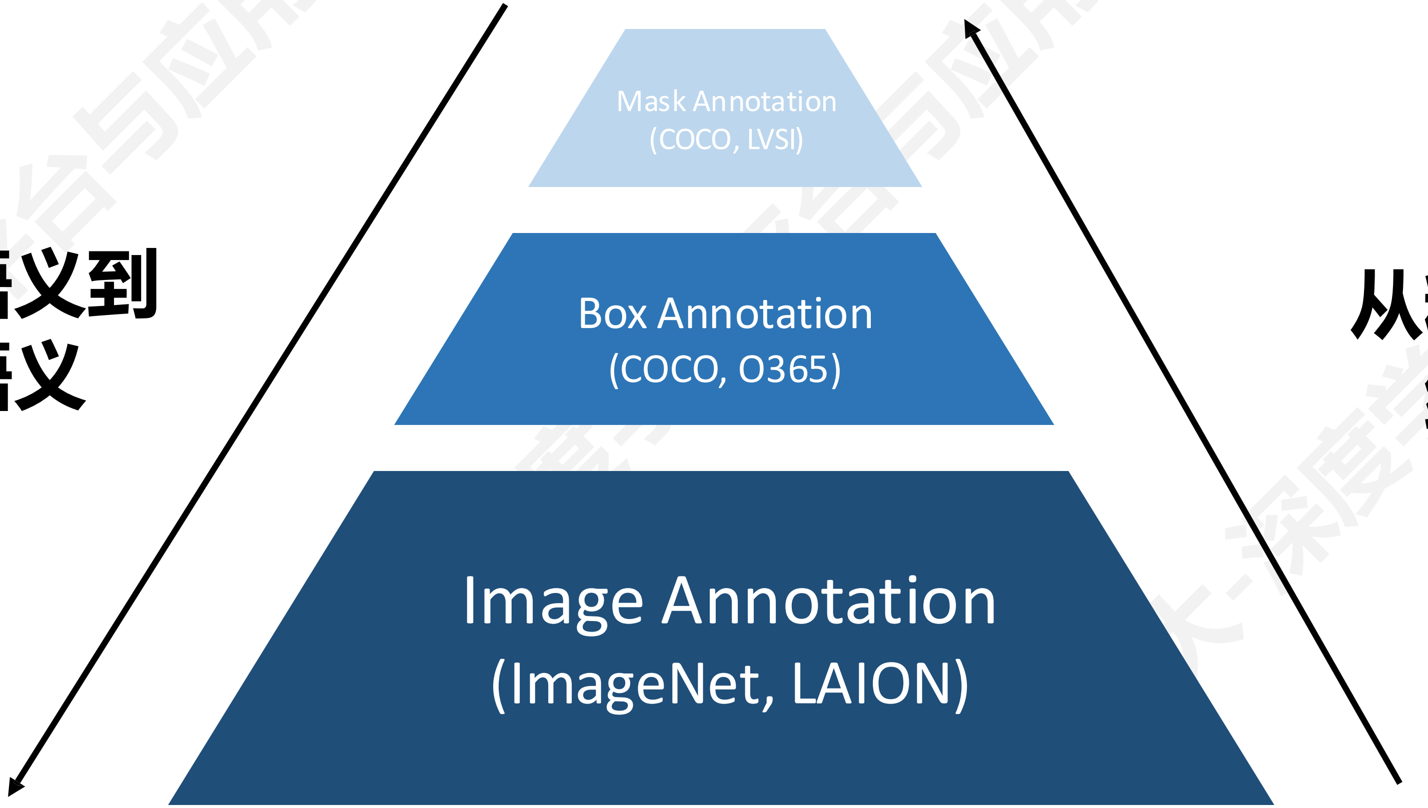
Image-level: classification, captioning, etc.

Region-level: object detection, grounding, etc.

Pixel-level: segmentation, depth, SR, etc.

视觉中的挑战：数据

从低级语义到
高级语义



从粗粒度到
细粒度

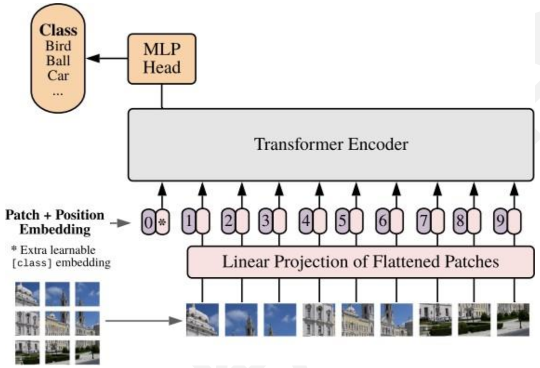
不同类型标注的数量差异很大

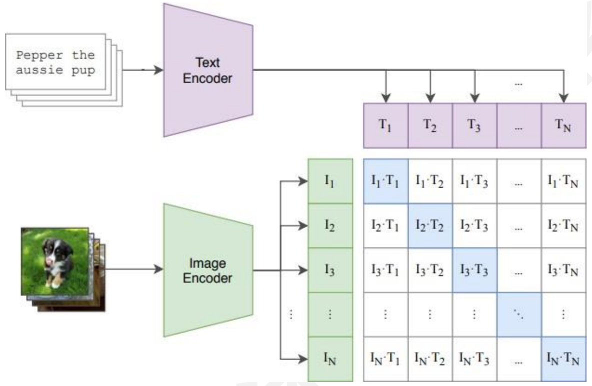
通用视觉模型的尝试

Closed-set Classification → Open-world Recognition

AlexNet[1], ResNet[2], ViT[3]

CLIP[4], ALIGN[5], FLORENCE[6]





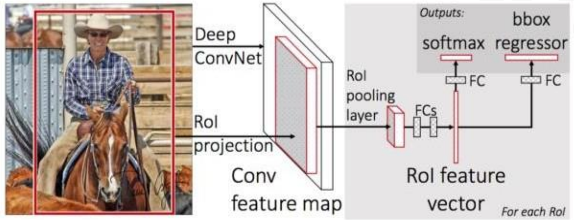
1 Krizhevsky et al. "Imagenet classification with deep convolutional neural networks.“. *NeurIPS* 2012
2 He et al. "Deep residual learning for image recognition." *CVPR* 2016.
3 Dosovitskiy et al. "An image is worth 16x16 words: Transformers for image recognition at scale.“ *ICLR 2021*.
4 Radford et al. Learning transferable visual models from natural language supervision, *ICML* 2021
5 Jia et al. "Scaling up visual and vision-language representation learning with noisy text supervision." *ICML* 2021.
6 Yuan et al. "Florence: A new foundation model for computer vision." *arXiv 2021*.

通用视觉模型的尝试

Closed-set Classification → Open-world Recognition

Specialist Models → Generalist Models

Detection[1], Segmentation[2], VQA[3]



Pixel2Seqv2[4], UniTAB[5], OFA[6], Unified-IO[7], X-Decoder[8]

1 Girshick. "Fast r-cnn." *CVPR* 2015.
2 He et al. "Mask r-cnn." *ICCV* 2017.
3 Antol et al. "Vqa: Visual question answering." *ICCV* 2015.
4 Chen et al. "A unified sequence interface for vision tasks." *NeurIPS 2022*.
5 Yang et al. "Unitab: Unifying text and box outputs for grounded vision-language modeling." *ECCV 2022*.
6 Wang et al. "Ofa: Unifying architectures, tasks, and modalities through a simple sequence-to-sequence learning framework." *ICML* 2022.
7 Lu et al. "Unified-io: A unified model for vision, language, and multi-modal tasks." *ICLR* 2022.
8 Zou et al. "Generalized decoding for pixel, image, and language." *CVPR* 2023.

通用视觉模型的尝试

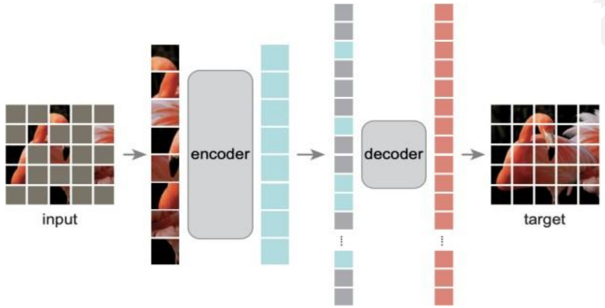
Closed-set Classification → Open-world Recognition

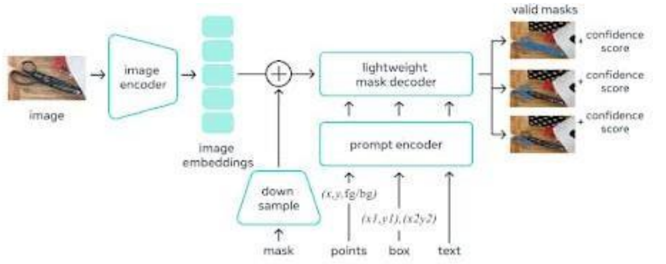
Specialist Models → Generalist Models

Representation Learning → Promptable Interface

BEIT[1], *MAE*[2], *DINO*[3]

SAM[4], *SegGPT*[5], *SEEM*[6]





1 Bao et al. BEiT: BERT Pre-Training of Image Transformers, ICLR 2022.
2 He et al. "Masked autoencoders are scalable vision learners." *CVPR* 2022..
3 Caron et al. "Emerging properties in self-supervised vision transformers." *ICCV* 2021.
4 Kirillov et al. "Segment anything." *arXiv* 2023.
5 Wang et al. "Seggpt: Segmenting everything in context." *arXiv* 2023.
6 Zou et al. "Segment everything everywhere all at once." *arXiv* 2023.

通用视觉模型的尝试

NANJING UNIVERSITY

Closed-set Classification → Open-world Recognition

Specialist Models → Generalist Models

Representation Learning → Promptable Interface

通用视觉模型的尝试

Open-world Recognition

Generalist Models

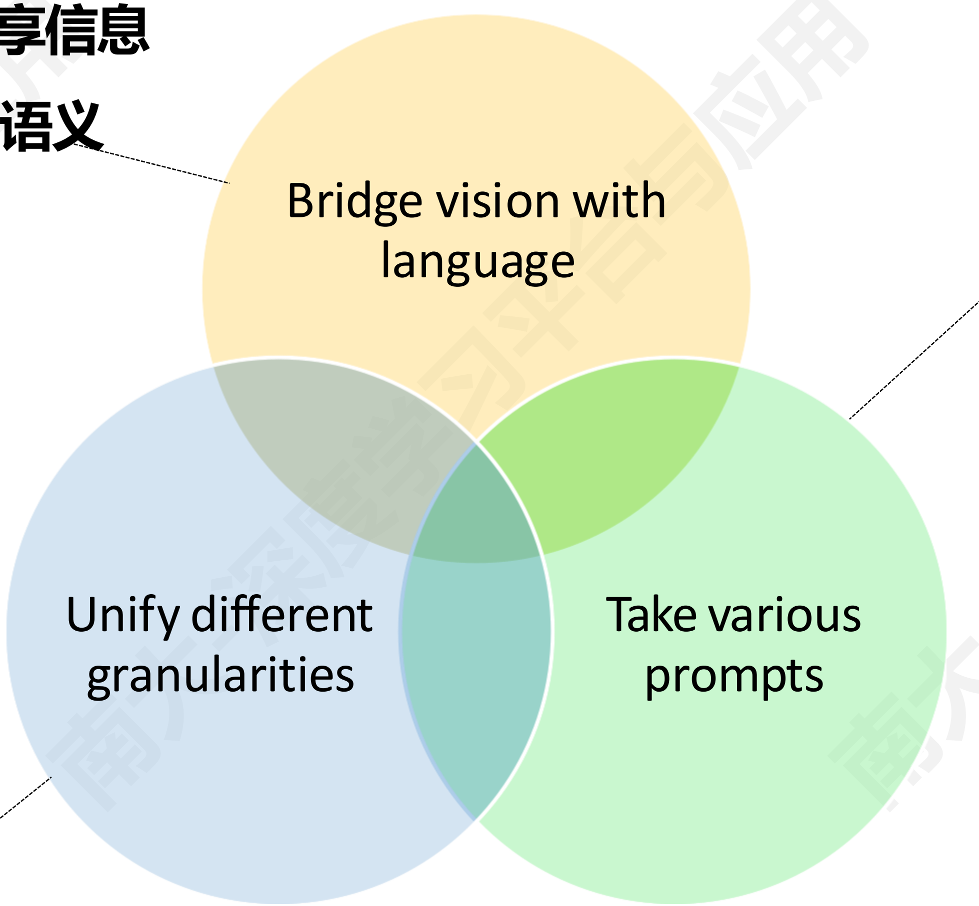
Promptable Interface

通用视觉模型的尝试

南京大學 NANJING UNIVERSITY

直觉: 语言特征空间存在共享信息

优势: 可以零样本迁移到新语义



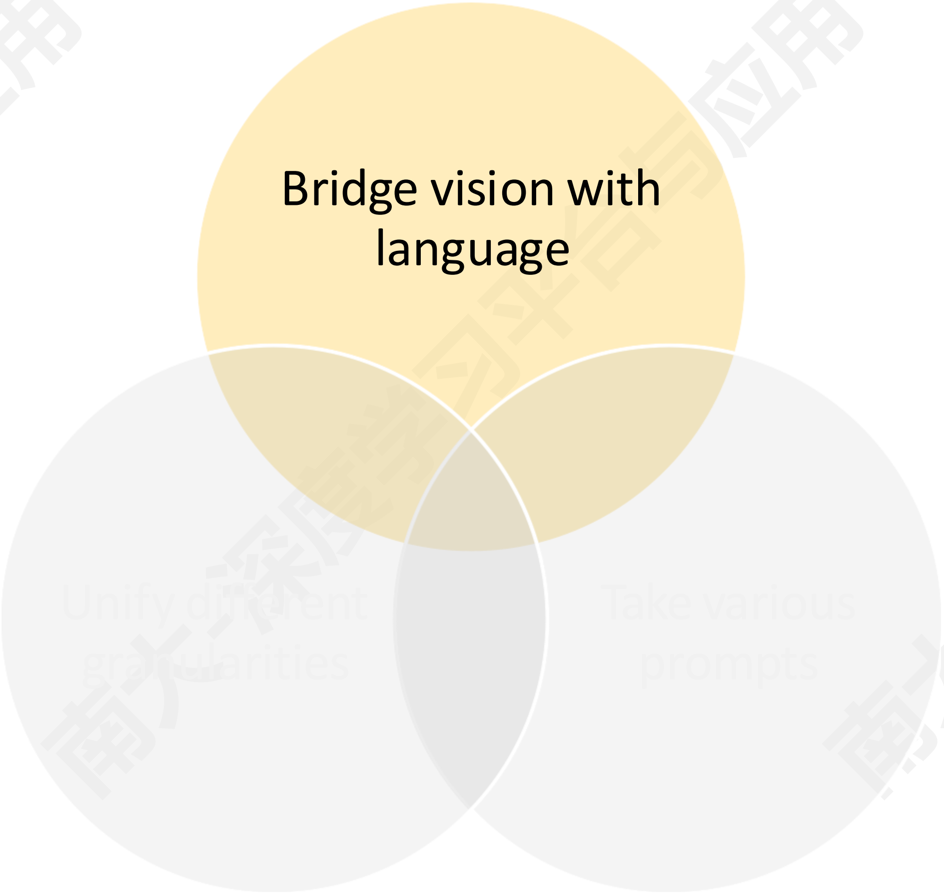
直觉: 语言、空间提示（prompt）等

优势: 降低模型理解人类意图的歧义

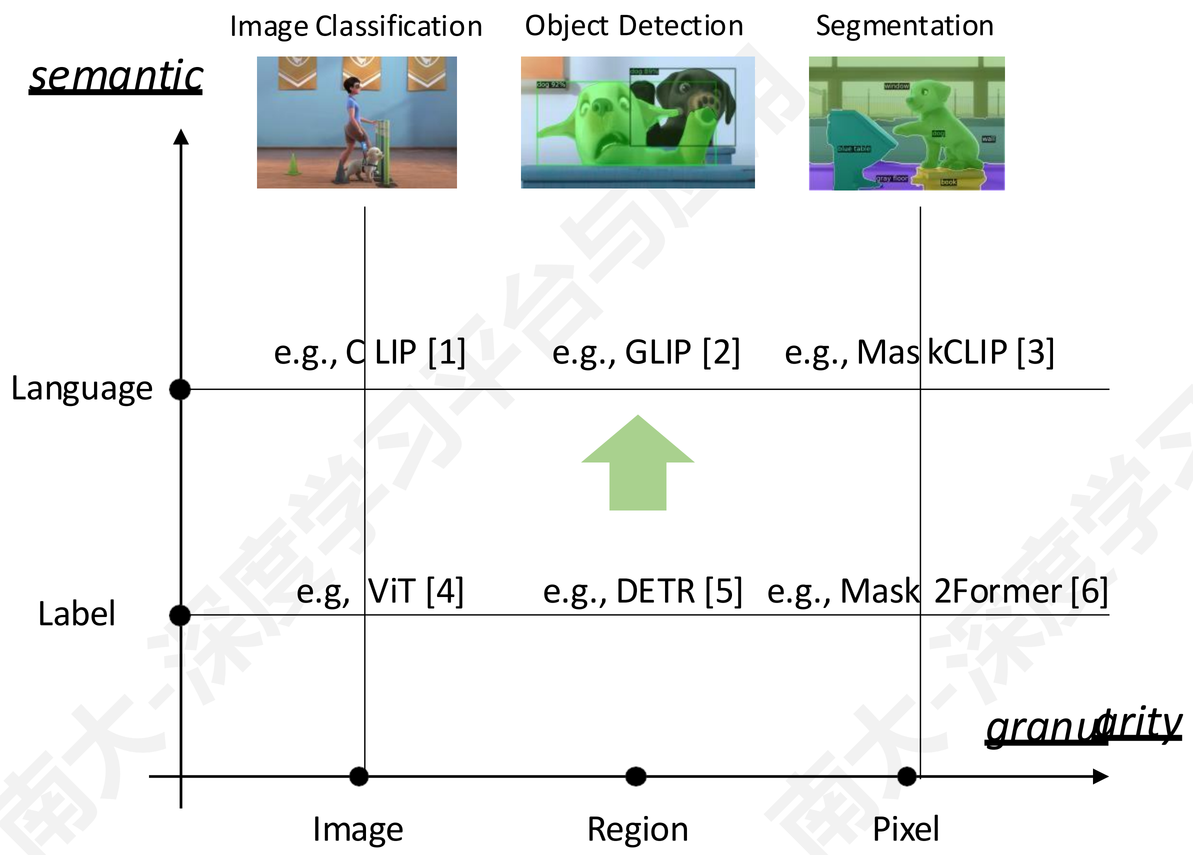
直觉: 视觉是多任务、多粒度的

优势: 不同粒度的视觉任务之间可以进行协同

建立视觉和语言的联系

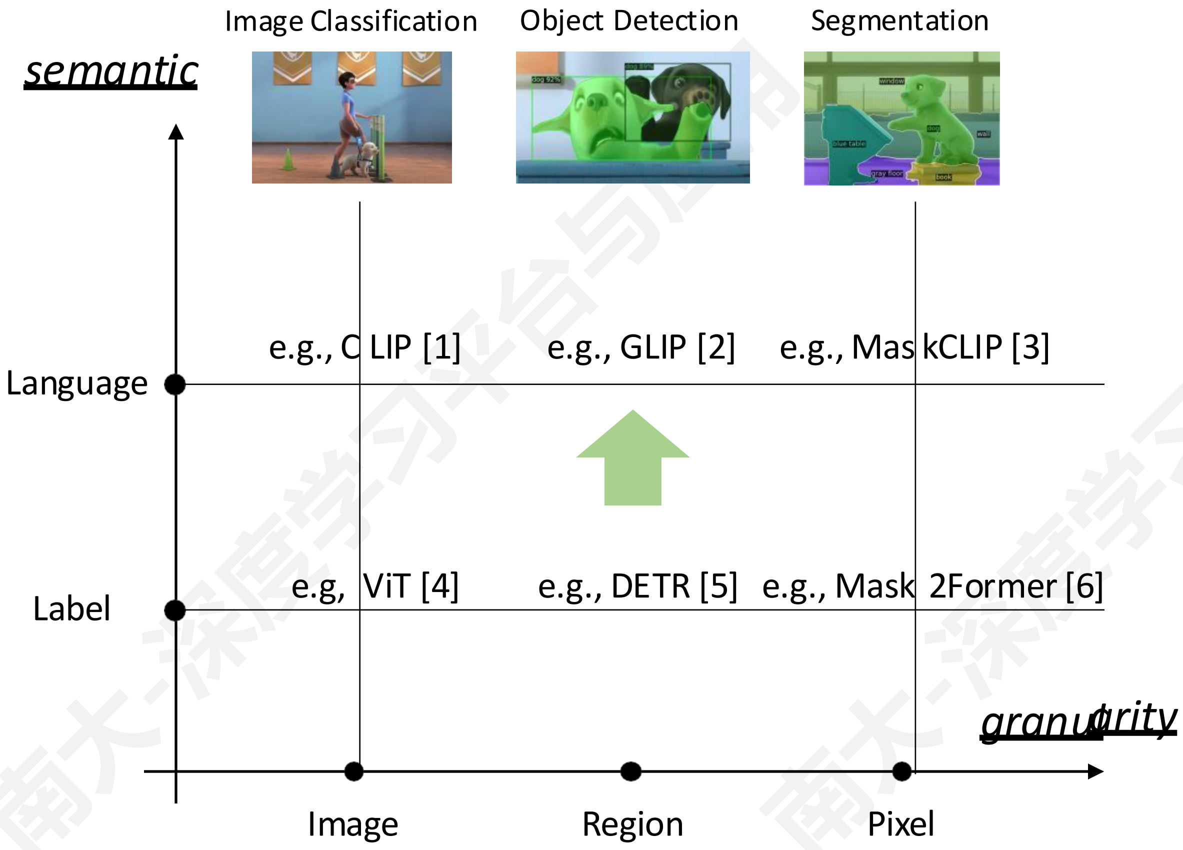

Bridge vision with language
Unify different granularities
Take various prompts

建立视觉和语言的联系



1 Radford et al. "Learning transferable visual models from natural language supervision." ICML, PMLR, 2021
2 Li et al. "Grounded language-image pre-training." CVPR, 2022
3 Zhou et al. "Extíact Fíee Dense Labels fíom CLIP." ECCV, 2022
4 Dosovitskiy et al. “An image is worth 16x16 words: Transformers for image recognition at scale.” ICLR, 2021
5 Carion et al. "End-to-end object detection with transformers." ECCV, 2020
6 Cheng et al. "Masked-attention mask transformer for universal image segmentation." CVPR. 2022

建立视觉和语言的联系



- 将图像标签转换为语言标签与任务粒度无关
- 粗粒度的知识可以迁移到细粒度的任务中

1 Radford et al. "Learning transferable visual models from natural language supervision." ICML, PMLR, 2021
2 Li et al. "Grounded language-image pre-training." CVPR, 2022
3 Zhou et al. "Extíact Fíee Dense Labels fíom CLIP." ECCV, 2022
4 Dosovitskiy et al. “An image is worth 16x16 words: Transformers for image recognition at scale.” ICLR, 2021
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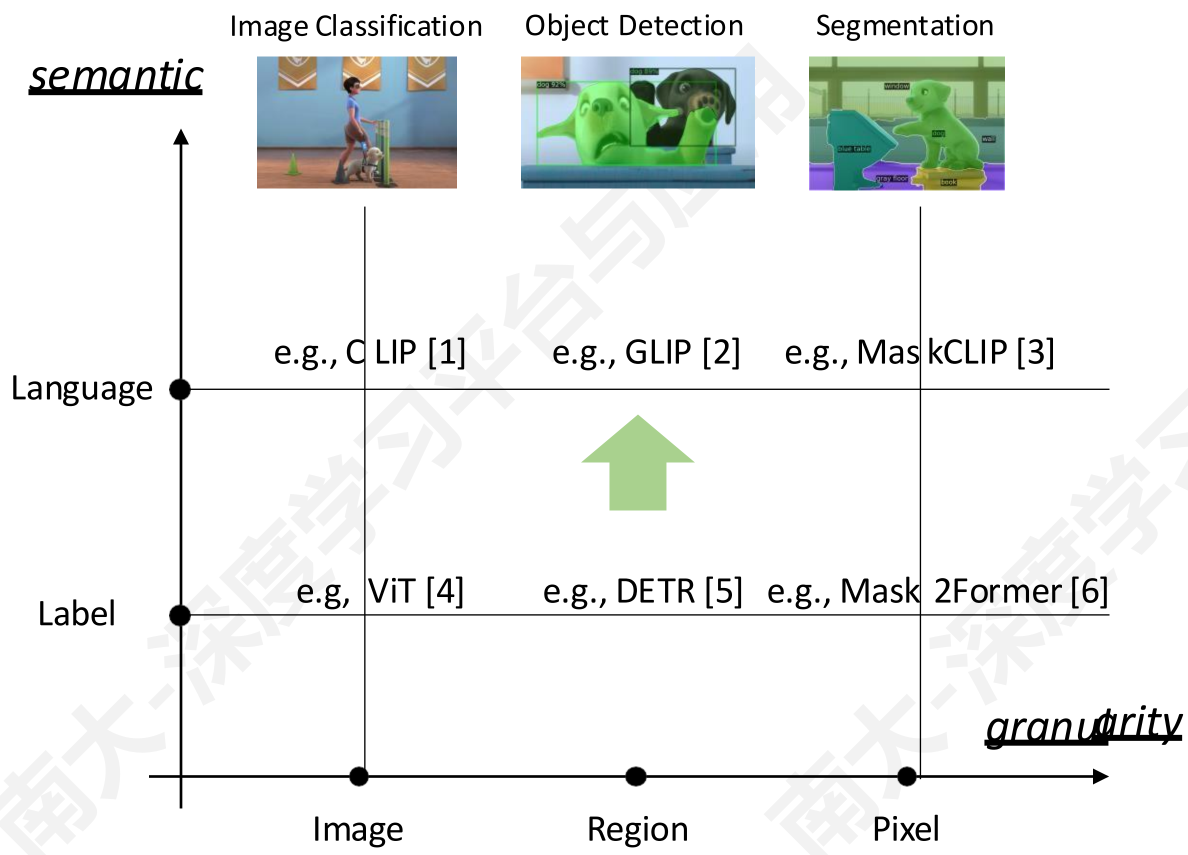
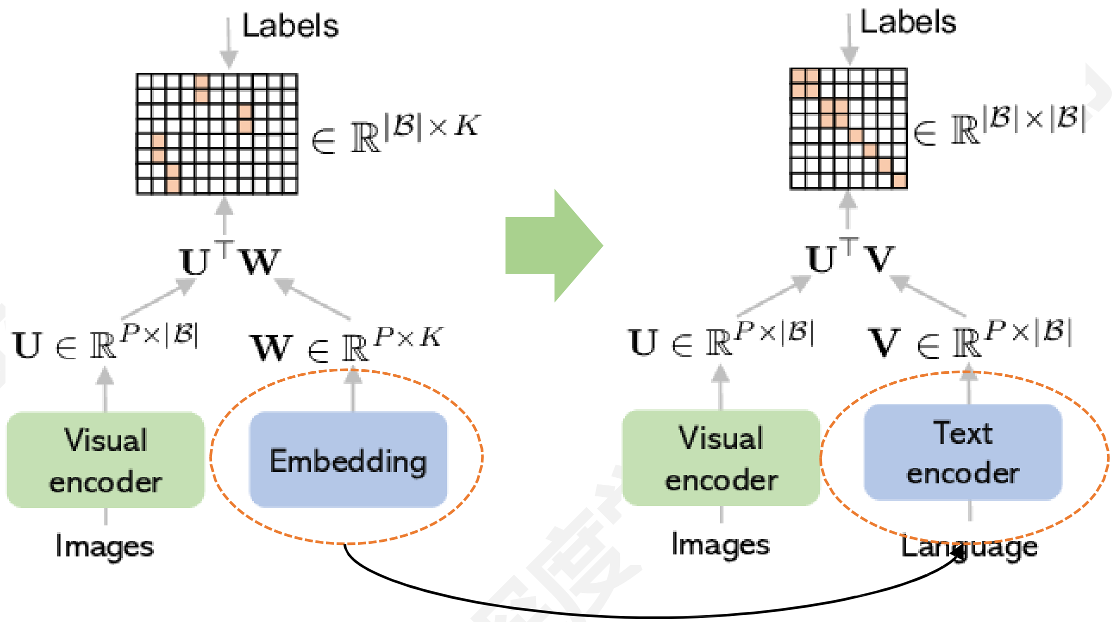
建立视觉和语言的联系





将标签替换为概念名称，并使用文本编码器对所有概念进行编码

1 Radford et al. "Learning transferable visual models from natural language supervision." ICML, PMLR, 2021
2 Li et al. "Grounded language-image pre-training." CVPR, 2022
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建立视觉和语言的联系

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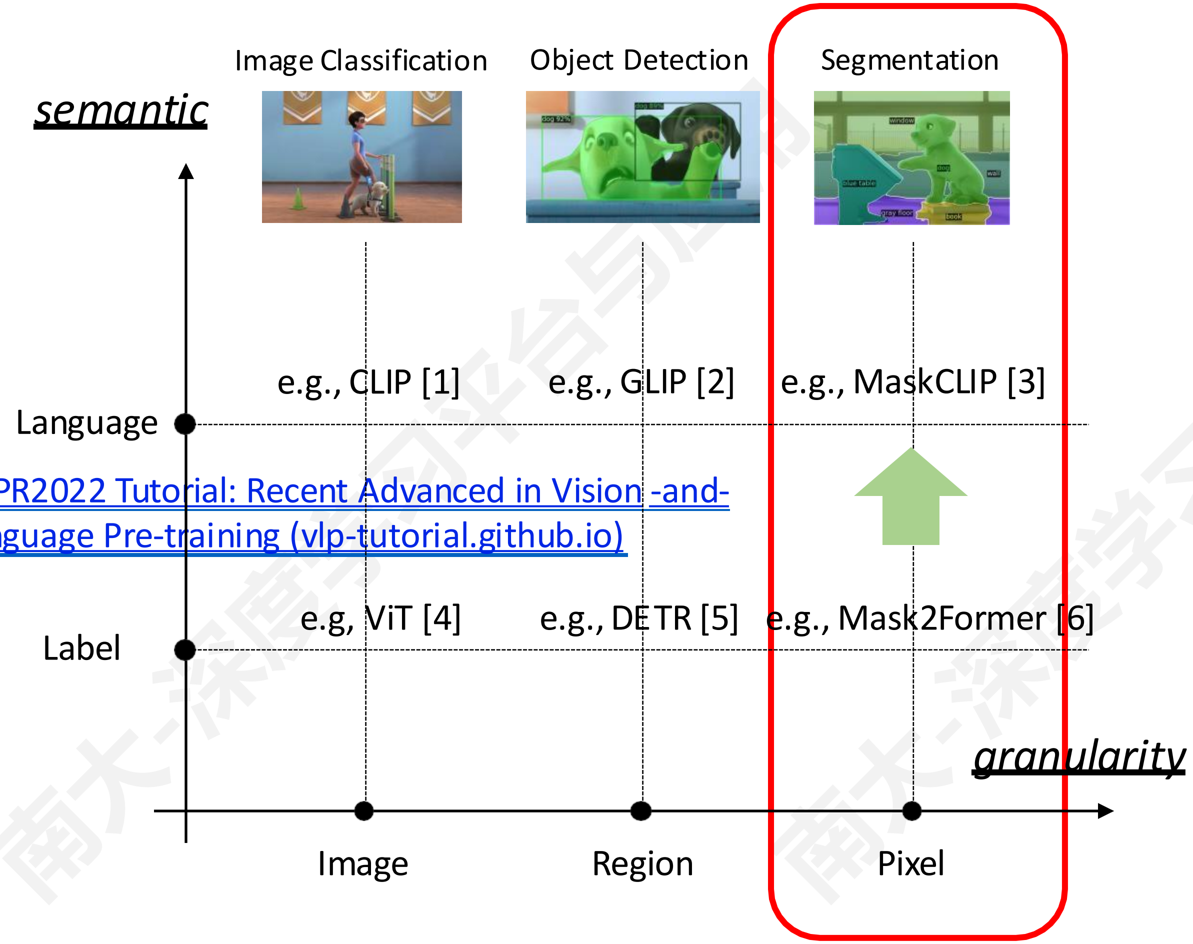
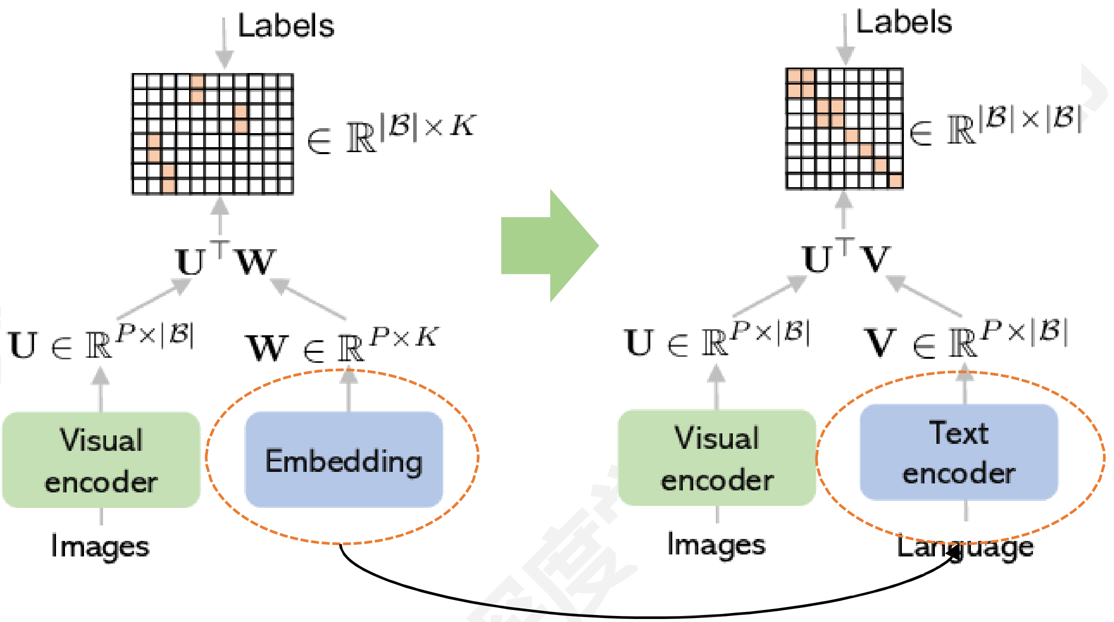


[CVPR2022 Tutorial: Recent Advanced in Vision -and-Language Pre-training (vlp-tutorial.github.io)](vlp-tutorial.github.io)



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1 Radford et al. "Learning transferable visual models from natural language supervision." ICML, PMLR, 2021
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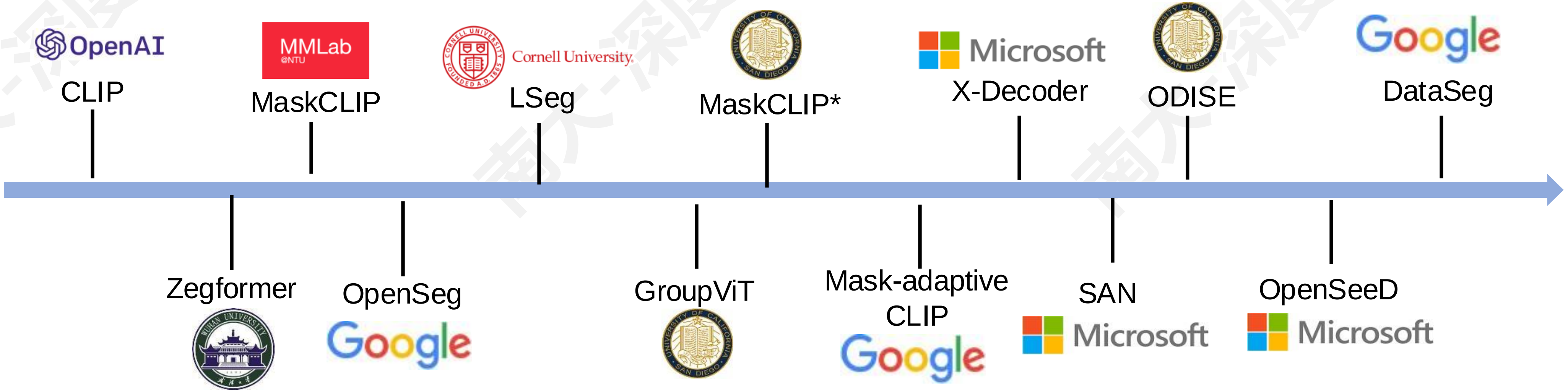
5 Carion et al. "End-to-end object detection with transformers." ECCV, 2020
6 Cheng et al. "Masked-attention mask transformer for universal image segmentation." CVPR. 2022

建立视觉和语言的联系（图像分割）

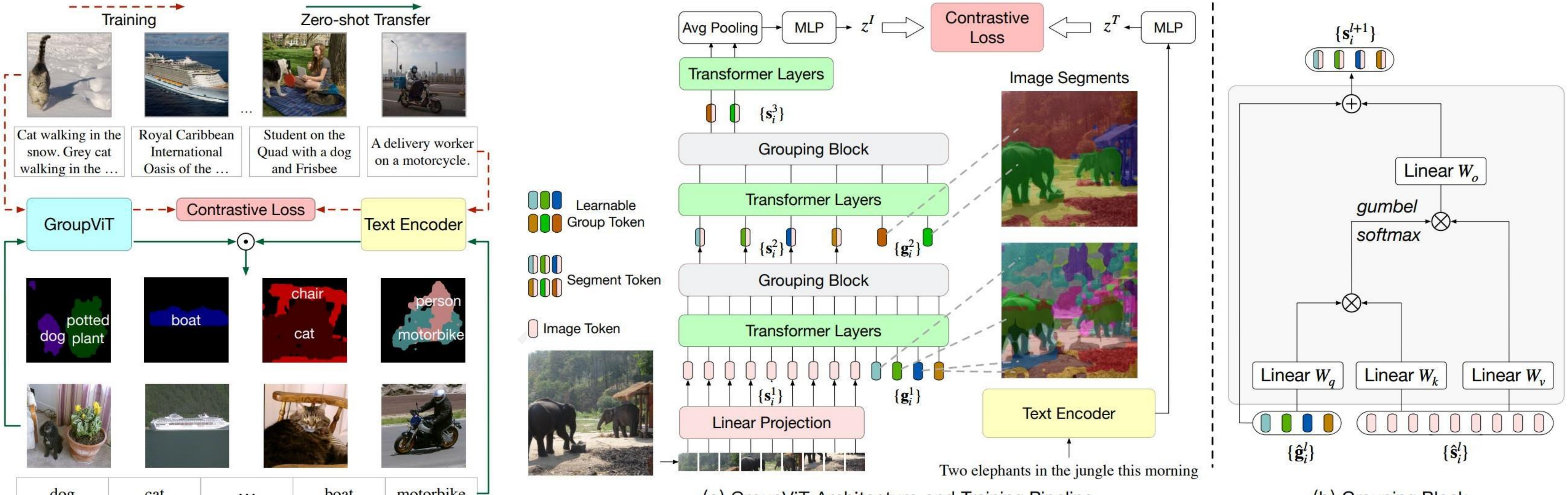
- **分割任务：**
  - **Generic segmentation (semantic/instance/panoptic)**
  - **Referring segmentation (根据特定的文本短语分割图像)**

- **方法：**
  - **CLIP 初始化 v.s. 从头训练**
  - **弱监督训练 v.s. 全监督训练**
  - **两阶段训练 v.s. 端到端训练**



建立视觉和语言的联系（图像分割）

- **GroupViT: 从头开始学习图像-文本对**
  - **学会对语义相似的区域进行 grouping（自下向上 grouping，自上向下图像-文本对监督）**

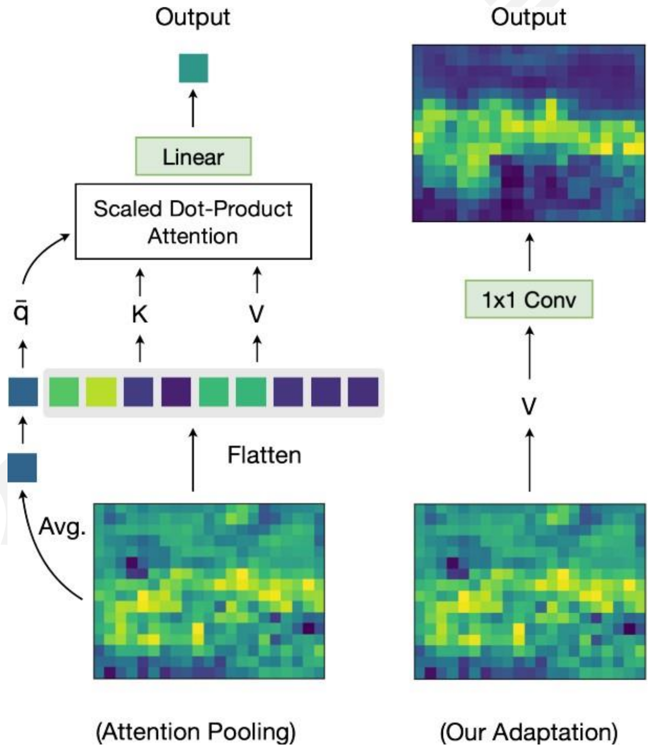


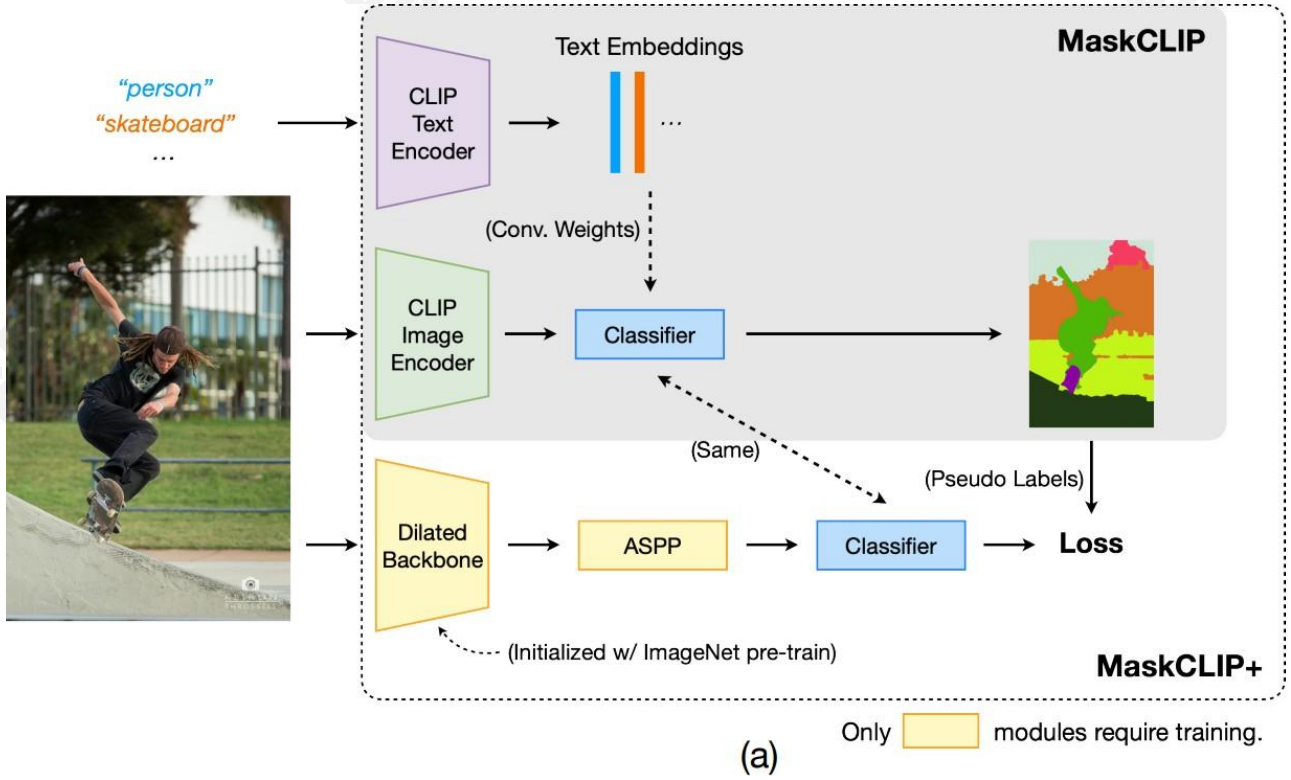
(a) GroupViT Architecture and Training Pipeline

(b) Grouping Block

建立视觉和语言的联系（图像分割）

- **MaskCLIP: “免费”使用 CLIP 进行分割**
  - **MaskCLIP+: 冻结 CLIP，使用 CLIP 作为生成伪标签的教师模型**

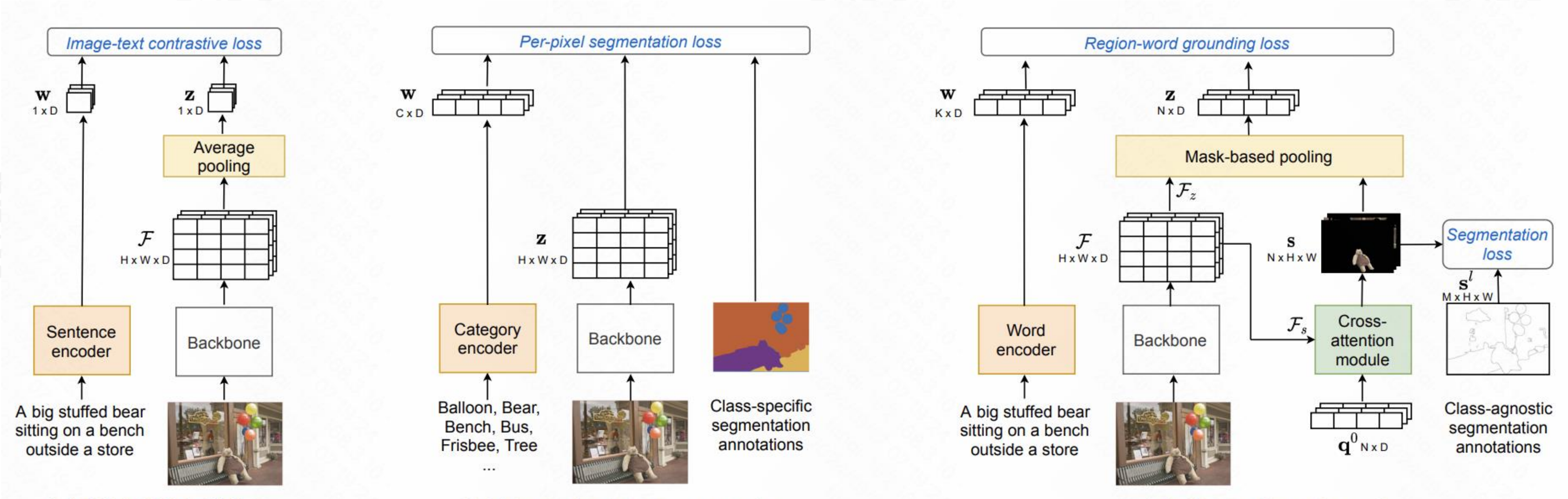




[1] Zhou et al. "Extíact Fíee Dense Labels fíom CLIP." ECCV, 2022

建立视觉和语言的联系（图像分割）

- **OpenSeg: 弱监督学习-将文本特征与图像区域特征对齐**
  - 需要使用预训练好的分割模型，从而提取图像区域
  - 从图像描述中学习图像区域与单词之间的弱监督对齐

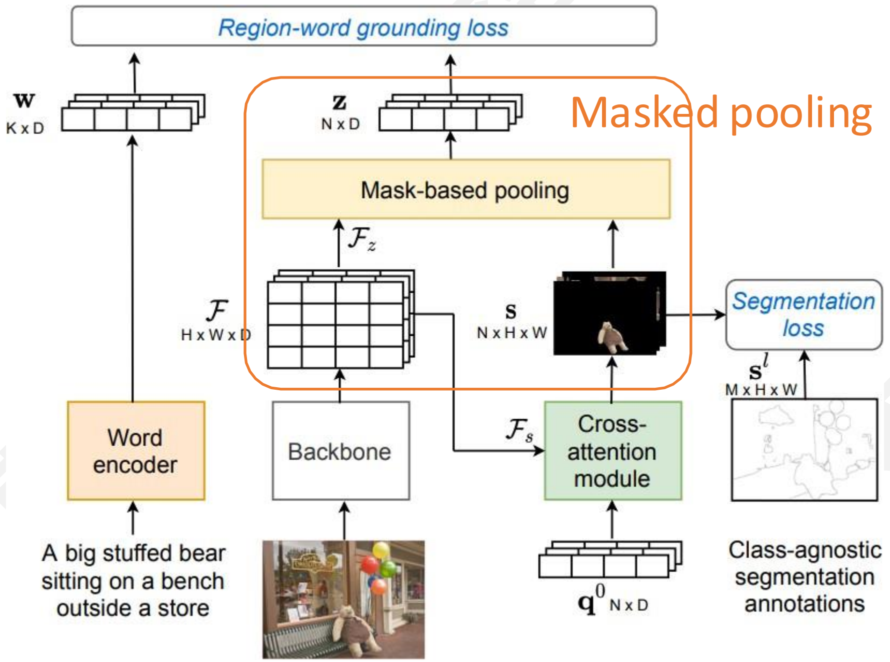


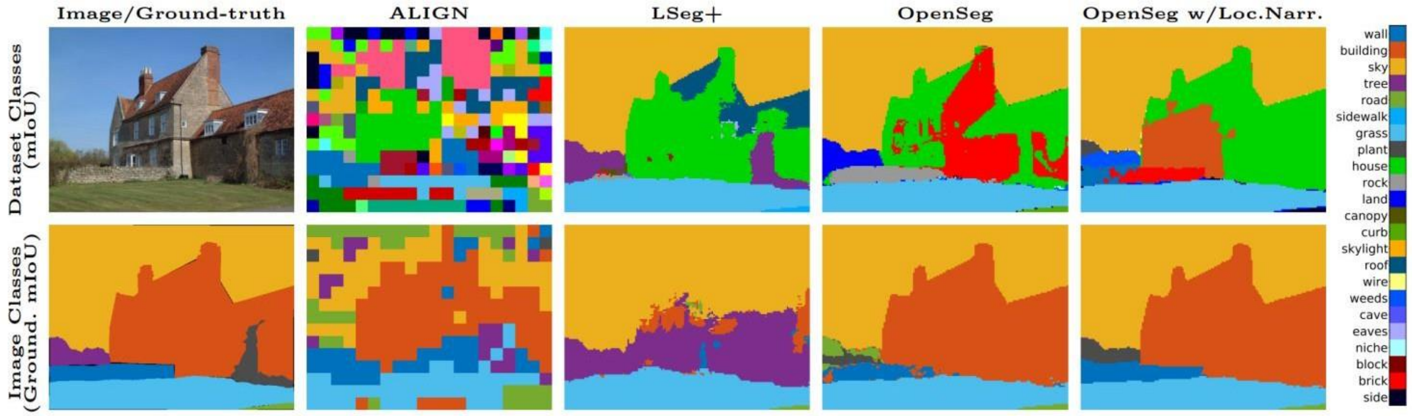
(a) ALIGN / CLIP (b) Per-pixel segmentation (c) OpenSeg (ours)

建立视觉和语言的联系（图像分割）

OpenSeg: 弱监督学习-将文本特征与图像区域特征对齐

- 需要使用预训练好的分割模型，从而提取图像区域
- 从图像描述中学习图像区域与单词之间的弱监督对齐

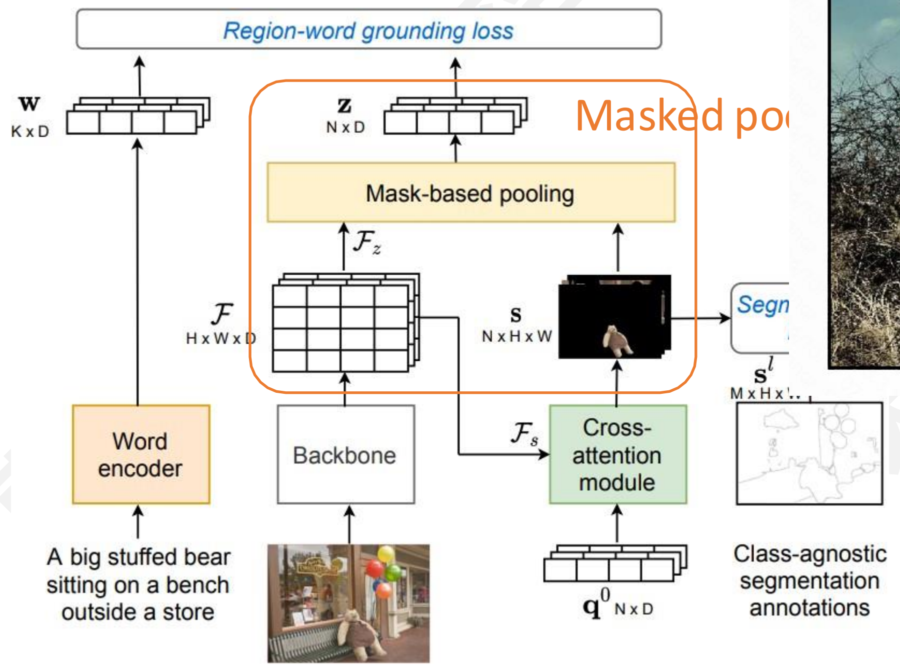




	COCO Train			mIoU					Grounding mIoU				
	label	mask	cap.	A-847	PC-459	A-150	PC-59	COCO	A-847	PC-459	A-150	PC-59	COCO
ALIGN	✗	✗	✗	4.8	3.6	9.7	18.5	15.6	17.8	21.8	25.7	34.2	28.2
ALIGN w/proposal	✗	✓	✗	5.8	4.8	12.9	22.4	17.9	17.3	19.7	25.3	32.0	23.6
LSeg+	✓	✓	✗	3.8	7.8	18.0	**46.5**	55.1	10.5	17.1	30.8	56.7	60.8
OpenSeg	✗	✓	✓	6.3	9.0	21.1	42.1	36.1	21.8	32.1	41.0	57.2	48.2
OpenSeg w/L. Narr.	✗	✓	✓	**6.8**	**11.2**	**24.8**	45.9	38.1	**25.4**	**39.0**	**45.5**	**61.5**	48.2

建立视觉和语言的联系（图像分割）

- **OpenSeg: 弱监督**
  - 需要使用预训练好的
  - 从图像描述中学习图



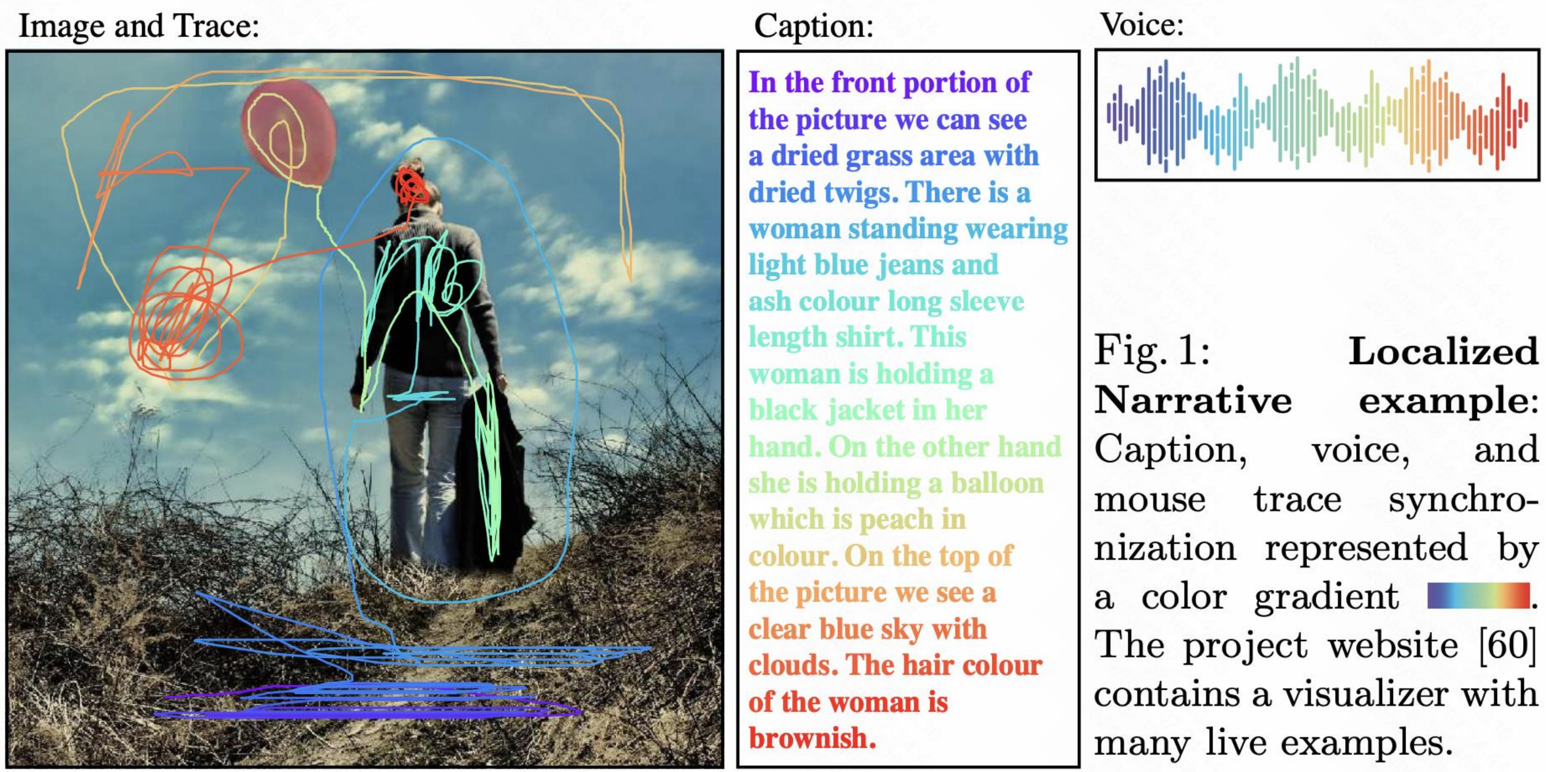


Fig. 1: Localized Narrative example: Caption, voice, and mouse trace synchronization represented by a color gradient. The project website [60] contains a visualizer with many live examples.

	COCO Train			mIoU					Grounding mIoU				
	label	mask	cap.	A-847	PC-459	A-150	PC-59	COCO	A-847	PC-459	A-150	PC-59	COCO
ALIGN	✗	✗	✗	4.8	3.6	9.7	18.5	15.6	17.8	21.8	25.7	34.2	28.2
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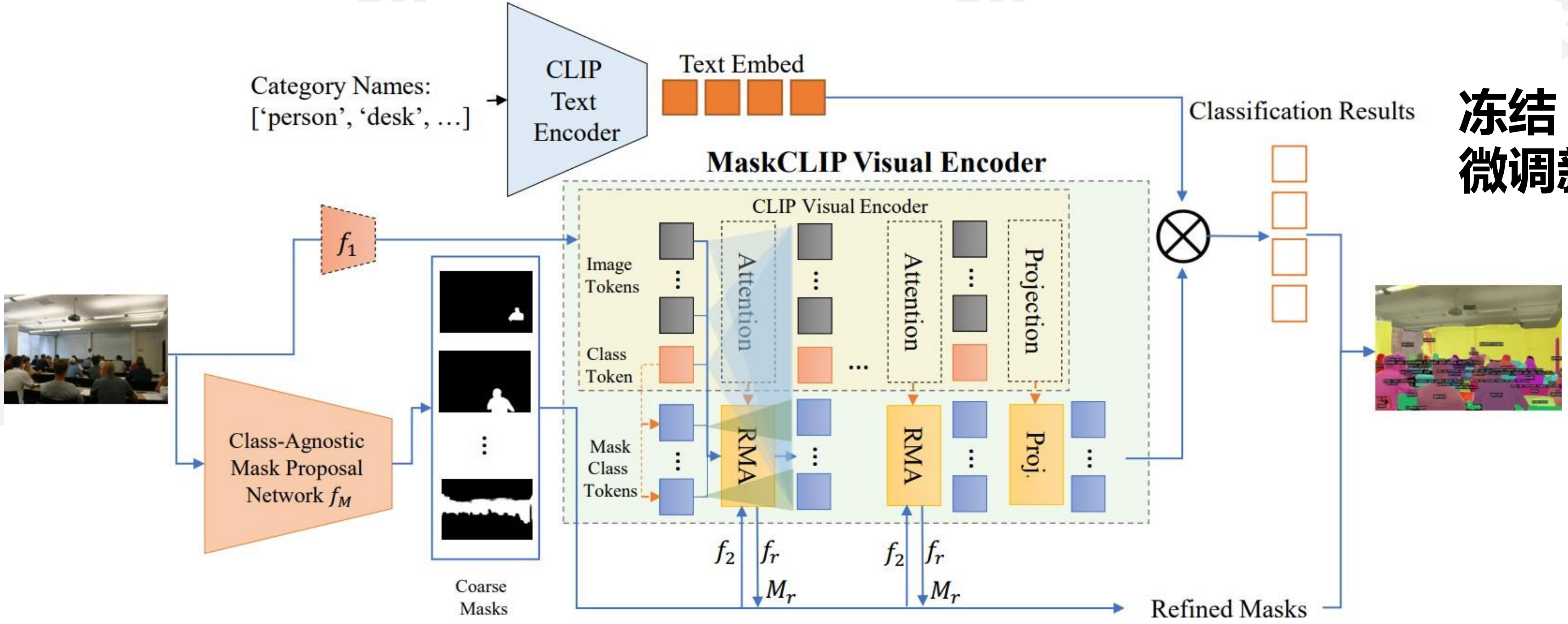
LSeg+	✓	✓	✗	3.8	7.8	18.0	**46.5**	55.1	10.5	17.1	30.8	56.7	60.8
OpenSeg	✗	✓	✓	6.3	9.0	21.1	42.1	36.1	21.8	32.1	41.0	57.2	48.2
OpenSeg w/L. Narr.	✗	✓	✓	**6.8**	**11.2**	**24.8**	45.9	38.1	**25.4**	**39.0**	**45.5**	**61.5**	48.2

带有局部描述额外数据可以进一步提升性能

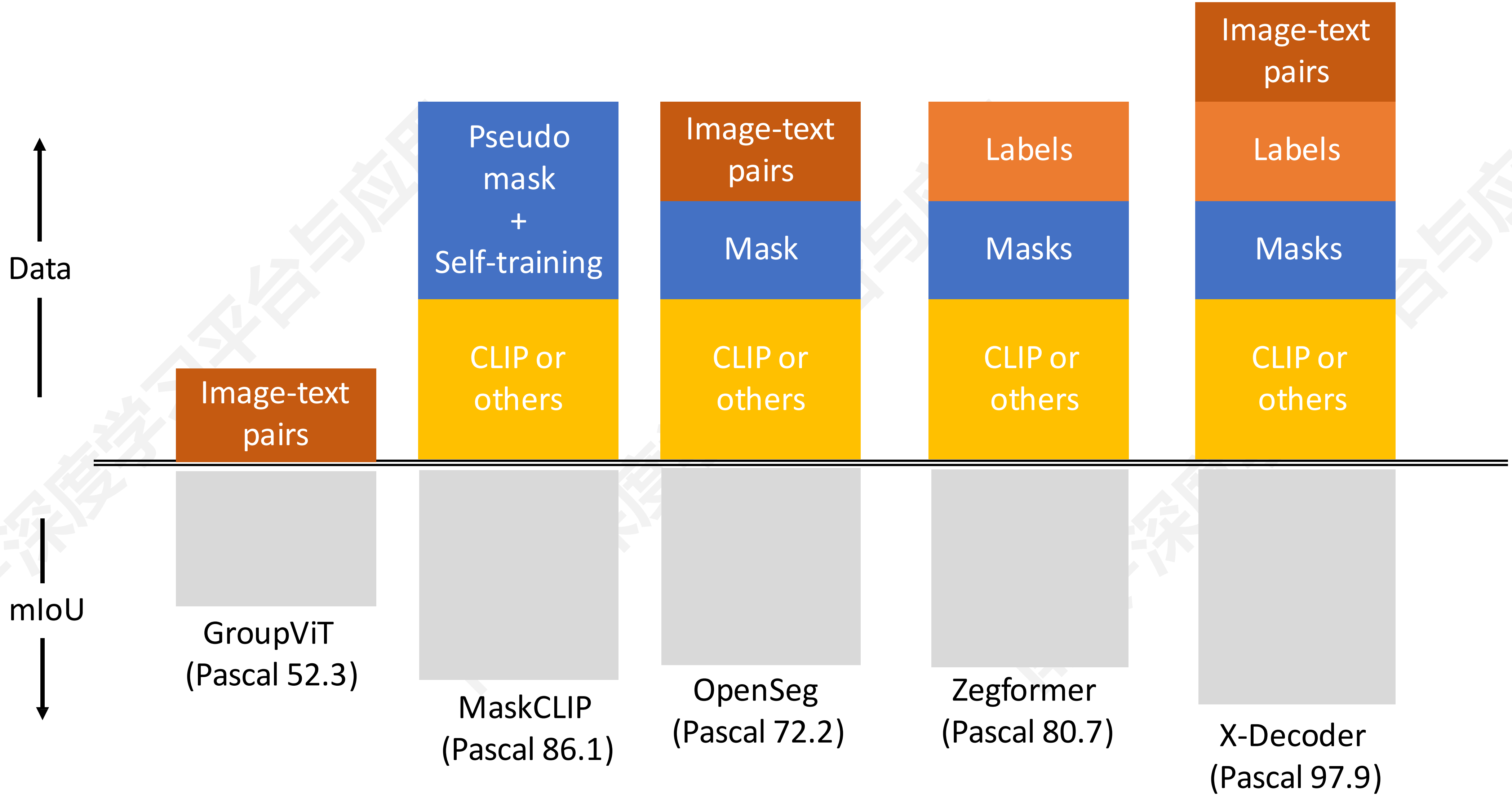
建立视觉和语言的联系（图像分割）

- **MaskCLIP(UCSD):** 使用 CLIP 作为初始化，在 COCO 上训练全景分割模型（两阶段训练：Mask proposal & CLIP adaptation）

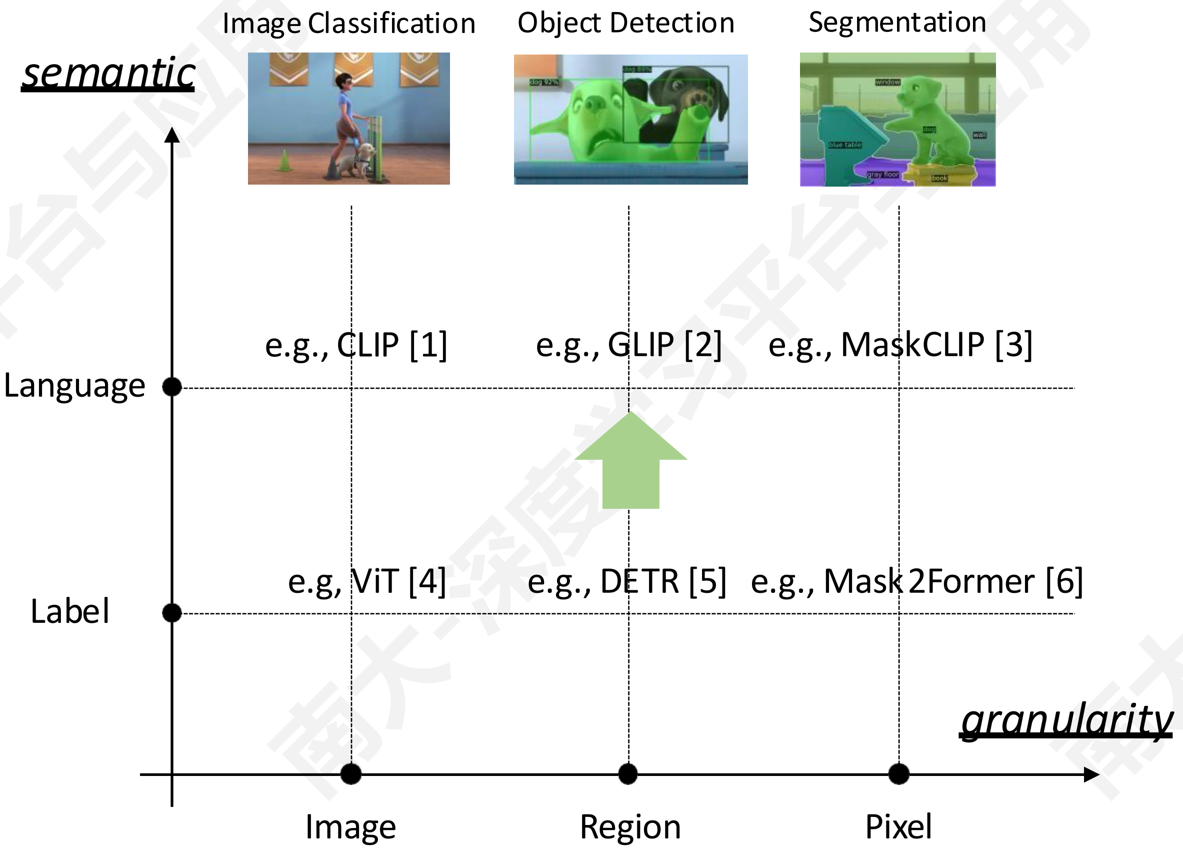


冻结 CLIP，微调新增参数

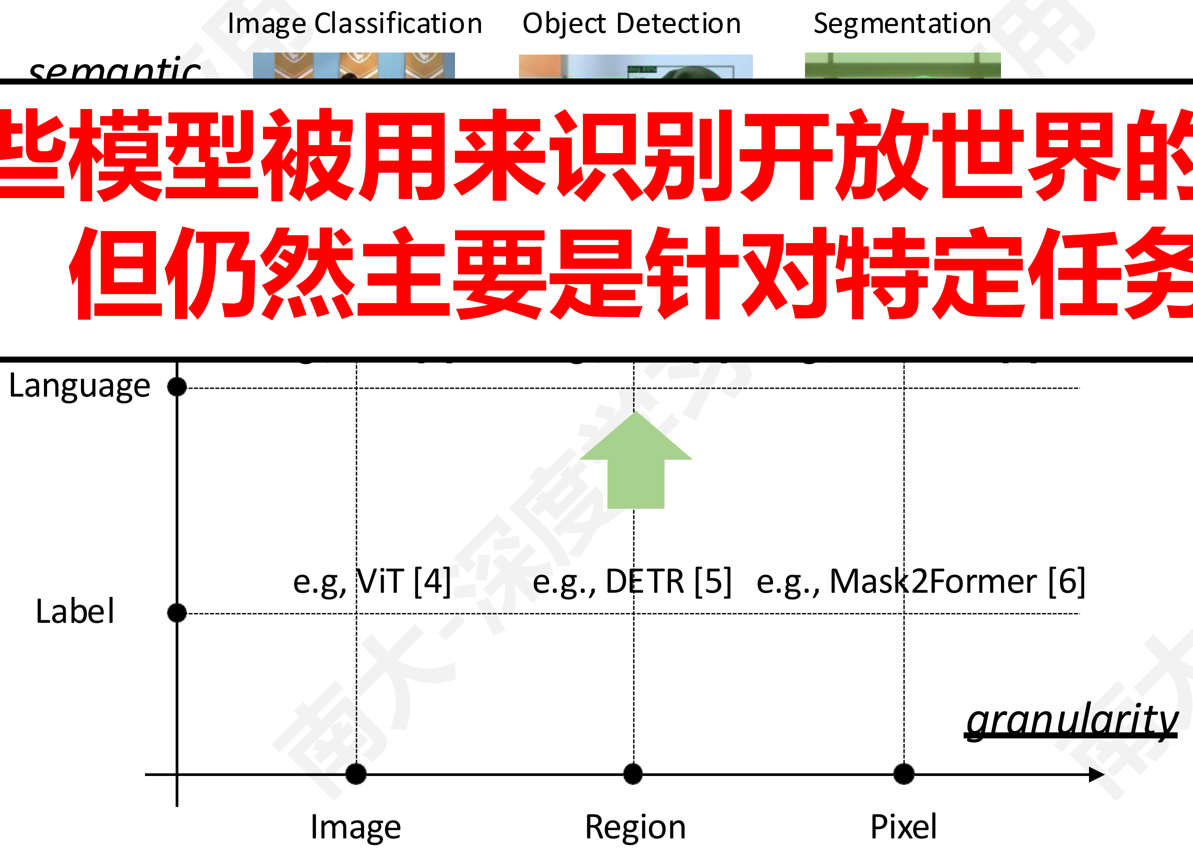
建立视觉和语言的联系（图像分割）



建立视觉和语言的联系（更多视觉任务）



建立视觉和语言的联系（更多视觉任务）



这些模型被用来识别开放世界的概念，但仍然主要是针对特定任务的

建立视觉和语言的联系（更多视觉任务）

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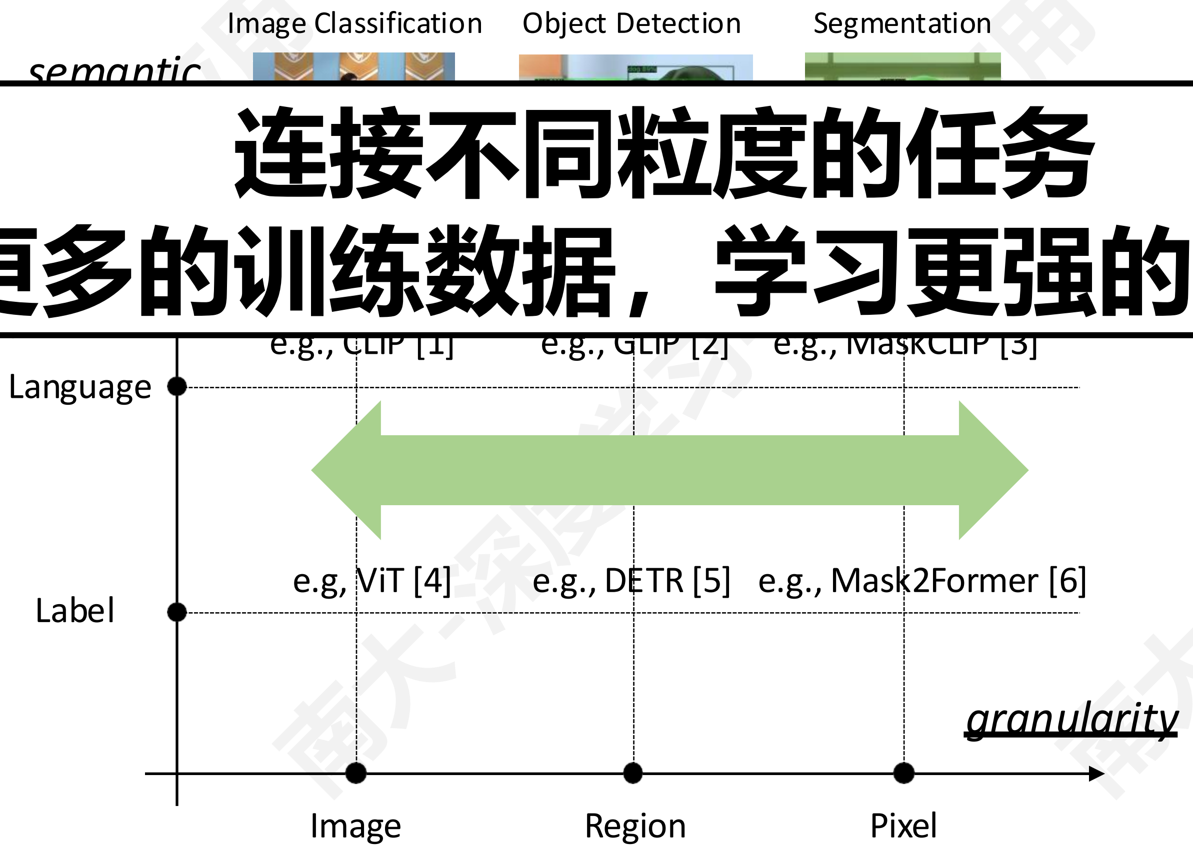
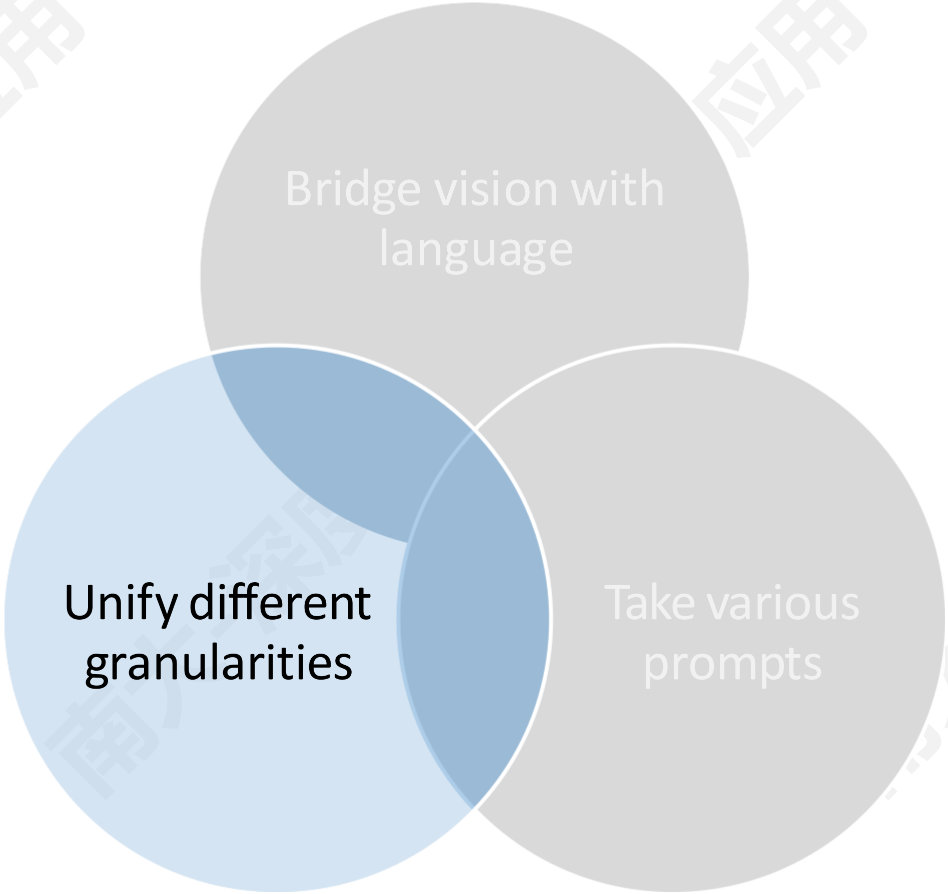


连接不同粒度的任务
更多的训练数据，学习更强的特征

统一不同任务粒度


Bridge vision with language
Unify different granularities
Take various prompts

统一不同任务粒度

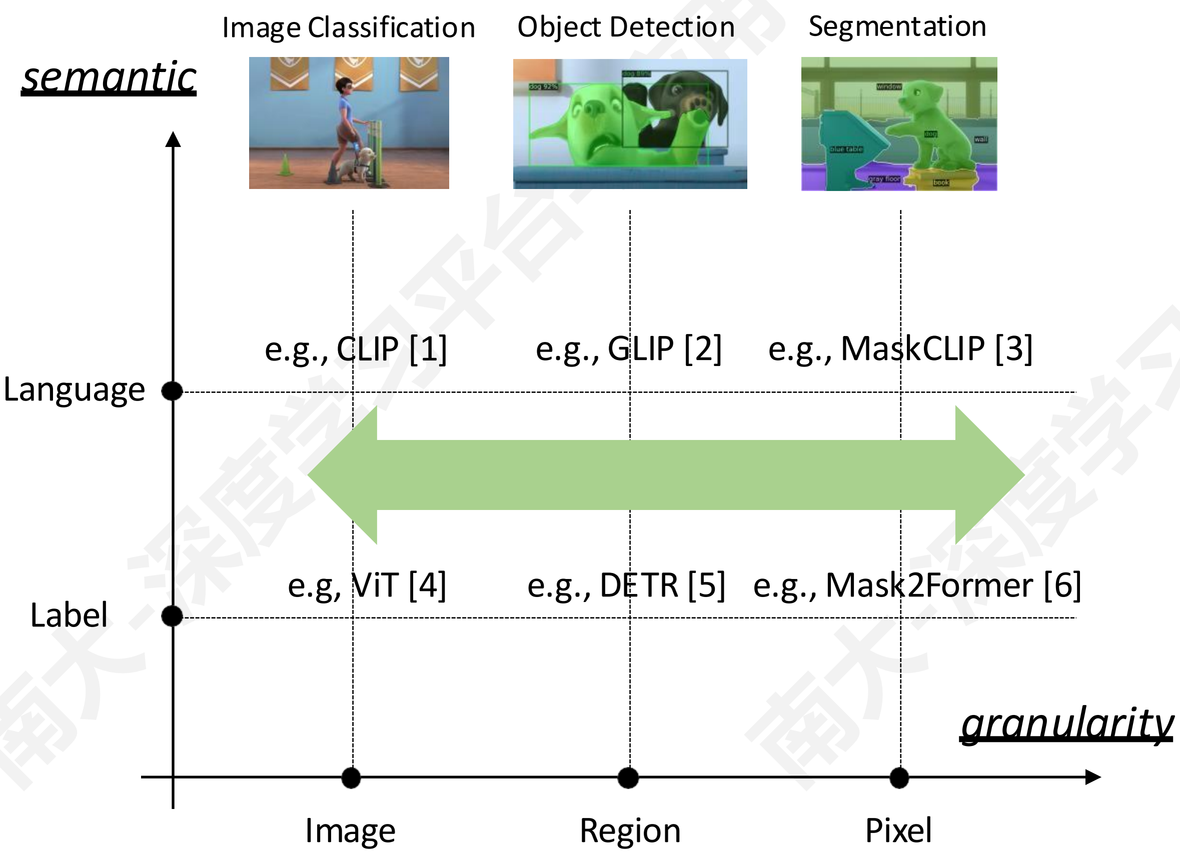
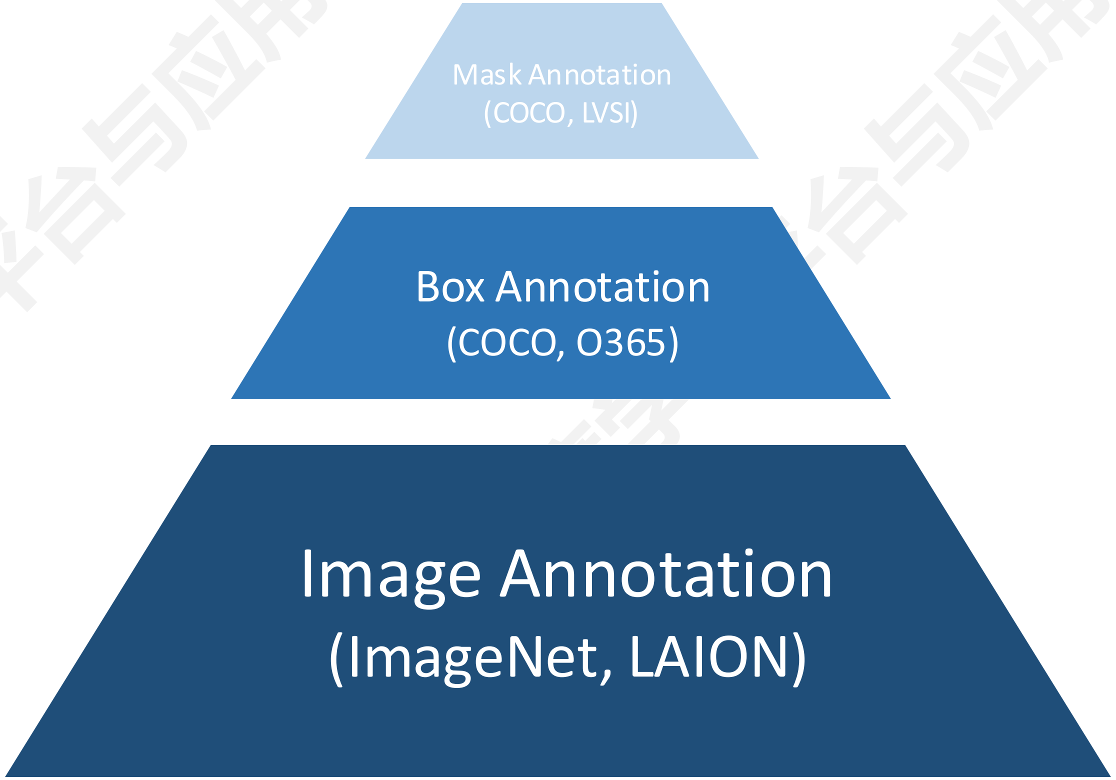
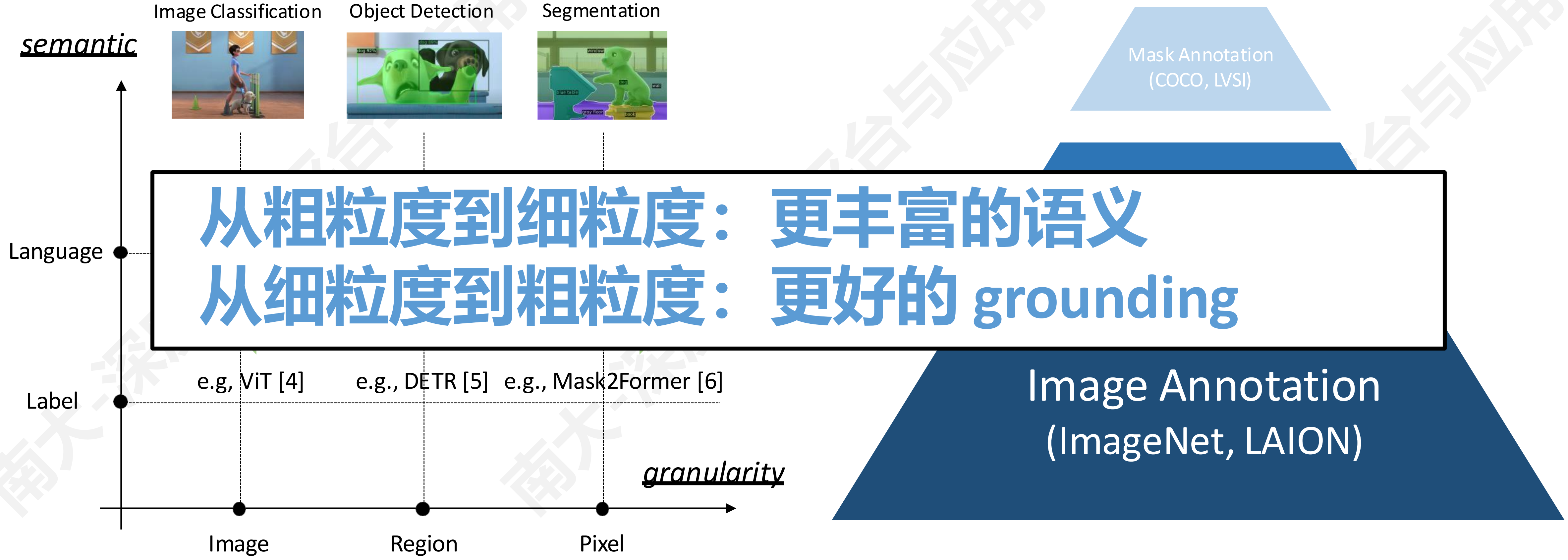


Image Classification
Object Detection
Segmentation
semantic
e.g., CLIP [1]
e.g., GLIP [2]
e.g., MaskCLIP [3]
Language
e.g, ViT [4]
e.g., DETR [5]
e.g., Mask2Former [6]
Label
granularity
Image
Region
Pixel



Mask Annotation
(COCO, LVSI)
Box Annotation
(COCO, O365)
Image Annotation
(ImageNet, LAION)

统一不同任务粒度



从粗粒度到细粒度：更丰富的语义

从细粒度到粗粒度：更好的 grounding

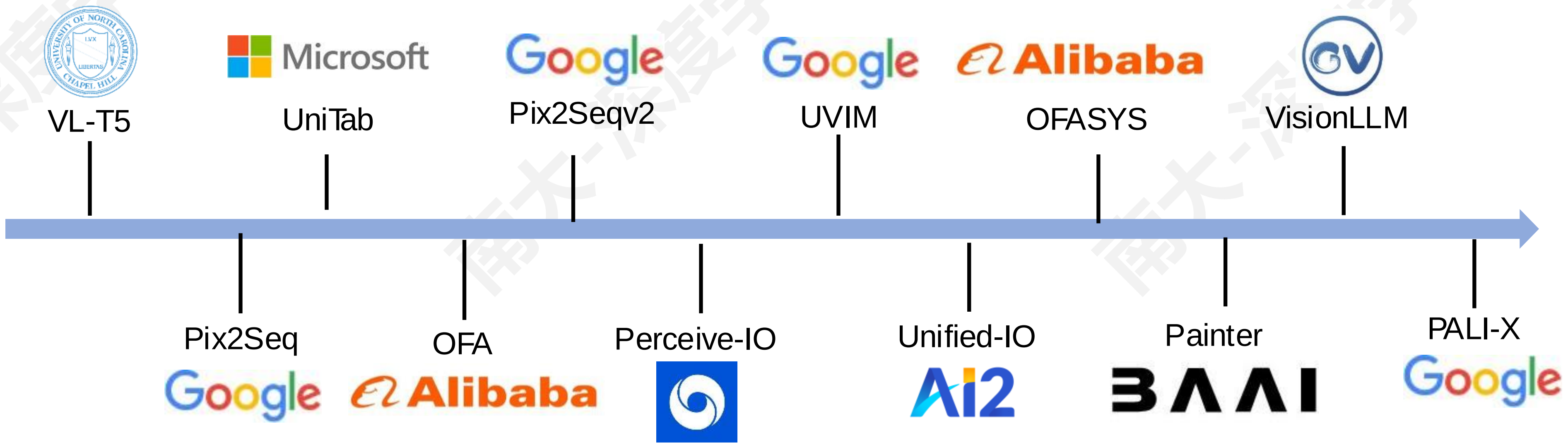
Grounding: 利用文本描述对图像中对应的视觉概念/物体进行定位

统一不同任务粒度

- 任务粒度：
  - 图像级: image recognition, image-text retrieval, image captioning, visual question answering, 等等
  - 区域级: object detection, dense caption, phrase grounding, 等等
  - 像素级: generic segmentation, referring segmentation, 等等
- 两种统一的方式：
  - 输出统一（+模型统一）：将所有任务的输出都转换为序列 tokens（输出的格式相同）
  - 模型统一（输出不统一）：不同粒度的任务共享模型（输出的格式不同）

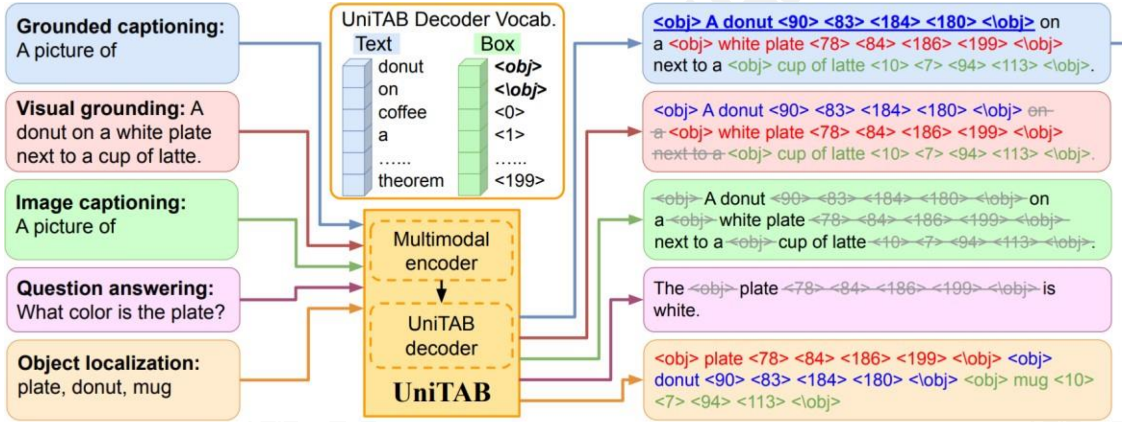
输出统一

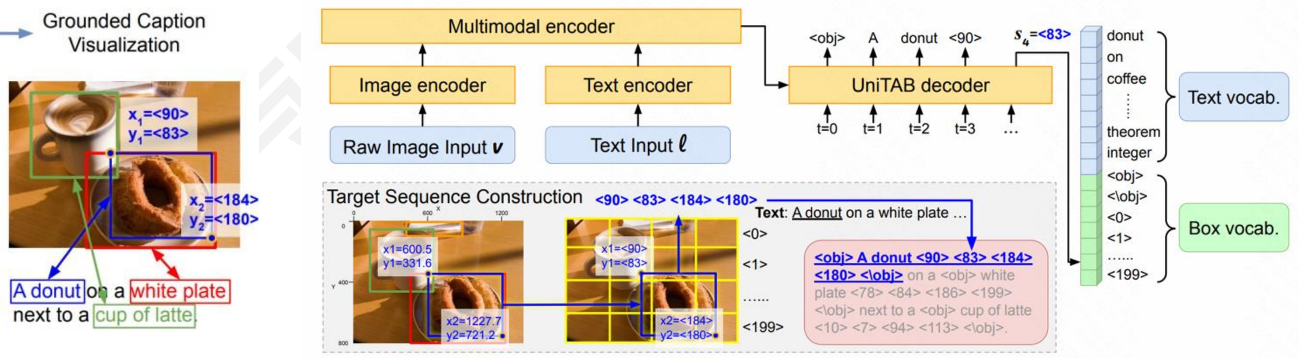
- **将输入和输出转换为序列：**
- **输入**：输入文本或添加一些前缀；图像特征
- **输出**：Box：坐标序列（左上+右下）；Mask：mask 多边形坐标序列；Keypoint：坐标序列

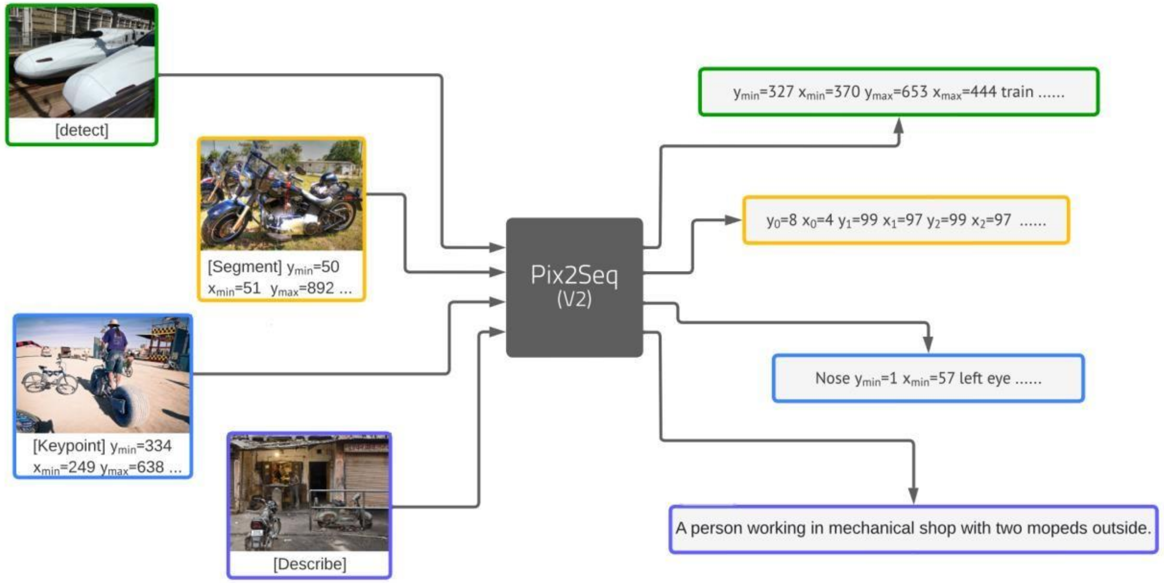


输出统一

UniTAB and Pix2Seqv2: 统一输出格式



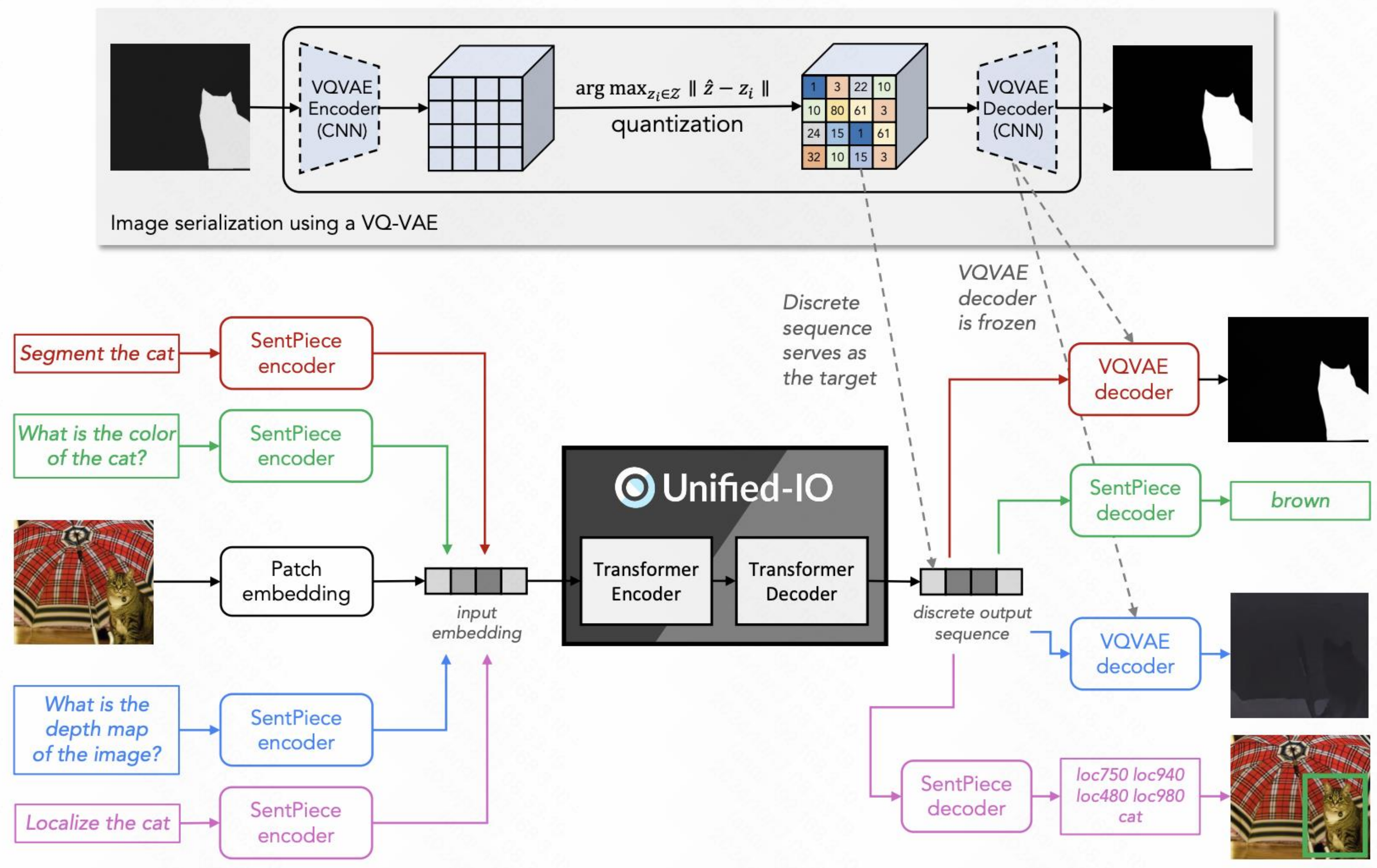




- **共享的词汇表**: 文本和坐标都转为 token 序列之后放进相同的词汇表中
- **任务前缀**: 使用任务前缀来确定模型正在处理的任务

输出统一

- **Unified-IO**
- **统一多种任务**
- **输入和输出都统一**
- **Step 1: 预训练 VQVAE**
- **Step 2: 在多个任务上进行联合训练**

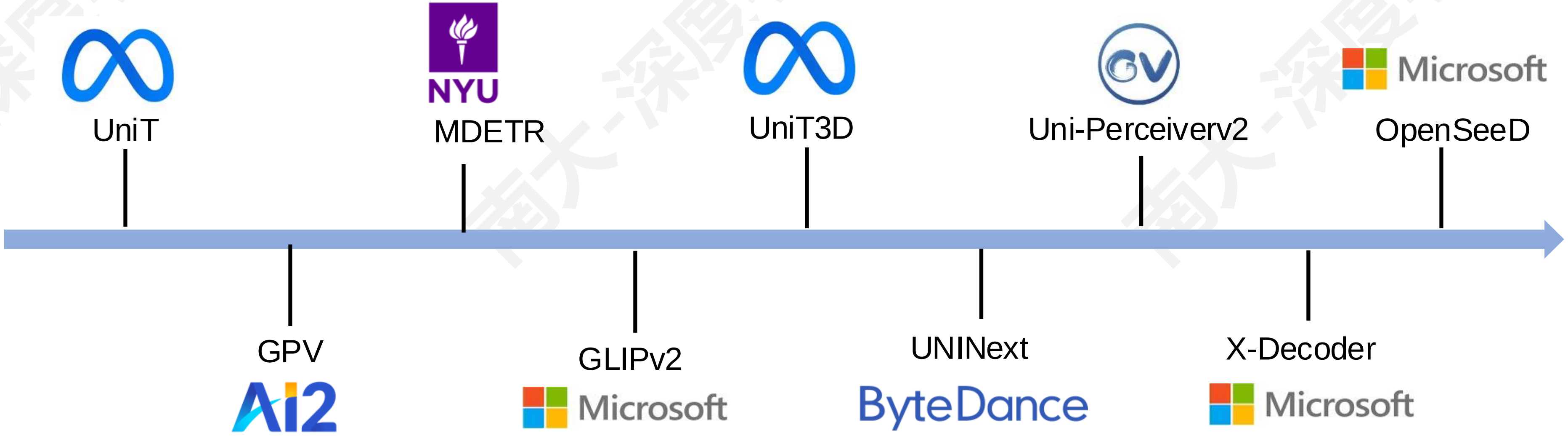


输出统一

- 其他的工作例如 VisionLLM 使用 LLM 作为输出端口
- 统一了多样的视觉任务，可以使用一个 encoder-decoder 模型进行端到端训练
  - 需要 task-specific decoder 将序列解码为最终输出
    - 例如，将坐标并转换为 box，将多边形/输出图转换为 mask
  - 可能很难解释不同粒度的不同任务之间的相互作用
  - 可能无法建立强大的跨任务协同效应

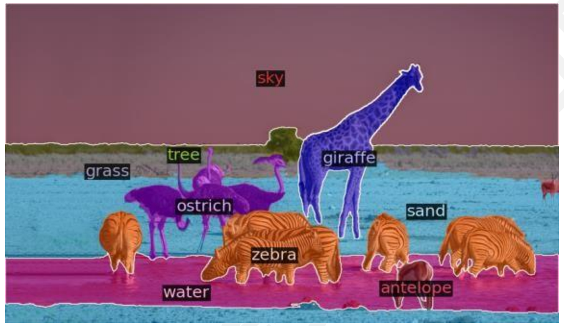
模型统一

- **视觉任务之间是有联系的：**
- **Box 预测: generic object detection, phrase grounding, regional captioning**
- **Mask 预测: instance/semantic/panoptic segmentation, referring segmentation**
- **Semantic 预测: image classification, image captioning, regional captioning, detection, segmentation, visual question answering, image-text retrieval**



模型统一

■ X-Decoder: 像素、图像和语言共享模型

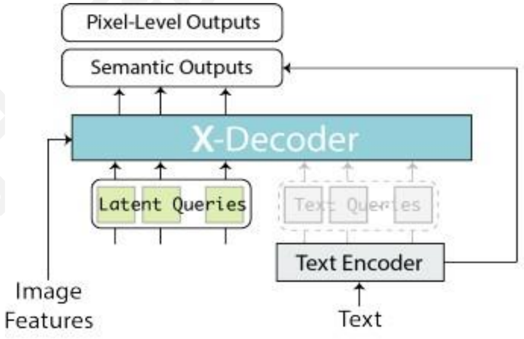


Query:Zebra,antelope,giraffe,ostrich,sky,water,grass,sand,tree

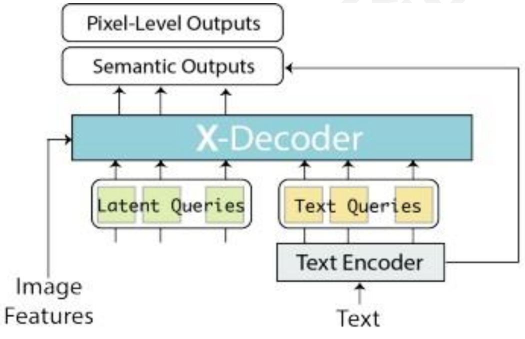
Query: Owl on the left

Query: The tangerine on the plate.

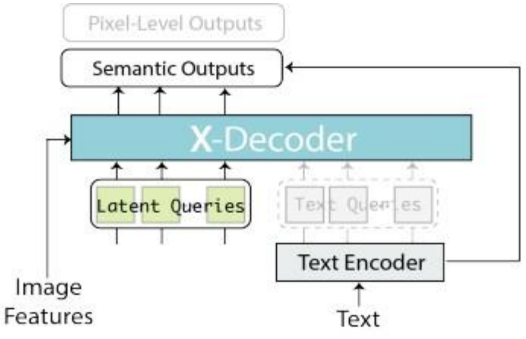
Cap: river in the mountains near the town



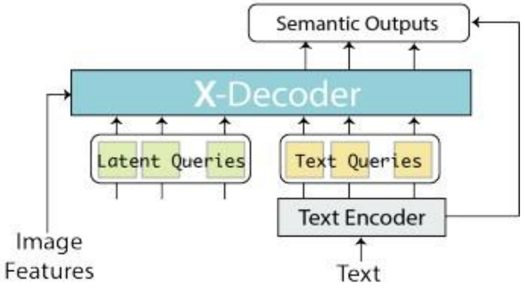
(a) Generic Segmentation



(b) Referring Segmentation



(c) Image-Text Retrieval



(d) Image Captioning/VQA

开放世界中的计算机视觉任务

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Image Classification in the Wild

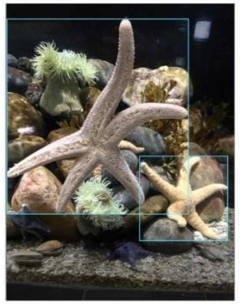
Example of knowledge sources

- Concept name: risotto
- Def_wik: An Italian savoury dish made with rice and other ingredients
- Def_wn: rice cooked with broth and sprinkled with grated cheese
- Path_wn: [risotto, dish, nutriment, food, substance, matter, physical_entity, entity]
- GPT3: ["A rice dish made with arborio rice and typically served with meat or fish.", "A rice dish made by stirring rice into a simmering broth]

Task preview

Object Detection in the Wild

Example of knowledge sources



- Concept name: starfish
- Def_wik: Any of various asteroids or other echinoderms (not in fact fish) with usually five arms, many of which eat bivalves or corals by everting their stomach.
- Def_wn: echinoderms characterized by five arms extending from a central disk
- Path_wn: [starfish, echinoderm, invertebrate, animal, organism, living_thing, whole, object, physical_entity, entity]
- GPT3: A marine animal of class Asteroidea, typically having a central disk and five arms.

Task preview

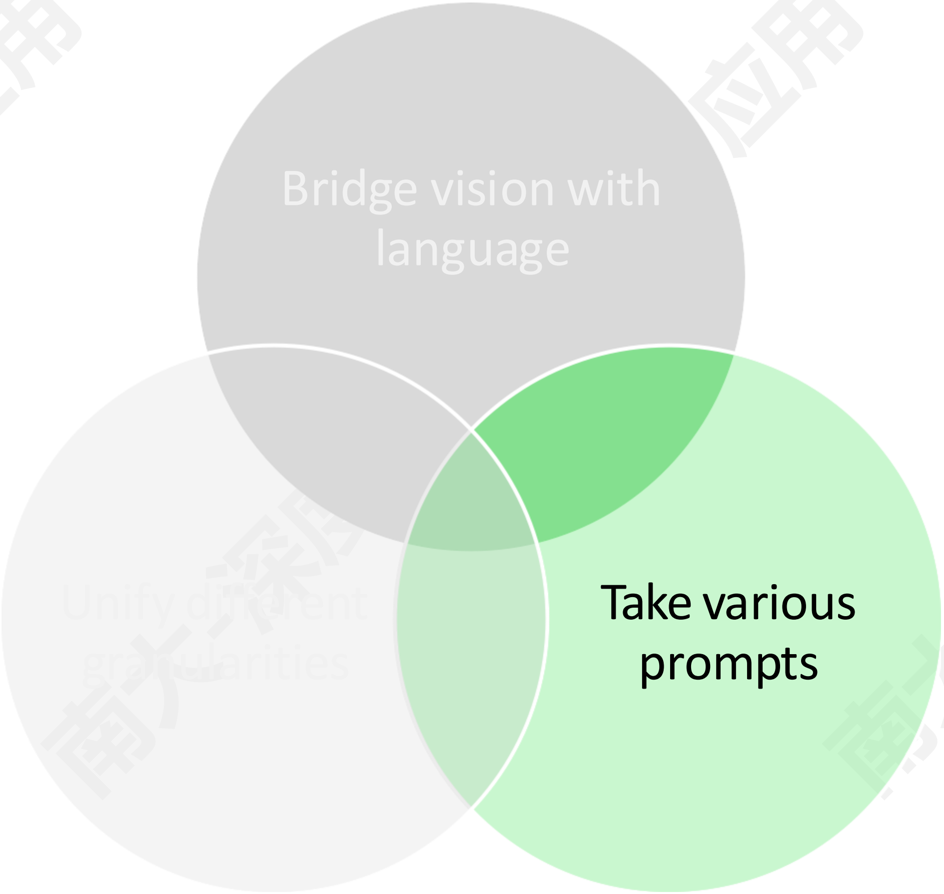
Segmentation in the Wild

Exemplar images in SGinW Benchmark

Task preview

2nd Workshop on Computer Vision in the Wild, East Ballroom B, June 19th full day

不同的提示



如何让视觉模型与人“交谈”？

Decoder LLMs (e.g., GPT) —Human-AI Interaction→ Conversational AI (e.g., ChatGPT)

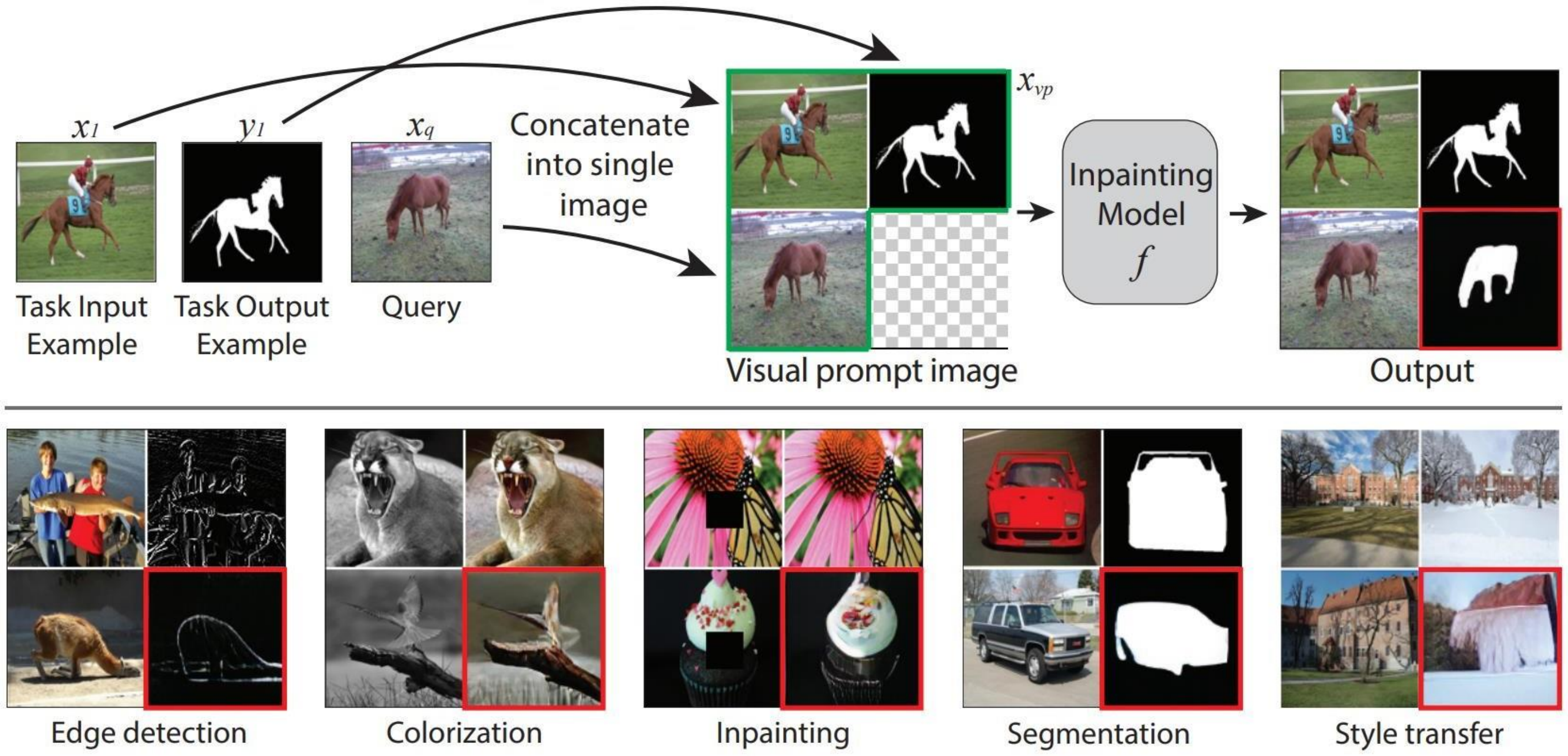
Generalist Vision Models —Human-AI Interaction→ ?

如何让视觉模型与人“交谈”？

- 需要构建一个具有两个重要属性的可提示接口：
- **便于 in-context learning**：与其微调模型参数，不如简单地提供一些上下文信息，使模型能够根据人类意图进行精准预测
- **交互式用户友好界面**：人类和人工智能之间的多轮交互对于完成复杂的任务非常重要

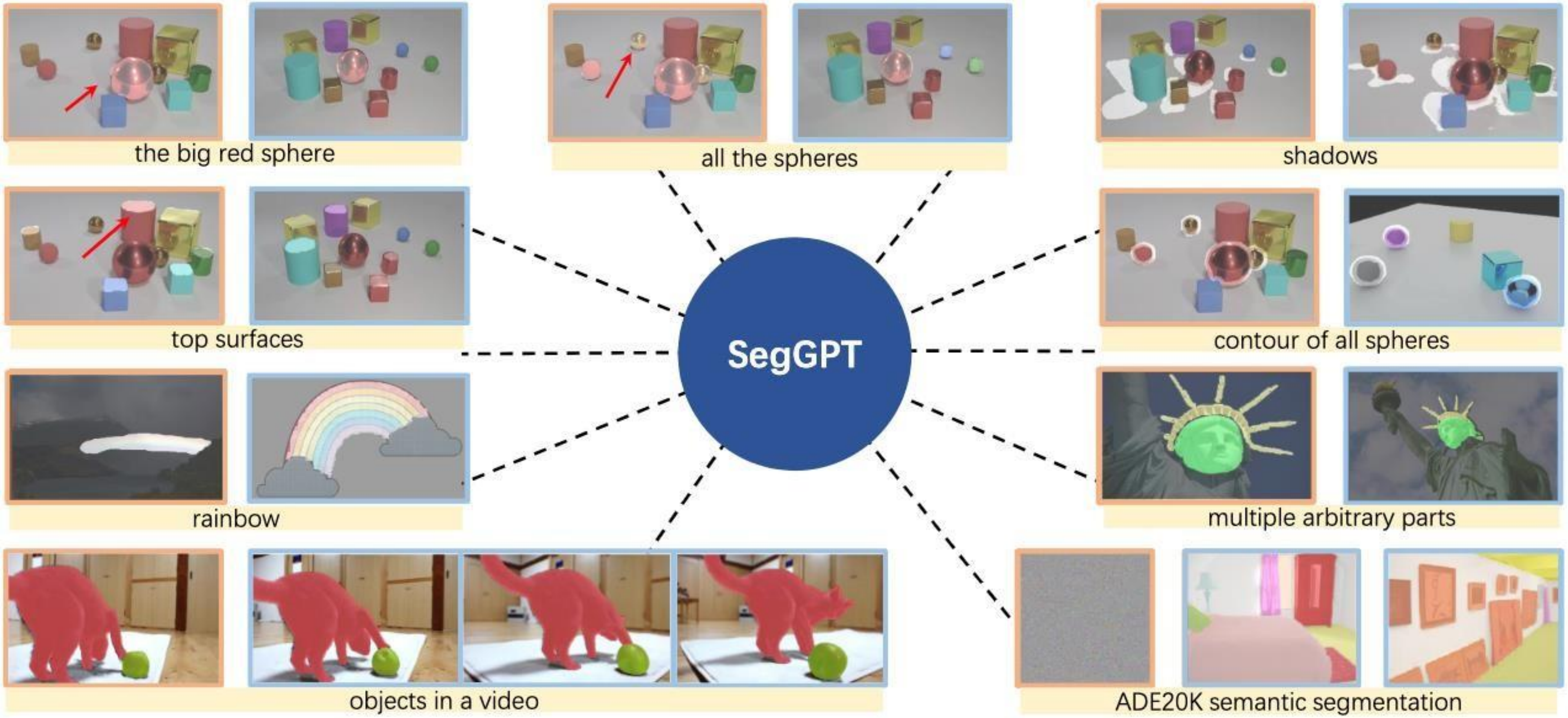
视觉中的 In-Context Learning

- 使用图像修复作为视觉提示（Visual Prompting）



视觉中的 In-Context Learning

- **SegGPT: 使用 In-Context Learning 进行分割**



视觉的可交互接口

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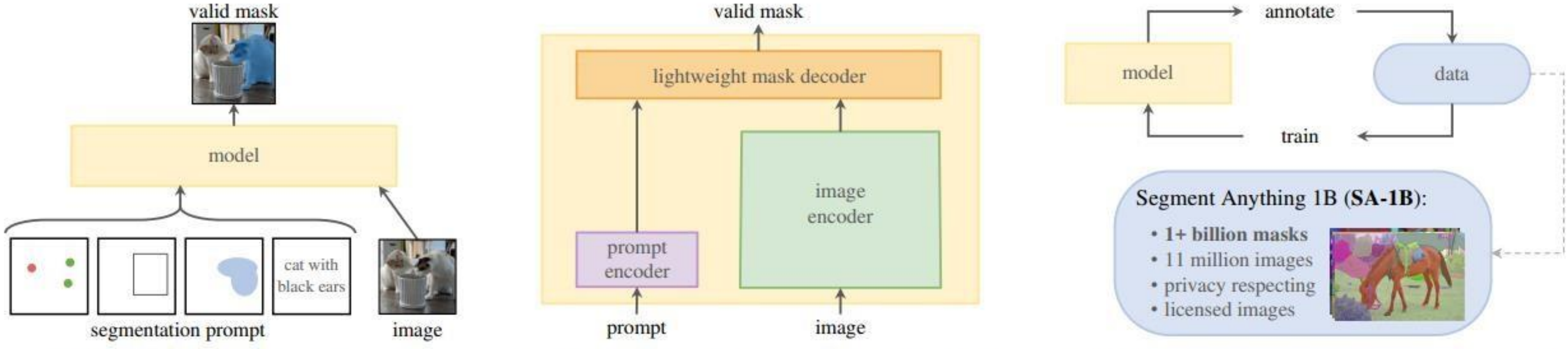
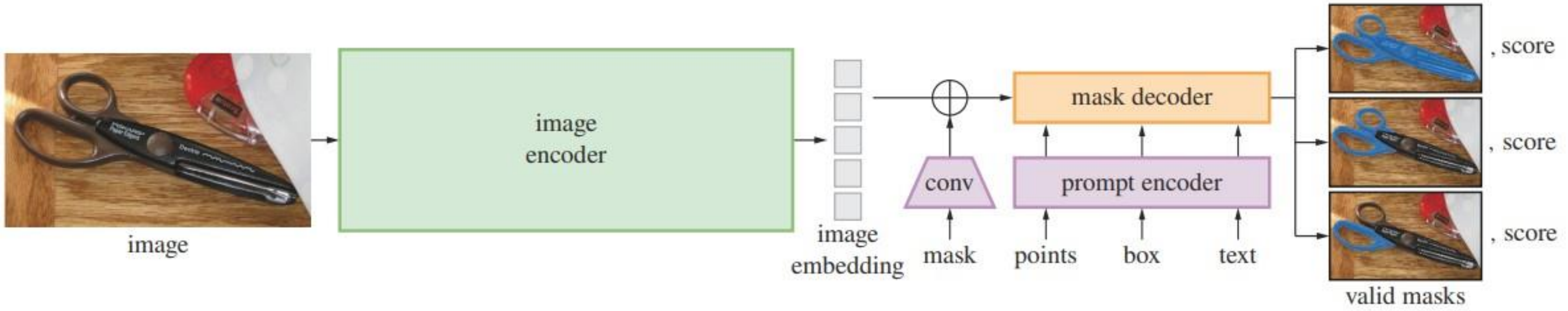
■ **SAM: 可交互的分割（~6000 citations)**



(a) **Task**: promptable segmentation (b) **Model**: Segment Anything Model (**SAM**) (c) **Data**: data engine (top) & dataset (bottom)



视觉的可交互接口

- **SEEM: Segment Everything Everywhere all at Once**
  - 数据集：MS COCO，引用量：370

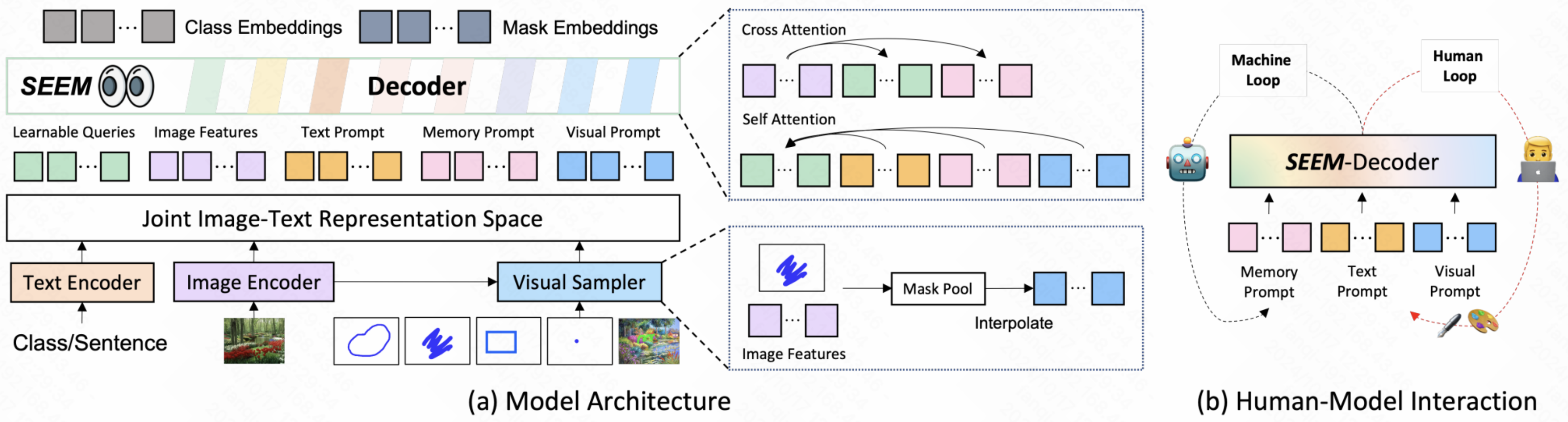
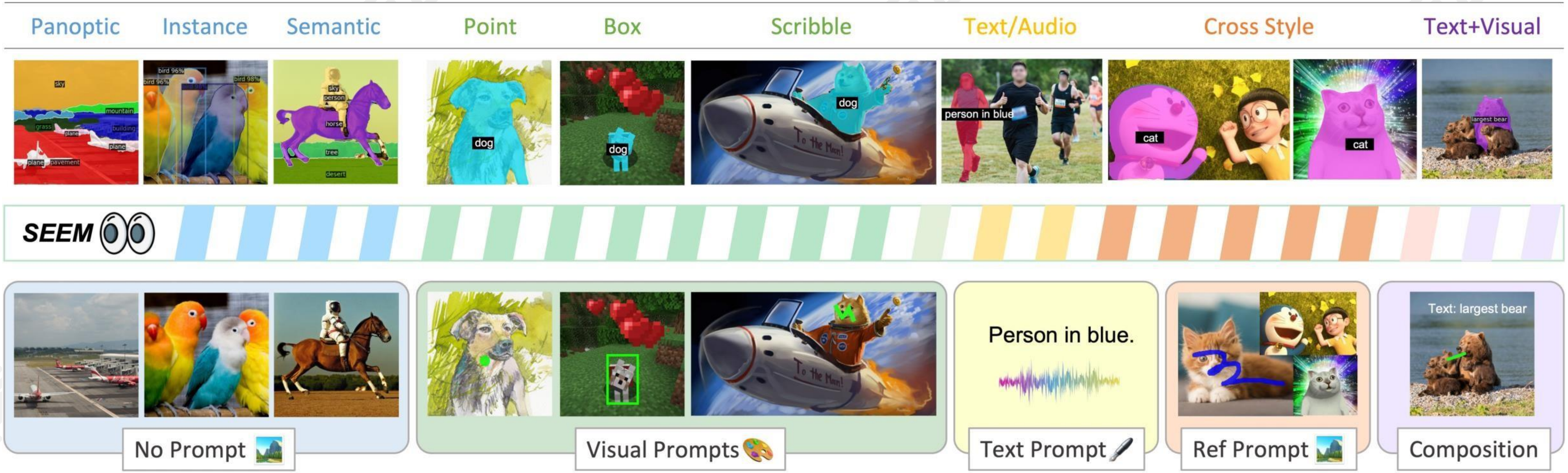


Figure 2: Overview of *SEEM*- Decoder. (a) *SEEM* encodes image, text, and human inputs into *joint visual-semantic space* as queries, features, and prompts, and then decodes queries to class and mask embeddings. (b) With the benefit of *SEEM* decoder, the machine loop enables memorizing history mask information, and the human loop provides new corrections to the next round.

视觉的可交互接口

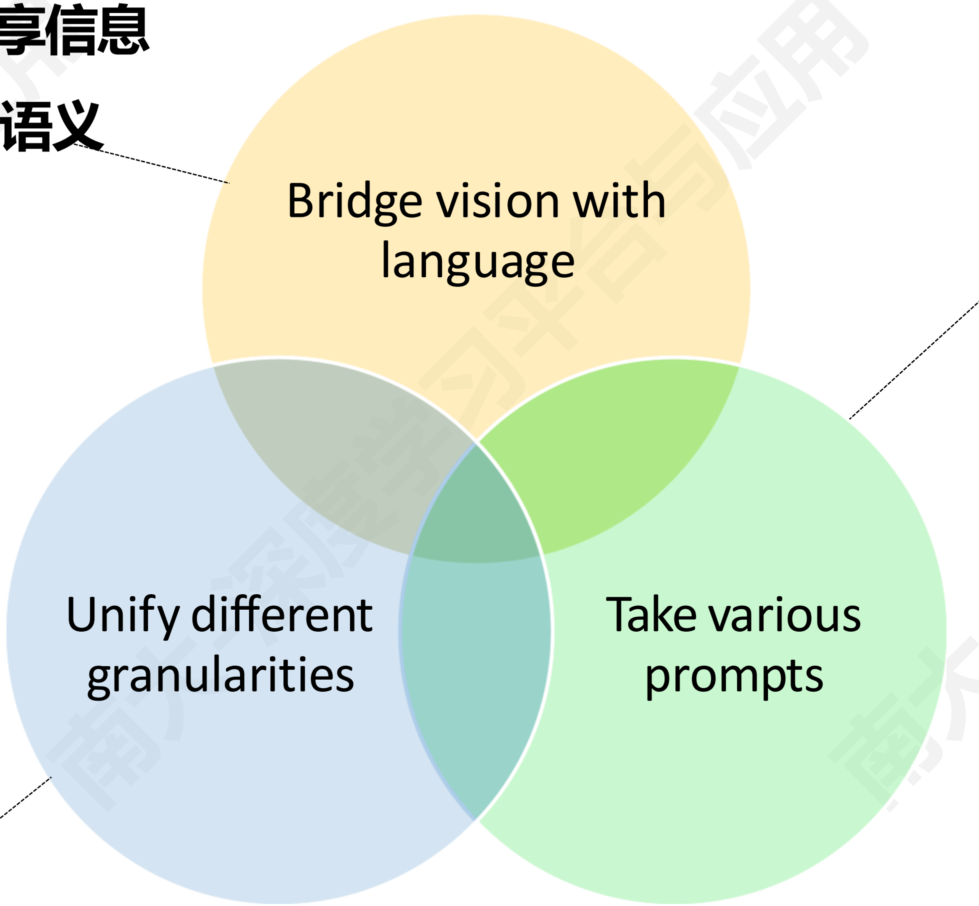
■ **SEEM: Segment Everything Everywhere all at Once**



回顾：通用视觉模型的尝试

直觉: 语言特征空间存在共享信息

优势: 可以零样本迁移到新语义



直觉: 语言、空间提示（prompt）等

优势: 降低模型理解人类意图的歧义

直觉: 视觉是多任务、多粒度的

优势: 不同粒度的视觉任务之间可以进行协同

启发通用人工智能（AGI）

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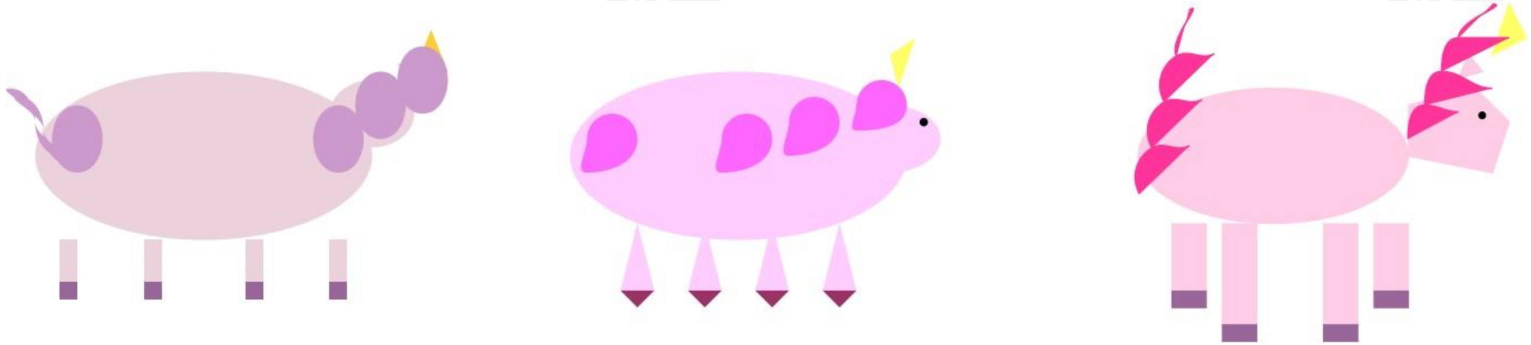
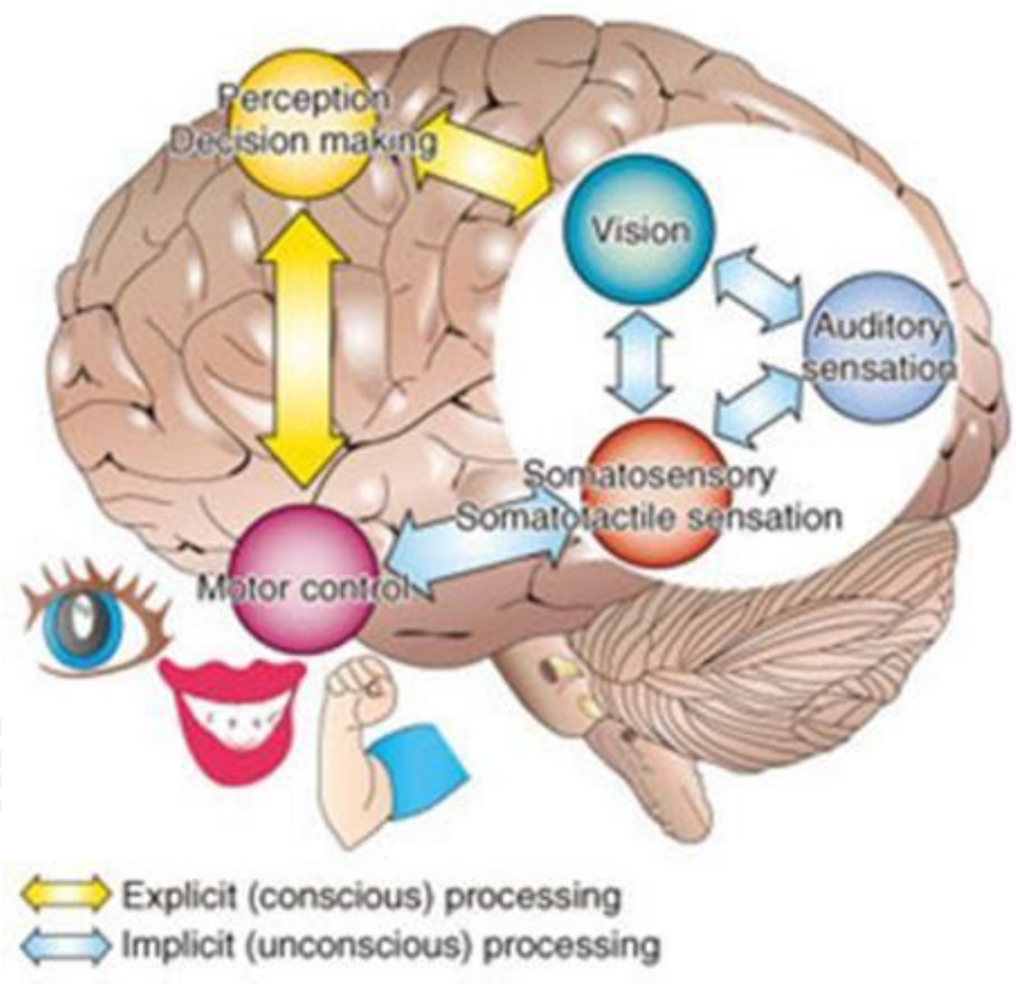
Sparks of Artificial General Intelligence: Early experiments with GPT-4

Sébastien Bubeck　Varun Chandrasekaran　Ronen Eldan　Johannes Gehrke
Eric Horvitz　Ece Kamar　Peter Lee　Yin Tat Lee　Yuanzhi Li　Scott Lundberg
Harsha Nori　Hamid Palangi　Marco Tulio Ribeiro　Yi Zhang

Microsoft Research

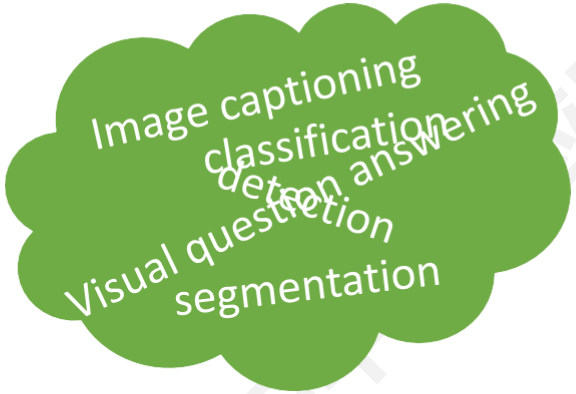


通用人工智能（AGI）



- Natural Language Processing
- Computer Vision
- Auditory sensation - Speech
- Motor control - Action
- …

通用视觉智能



Vision

— Unification → …… — Human-AI Interaction → ……

还有很大的想象空间！

与大语言模型合作完成物理世界的任务：

赋予 GPT, ChatGPT, BioGPT 视觉能力

增强对图像/视频的处理能力：

让DALLE-1/2不仅能生成图像，而且符合现实

实现多感官通用智能 agent：

一个真正能看、能说、能行动的 agent